

Surpac Tutorial

CONTENTS

Building of geological database in SURPAC

1. Introduction.....	02
1.1 Geological database in SURPAC.....	03
1.2 Mandatory tables.....	04
1.3 Optional tables.....	04
2. Creating database Tutorial.....	04-22
2.1. Creating database structure in SURPAC.....	04
2.1.1 Setting up your work directory.....	04
2.1.2 Creating a new database.....	06
2.2. Importing drill-hole data in SURPAC.....	11
2.2.1 Displaying the drill holes.....	15
2.2.2 Drillhole display styles.....	17
2.2.3 Labelling and Depth Marking.....	20
2.2.3.1 Depth Marking.....	20
2.2.3.2 Labelling.....	22
3. Drill hole layout.....	23
4. Creating a Topographical DTM.....	24
5. Extracting Surface Boundary.....	25
5.1 Expanding The Boundary.....	27
6. Contouring.....	29
7. Solid modelling.....	32-41
7.1 Sectioning and Digitising.....	32
7.1.1 Normalising the sections.....	35
7.1.2 cleaning the sections.....	35
7.1.3 Smoothing.....	36
7.2 Changing the orientation of the segments.....	37
7.3 Creating a Solid DTM.....	38
7.3.1 Validating the solid.....	40
7.3.2 Solid Volume Calculation.....	41
8. Compositing.....	43
8.1 Downhole compositing.....	43
9. Statistical analysis.....	46-49
9.1 Basic statistics.....	46
9.1.1 Probability curve.....	48
9.1.2 Normal Distribution Curve.....	48
9.1.3 Mean and co-efficient variation curve.....	49
10. Variogram modelling.....	50-56
10.1 Variogram map.....	52
10.2 Variogram model.....	56
11. Block modelling.....	58-75
11.1 Creating a new block model.....	58
11.1.1 Attributes.....	60
11.1.2 Partial Percentage.....	60
11.1.3 Specific Gravity.....	65
11.1.4 Fe attribute (Assay Attribute).....	65
11.2 Adding high grade ore Constraint.....	70
11.3 Volume calculation of block model.....	72
11.3.1 Reserve volume calculation.....	72

11.3.2 Resource volume calculation.....	75
11.4 Extracting bench slices from block model.....	75-78
12. PIT DESIGN	78-90
11.5.1 Set slope Gradient.....	79
11.5.2 Creating a ramp.....	81
11.5.3 Expanding by bench height.....	83
11.5.4 Expanding by berm width.....	84
11.5.5 Integrated pit configuration.....	87
11.5.6 Integrated pit configuration clipped with topography.....	90

Surpac Tutorial

Building of geological database in SURPAC

2. Introduction

Drill hole data is the starting point of the mine planning process. Geological data are generated from the field exploration as per an exploration plan (Figure 1)

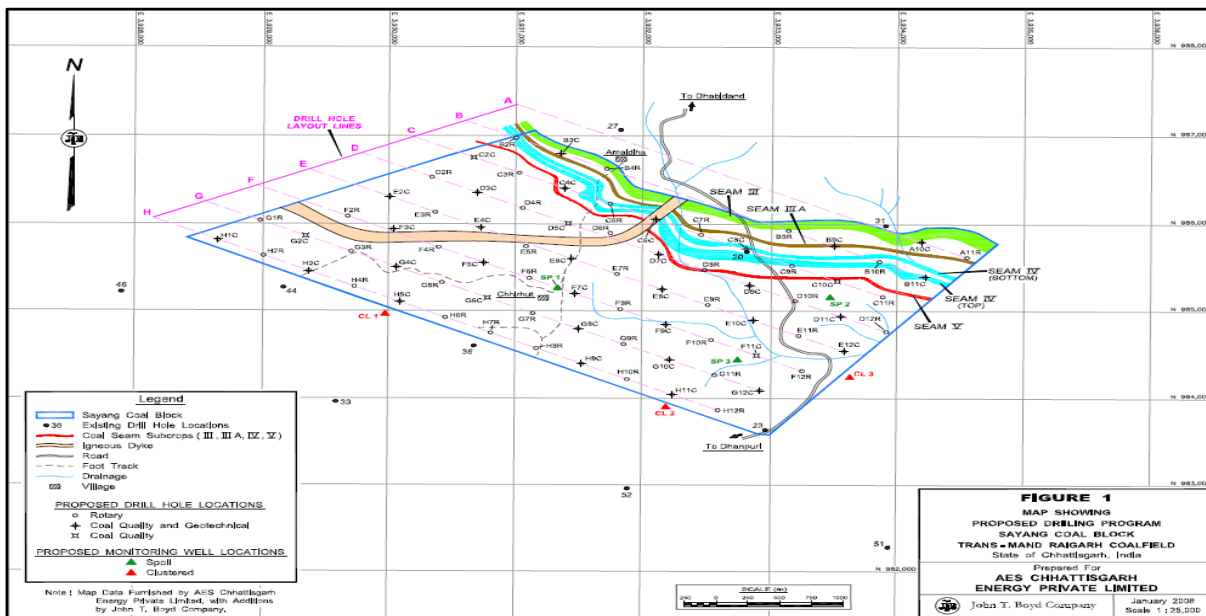


Figure 1. A typical geological exploration plan.

Geological data is typically received from field exploration in the following formats.

Table 1. Logging/Structure data

BORE HOLE NO.	SBH - 10/1	CO-ORDINATES	North h: 2499.393	East: 1075.995	R.L OF BH COLLAR :	609.892	DEPTH mtr	DRILLED :	88.10 mtr		
AZIMUTH : 0.00	TYPE OF DRILLING : 100% WET			Dip: -90°	DATE OF COMMENCEMENT : 05.12.2010	DATE OF CLOSURE :	11.12.2010	LOGGED BY Sanjiv & Srikant			
FROM (mtr)	TO (mtr)	length (mtr)	Lith 1	Lith 1 %	Lith 2	Lith 2 %	Weathering	Alteration	Mineralisation	Colour	Comments
1	2	3	4	5	6	7	8	9	10	11	12
0.00	2.10	2.10	CLAY BAND	60.00	HLIO	40.00	W4		HMT+IH	Light Brown	HLIO + Clay bands occurs as OB
2.10	5.10	3.00	SLIO	60.00	CLAY BAND	40.00	W4		HMT+IH	Dark Gray	SLIO+Laterite occur Ore occurrence
5.10	8.10	3.00	SLIO	90.00	LAT	10.00	W4		HMT+IH	Dark Gray	SLIO, Ore Continues
8.10	11.10	3.00	SLIO	100.00			W4		HMT	Dark Gray	SLIO, Ore Continues
11.10	14.10	3.00	SLIO	100.00			W4		HMT	Dark Gray	SLIO, Ore Continues
14.10	17.10	3.00	SLIO	100.00			W3		HMT	Dark Gray	SLIO, Ore Continues
17.10	20.10	3.00	SLIO	100.00			W3		HMT	Dark brown	SLIO, Ore Continues

Table 2. Quality / Assay data.

BORE HOLE NO.		SBH -10/1		BORE HOLE TYPE:		VERTICAL												
DATE SAMPLED:		16/12/2010		SAMPLER:		Dhiren		GEOLOGIST <u>Sanjiv & Srikant</u>										
Sample Number	FROM	TO	Interval Length	Condition	Weight	Type	Method	Fe	LOI	SiO ₂	Al ₂ O ₃	Assay Results (%)						Total
1	2	3	4	5	6	7	8	9	10	11	12	P	Mn	TiO ₂	CaO	MgO	S	19
SBH 10/1-1	0.00	2.10	2.10			CORE	1/2 CORE	49.69	7.96	9.28	7.31	****	****	****	****	****	****	
SBH 10/1-2	2.10	5.10	3.00			CORE	1/2 CORE	60.80	1.92	3.39	4.29	****	****	****	****	****	****	
SBH 10/1-3	5.10	8.10	3.00			CORE	1/2 CORE	65.96	1.43	1.61	1.89	****	****	****	****	****	****	
SBH 10/1-4	8.10	11.10	3.00			CORE	1/2 CORE	65.28	1.79	2.22	2.02	****	****	****	****	****	****	
SBH 10/1-5	11.10	14.10	3.00			CORE	1/2 CORE	67.30	1.21	0.76	1.49	****	****	****	****	****	****	
SBH 10/1-6	14.10	17.10	3.00			CORE	1/2 CORE	66.85	1.58	0.69	1.70	****	****	****	****	****	****	
SBH 10/1-7	17.10	20.10	3.00			CORE	1/2 CORE	66.96	1.30	0.61	1.97	****	****	****	****	****	****	

Geological database in SURPAC:

A geological database is an organized collection of drill-hole data. In SURPAC, drill-hole data are organized according to the relational model. The database is organized in the form of mandatory tables and optional tables (Figure 2).

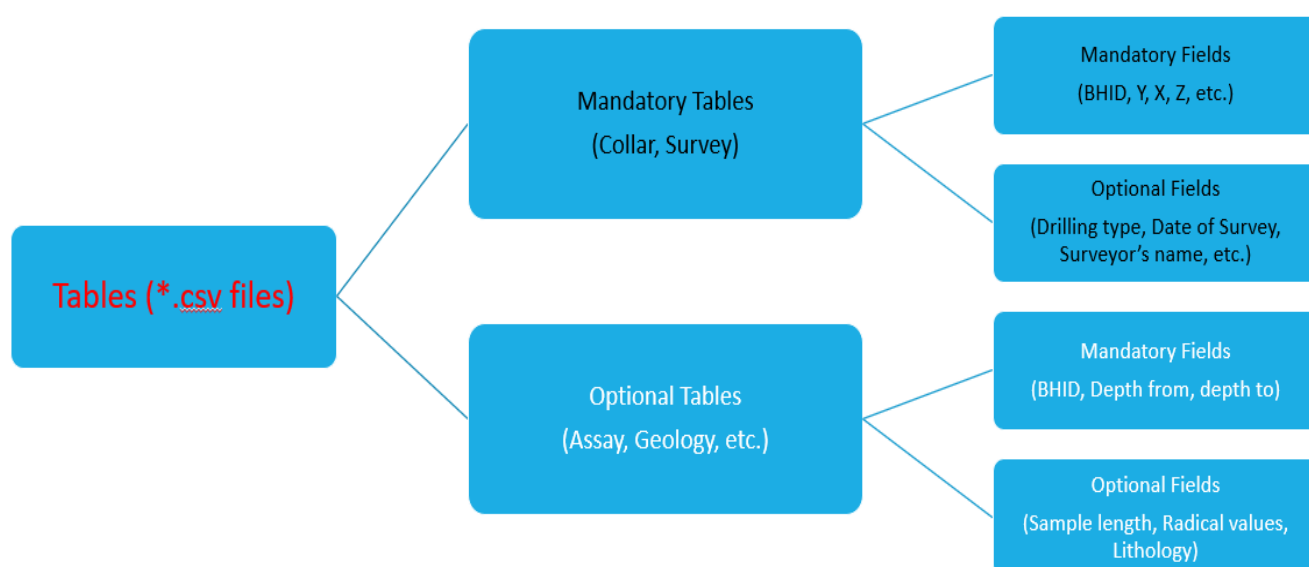


Figure 2. SURPAC tables

Mandatory tables:

- **Collar Table** – Shows the actual position of the drill holes in the form of “***hole I’d, Y, X, RL, and Maximum Depth***” as the mandatory fields.
- **Survey Table** – The table contains, “***hole I’d, Depth (0 – Max.), Dip, and Azimuth***”.
- **Translational Table** – This table is added by Surpac automatically during the build-up process.

Optional tables:

- **Assay table** – This table contains the fields for, “***hole I’d, Sample I’d, To, From, Grade, Sample type and so on***”.
- **Geology table** – This table contains the fields for, “***hole I’d, to, from, lithology 1. lithology 2 and so on***”.
- **Styles table** – This table is again created by SURPAC automatically by itself.

Tutorial # 2. Creating a database Tutorial

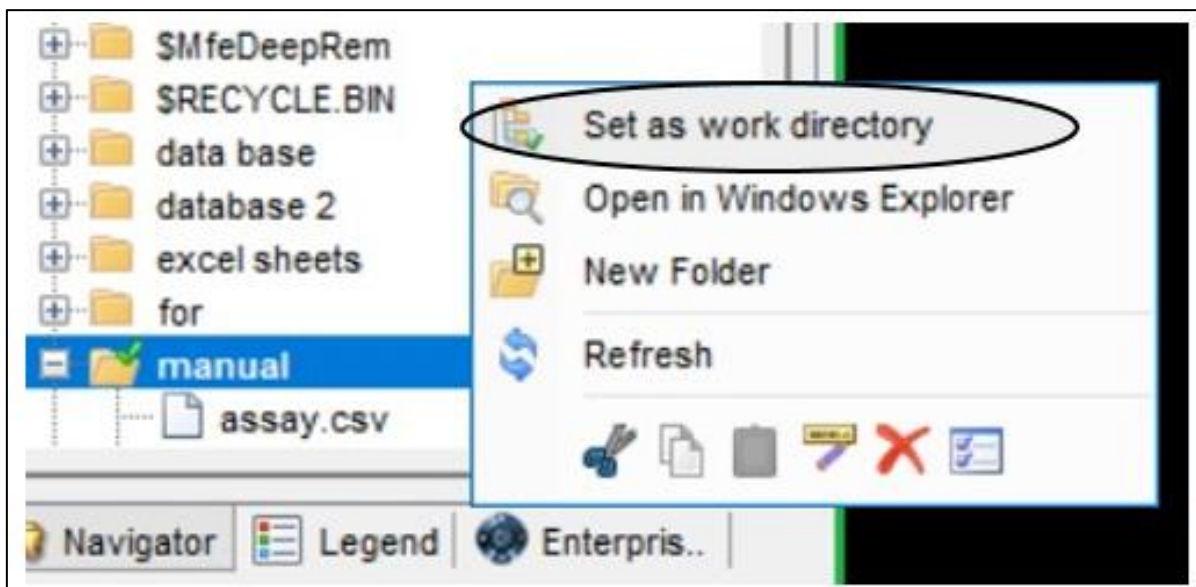
2.1. Creating database structure in SURPAC

Creating the database is the first thing to do before working on your exploration data. Surpac reads your exploration data in the form of a database only having a file extension (.ddb). One must be very careful during creating and importing the database because any missing column or data from the tables can create a malfunction in the project.

Step 1: Setting up your work directory

Work directory is the location on your device in which Surpac saves the file, once you are done with setting up a directory, the new files of whatever you performed in Surpac will be saved in that folder/location.

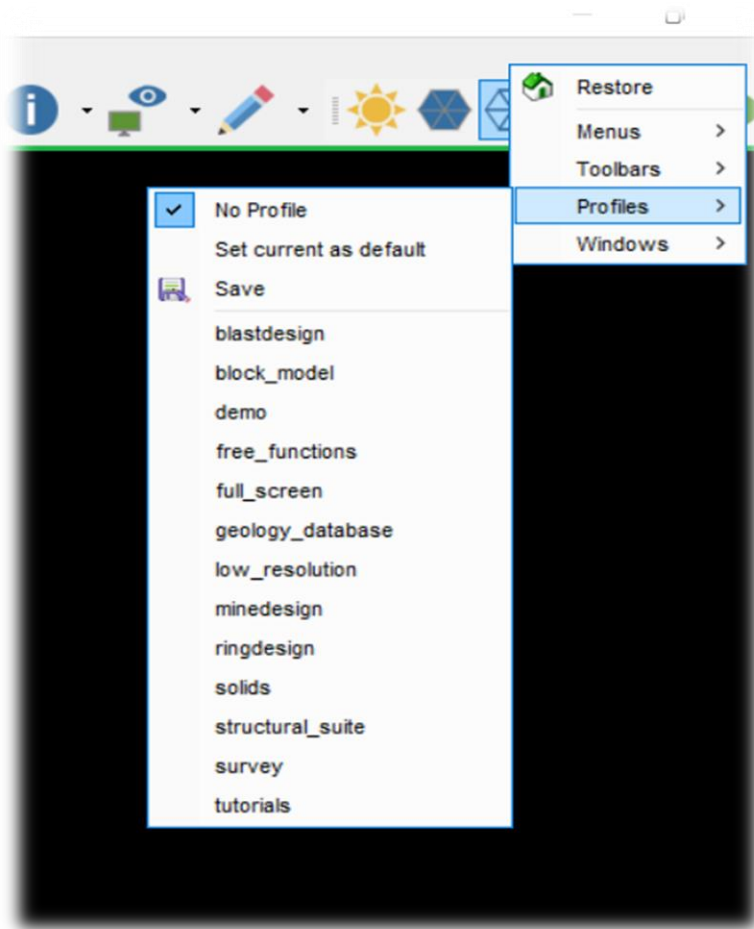
- **Click on Select the folder >> right click >> set as work directory**



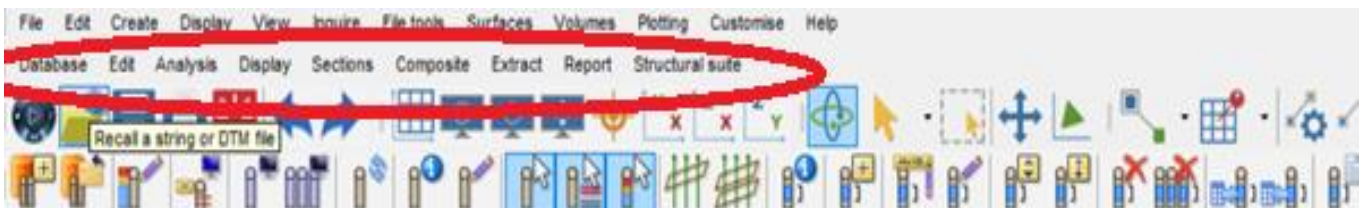
After setting up the working directory a tick mark should appear on the selected folder as shown in the above figure.

- The next step would be bringing up the geology database to your screen.

- Right click anywhere on the top >> Profiles >> **geology_database**

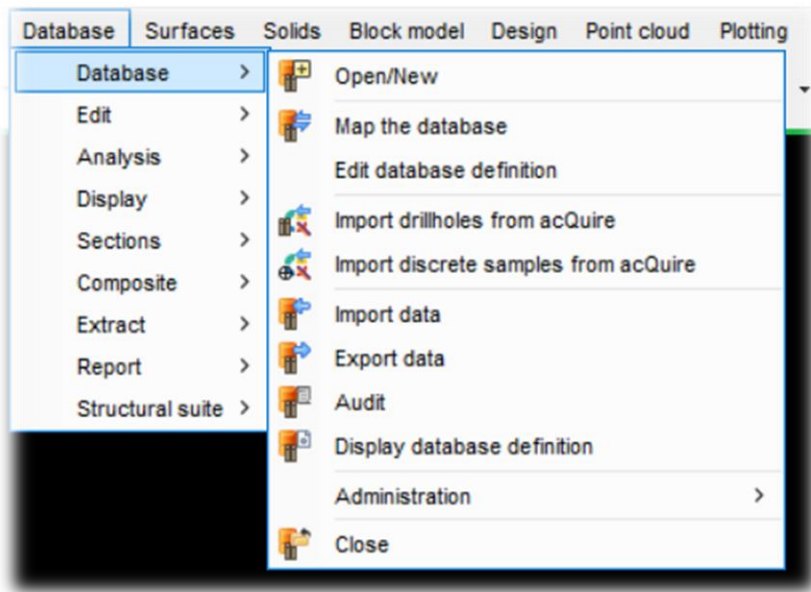


The geology_database would appear as shown in the figure below.

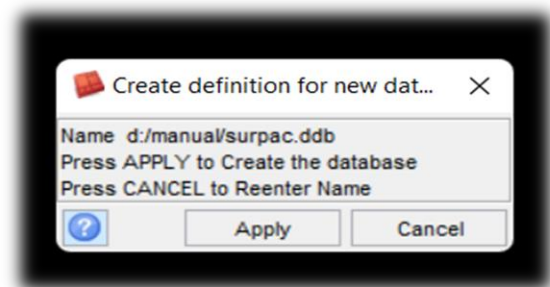
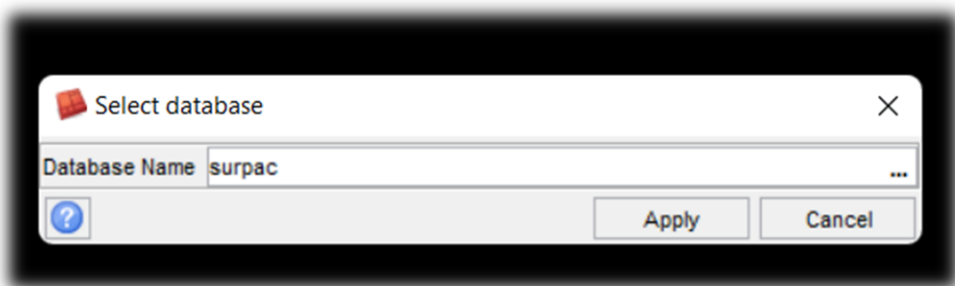


Step 2: Creating a new database

➤ Click on Database >> Open/New

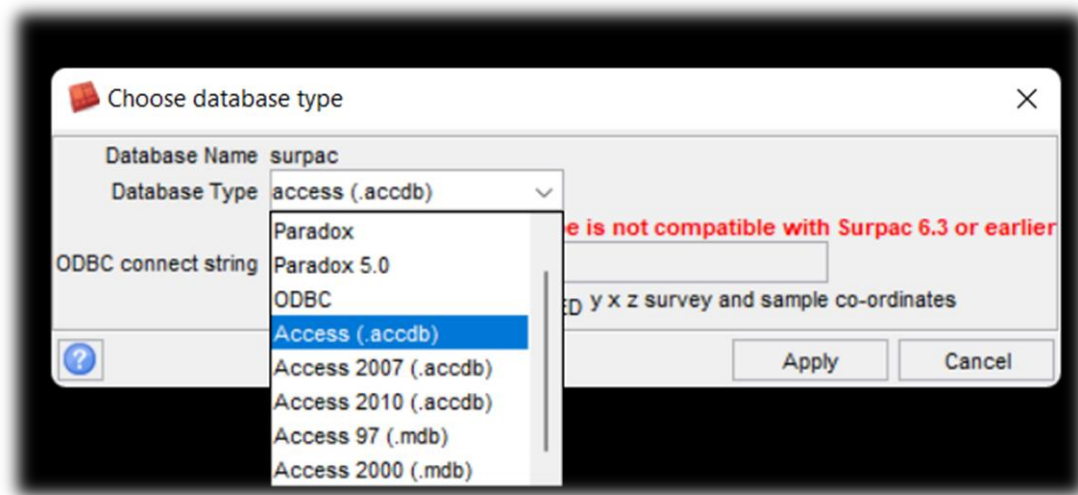


- Give a Name to the database that you want to create.

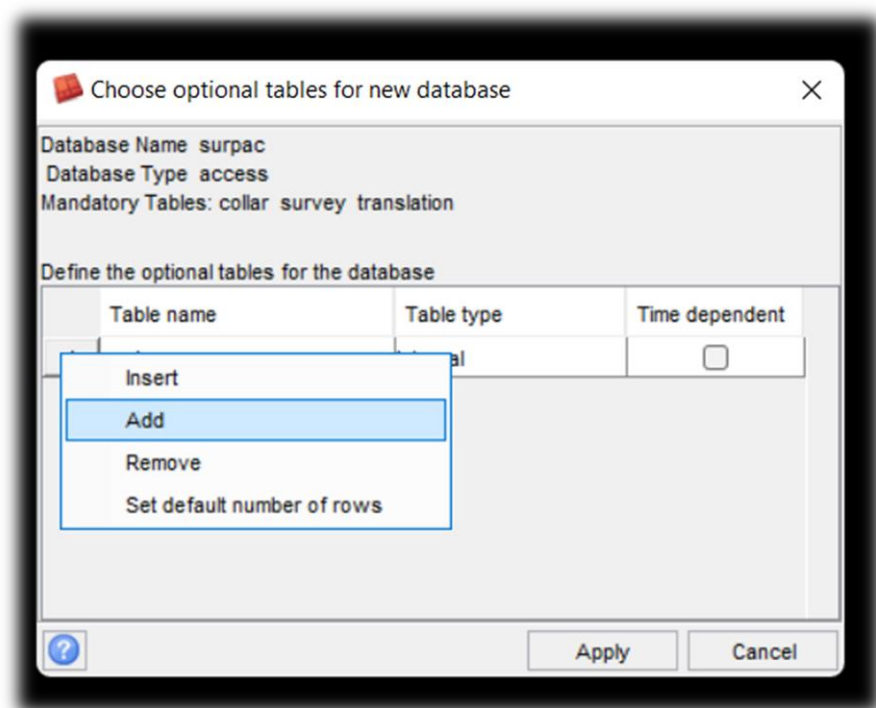


- Then click on **apply >> apply**.

- Next, a pop-up will appear on your screen to choose database type. There are many types of databases are there but the most common is **Access(.accdB)**



- The next pop-up would ask you to choose optional tables for the new database.
- **Right click >> Add >> add 2 tables and give them names as Assay and geology.**



Choose optional tables for new database

Database Name: **surpac**
 Database Type: **access**
 Mandatory Tables: **collar survey translation**

Define the optional tables for the database

	Table name	Table type	Time dependent
1	geology	interval	☐
2	assay	interval	☐

?

Apply Cancel

- The next step would be defining all fields for all tables.
- Before doing this you must know the names of columns in all the tables, that you can easily check by opening the **.Csv** files in excel.

Define all fields for all tables

collar survey translation styles geology assay

Mandatory Fields

	Field	Type	Nulls	Index	Length	No. Dec	Low Bound	High Bound
1	hole_id	character	☐	none	12	0	-999999	9999999
2	depth	real	☐	none	7	2	0	9999
3	y	real	☒	none	11	3	-999999	9999999
4	x	real	☒	none	11	3	-999999	9999999
5	z	real	☒	none	11	3	-999999	9999999
6	dip	real	☐	none	6	2	-90	90

Optional Fields

	Field	Type	Nulls	Length	No. Dec	Case	Low Bound	High Bound	Phys, Virt or Exp	Reference field or Expression	Va
1		character	☒	10	2	mixed	0	999	physical		

?

Apply Cancel

- Tick on Null if your table does not have that column/field.

Define all fields for all tables

collar survey translation styles geology **assay**

Mandatory Fields

	Field	Type	Nulls	Index	Length	No. Dec	Low Bound	High Bound
5	x_from	real	☒	none	11	3	-999999	9999999
6	z_from	real	☒	none	11	3	-999999	9999999
7	depth_to	real	☐	none	7	2	0	9999
8	y_to	real	☒	none	11	3	-999999	9999999
9	x_to	real	☒	none	11	3	-999999	9999999
10	z_to	real	☒	none	11	3	-999999	9999999

Optional Fields

	Field	Type	Nulls	Length	No. Dec	Case	Low Bound	High Bound	Phys, Virt or Exp	Reference field or Expression	Va
1	fe	character	☒	10	2	mixed	0	999	physical		

character
boolean
datetime
duration
integer
real
memo

Apply Cancel

- In some cases some tables will have extra fields other than mentioned, to add that extra fields **click on add >> type the field name >> choose the type of that field >> apply**

Define all fields for all tables

collar survey translation styles geology **assay**

Mandatory Fields

	Field	Type	Nulls	Index	Length	No. Dec	Low Bound	High Bound
5	x_from	real	☒	none	11	3	-999999	9999999
6	z_from	real	☒	none	11	3	-999999	9999999
7	depth_to	real	☐	none	7	2	0	9999
8	y_to	real	☒	none	11	3	-999999	9999999
9	x_to	real	☒	none	11	3	-999999	9999999
10	z_to	real	☒	none	11	3	-999999	9999999

Optional Fields

	Field	Type	Nulls	Length	No. Dec	Case	Low Bound	High Bound	Phys, Virt or Exp	Reference field or Expression	Va
1	fe	real	☐	10	2	mixed	0	999	physical		

Insert
Add
Remove
Set default number of rows

Apply Cancel

Define all fields for all tables

collar survey translation styles geology assay

Mandatory Fields

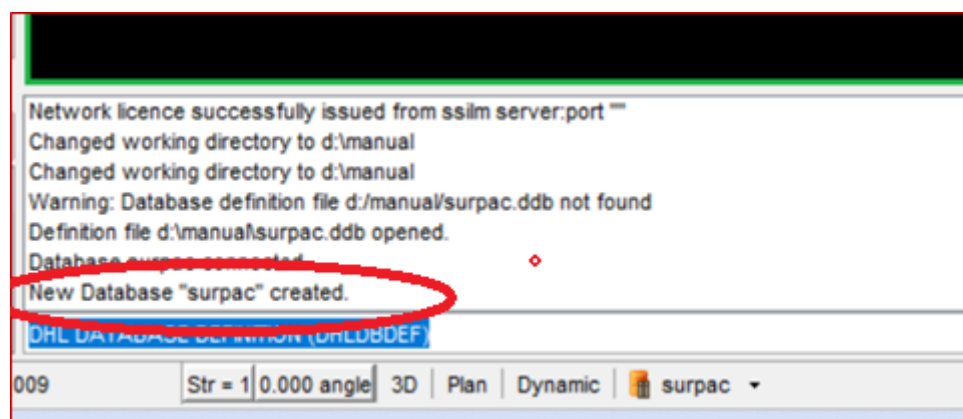
	Field	Type	Nulls	Index	Length	No. Dec	Low Bound	High Bound
5	x_from	real	☒	none	11	3	-999999	9999999
6	z_from	real	☒	none	11	3	-999999	9999999
7	depth_to	real	☐	none	7	2	0	9999
8	y_to	real	☒	none	11	3	-999999	9999999
9	x_to	real	☒	none	11	3	-999999	9999999
10	z_to	real	☒	none	11	3	-999999	9999999

Optional Fields

	Field	Type	Nulls	Length	No. Dec	Case	Low Bound	High Bound	Phys, Virt or Exp	Reference field or Expression	Va
1	fe	real	☐	10	2	mixed	0	999	physical		
2	sio2	real	☐	10	2	mixed	0	999	physical		
3	al2o3	real	☐	10	2	mixed	0	999	physical		
4	loi	real	☐	10	2	mixed	0	999	physical		
5	p	real	☐	10	2	mixed	0	999	physical		

Apply Cancel

- After defining all the tables click on apply and check for the comment new database created in description box.

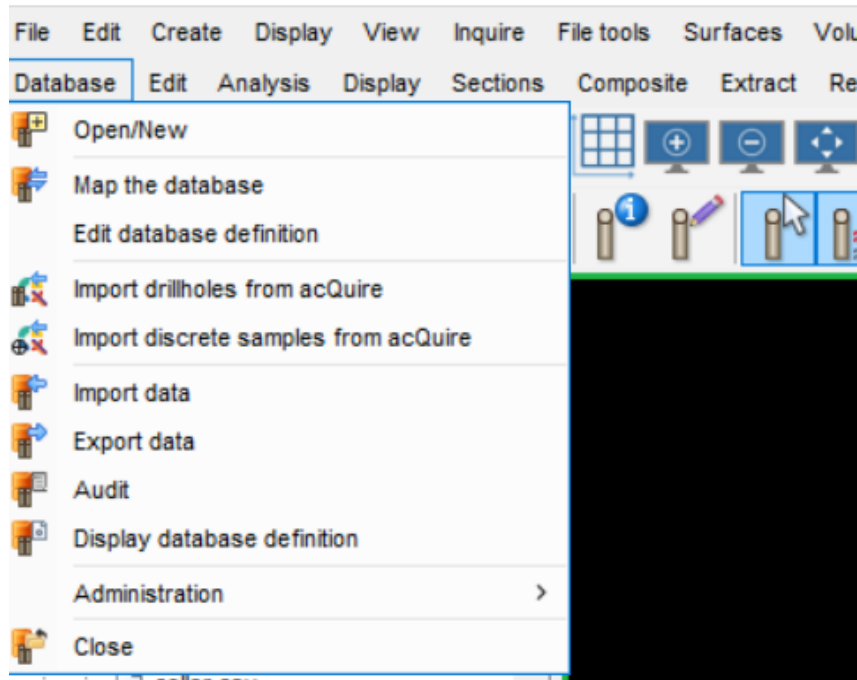


2.2. Importing drill-hole data in SURPAC

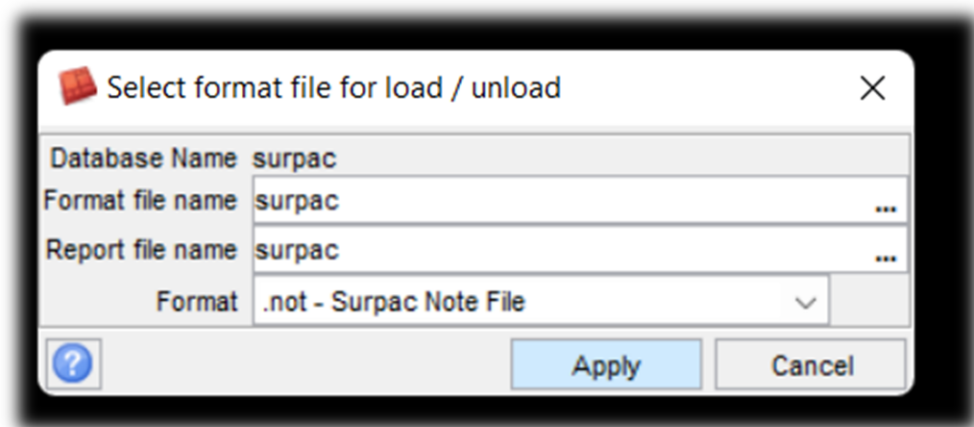
After creating the database, you need to import your drill hole data into that database.

To do that

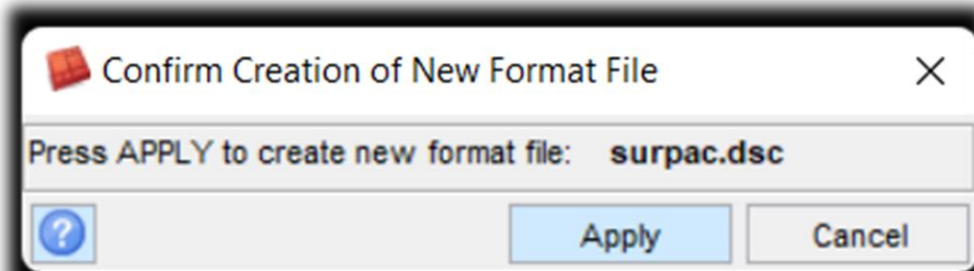
- Click on Database >> Import data



- In the next pop-up choose your format file name and report file name. Here, we have named it 'Surpac'.
- The format should always be the **.not** format while importing.



- Next click on apply in the next pop-up.



- In the next pop-up select all the tables you want to include in the format and click on Apply

Select database tables to include in format

Include all tables ☒

Space fill all tables ☒

	Table Name	Include	Format	Delimiter	Space Fill	Text Qual.
1	translation	☒	FREE	,	☒	None
2	collar	☒	FREE	,	☒	None
3	survey	☒	FREE	,	☒	None
4	assay	☒	FREE	,	☒	None
5	geology	☒	FREE	,	☒	None
6	styles	☒	FREE	,	☒	None

? Apply Cancel

- Then in the next pop-up select fields from every table to include in the format.
- If that table has that field tick on include, if that table doesn't have that field untick on that field in the include column.

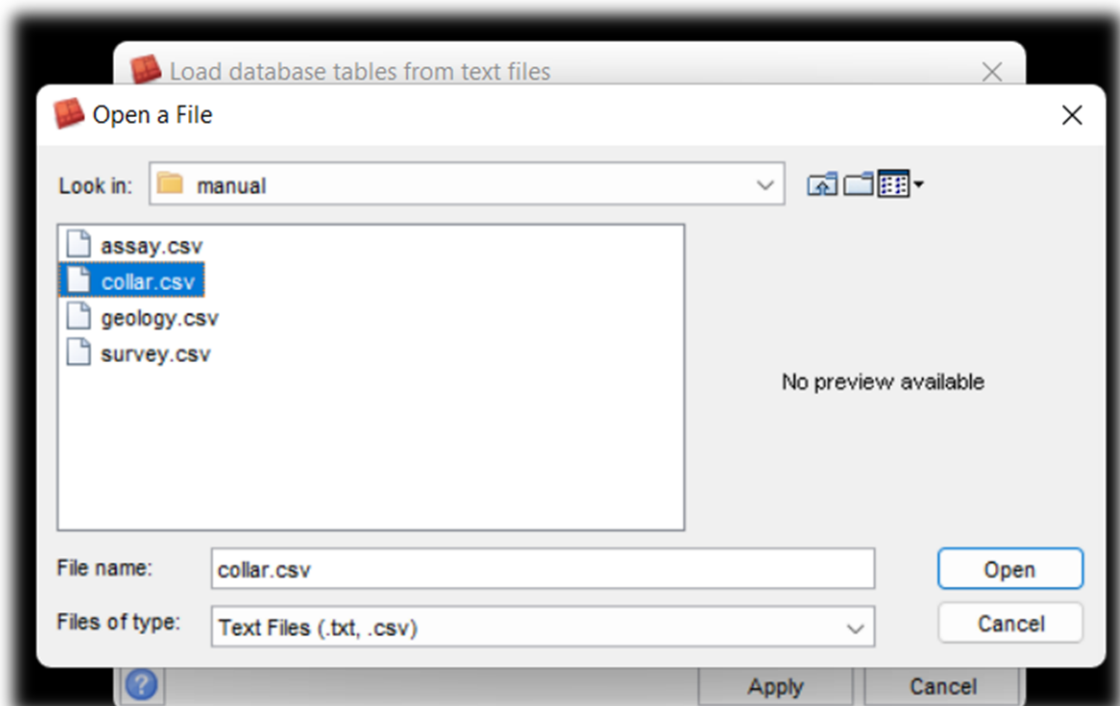
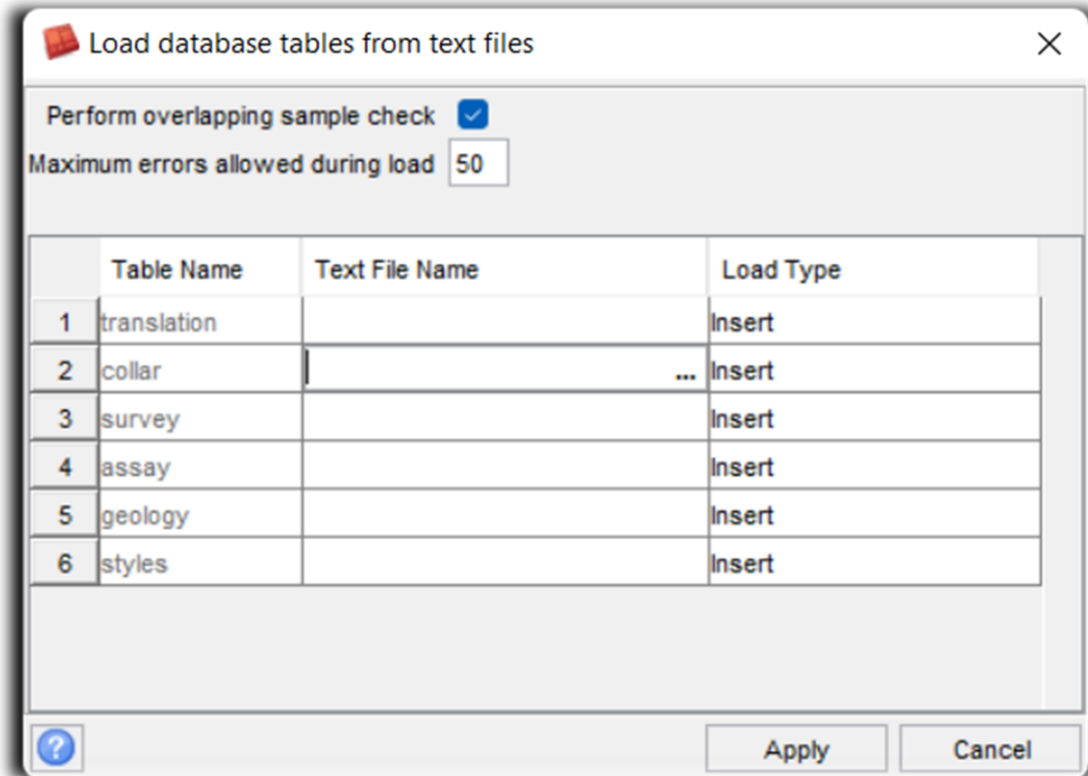
Select fields to include in format

	Table Name	Field Name	Include	Column	Length	Format
10		max_depth	☒	5	0	FREE
11		hole_path	☒	6	0	FREE
12	survey	hole_id	☒	1	0	FREE
13		depth	☒	2	0	FREE
14		y	☐	0	0	FREE
15		x	☐	0	0	FREE
16		z	☐	0	0	FREE
17		dip	☒	3	0	FREE
18		azimuth	☒	4	0	FREE
19	assay	hole_id	☒	1	0	FREE
20		samp_id	☒	2	0	FREE

? Apply Cancel

Click on [Apply](#).

- The next pop-up will appear to Load the database tables from text files.
- In the Text file name column upload the .Csv files with respect to their names in the order.



- After uploading all the files for all the tables, change the load type to Ins/Upd (Insert/Upload).

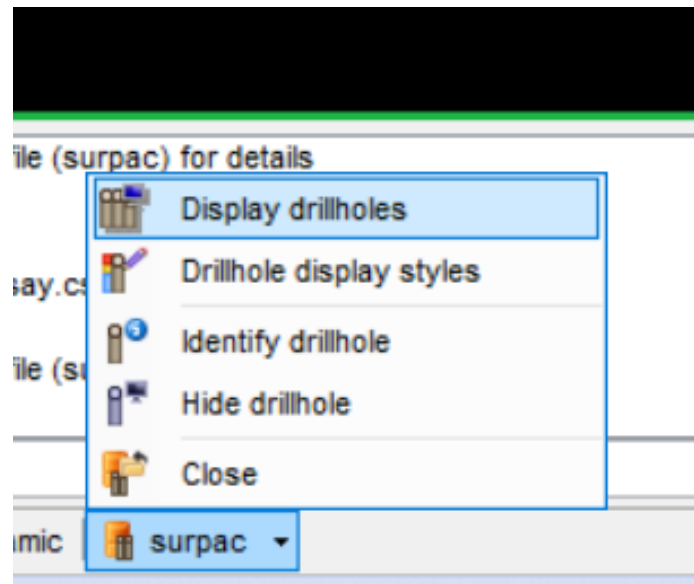
- Click on [Apply](#).

	Table Name	Text File Name	Load Type
1	translation		Insert
2	collar	collar.csv	Ins/Upd
3	survey	survey.csv	Ins/Upd
4	assay	assay.csv	Insert
5	geology	geology.csv	Ins/Upd
6	styles		Insert Update Ins/Upd

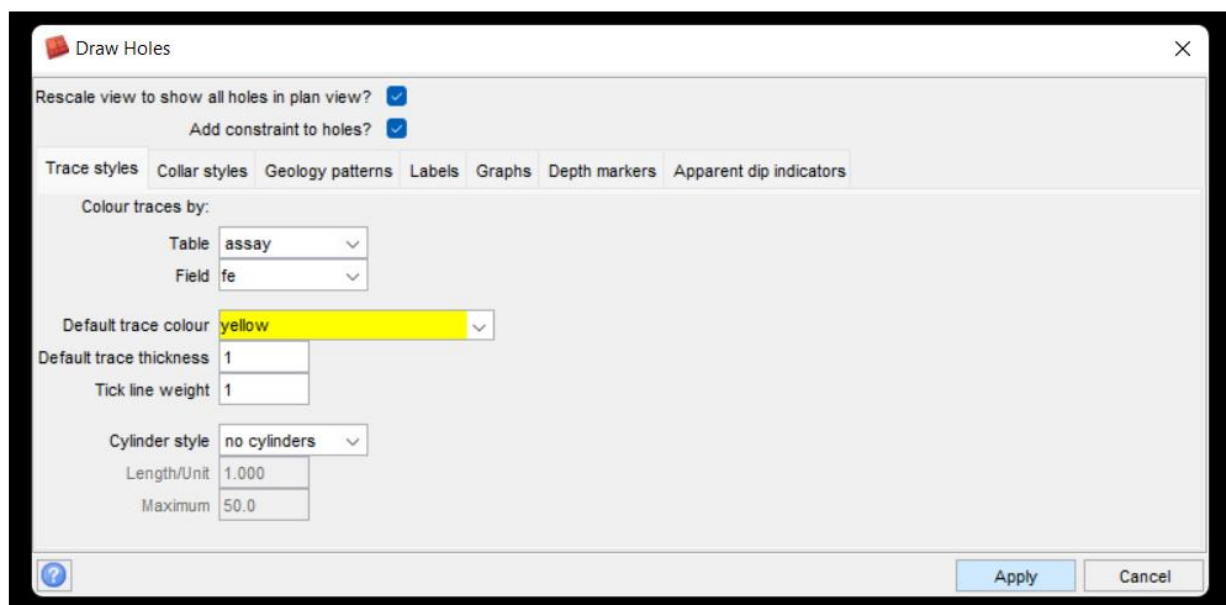
- After clicking on apply you'll get a [.not](#) report file of how many records were inserted, updated, and rejected from your data.

Displaying the drillholes:

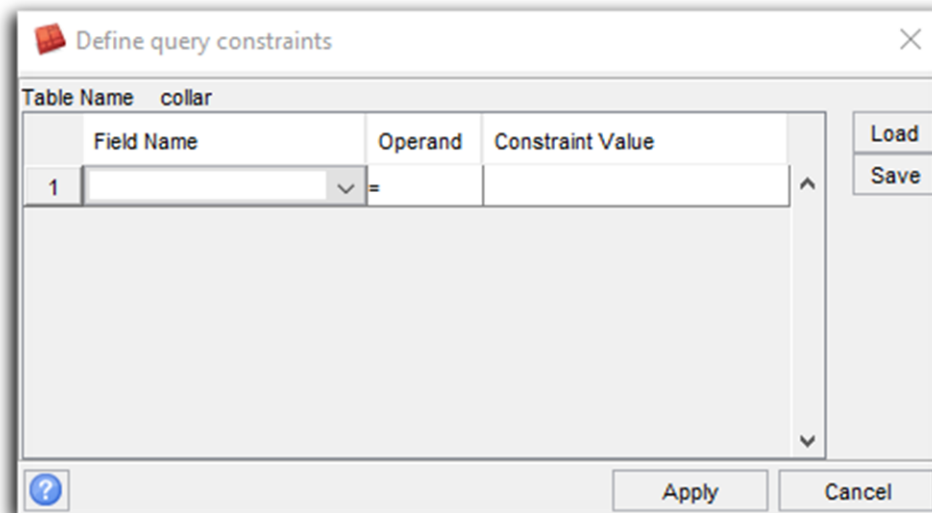
- After creating and importing the new database. Connect the database by dragging down or by clicking on open.



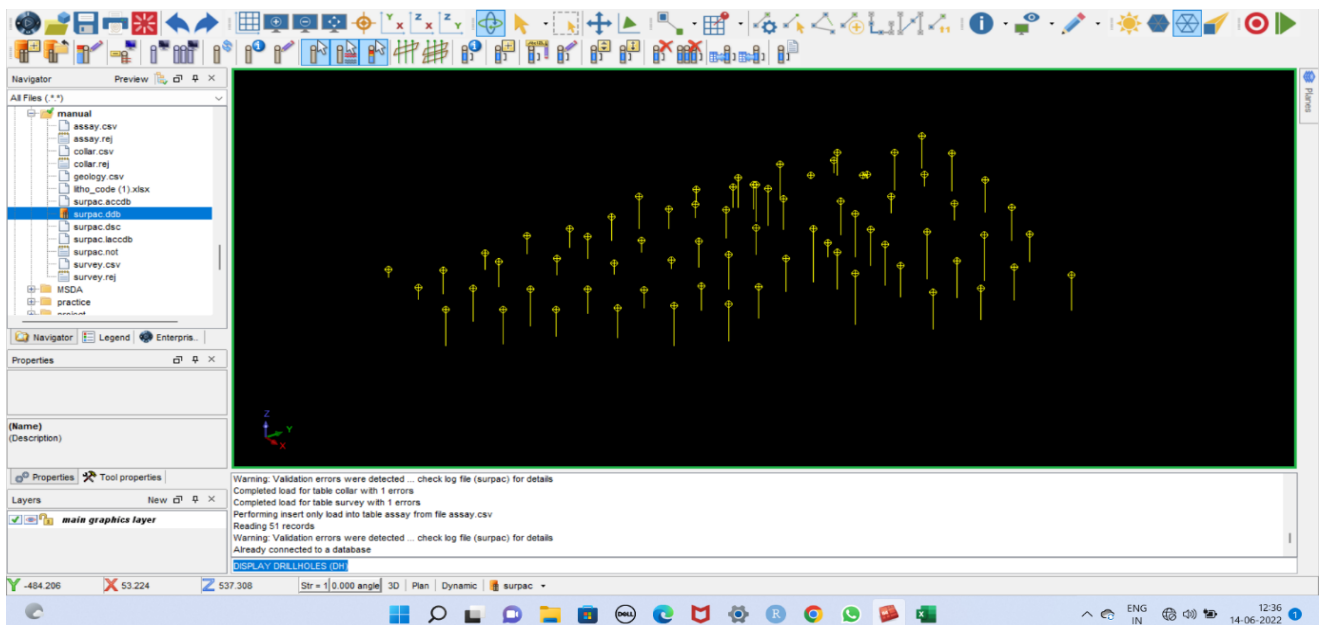
- Click on Display drillholes.
- In the Next pop-up choose the table and field of which you want to display the drillholes
- Give a colour of your choice. If you want to display the drillholes in cylinder style, choose fixed width from cylinder style and give a specific radius.
- Click on [Apply](#).



- Click on [Apply](#)

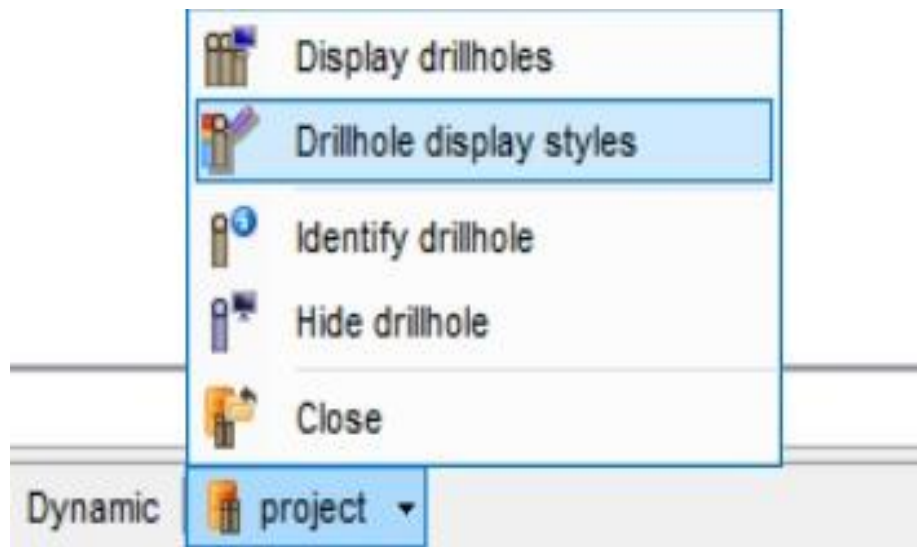


- Then your result should look something similar to the figure below.

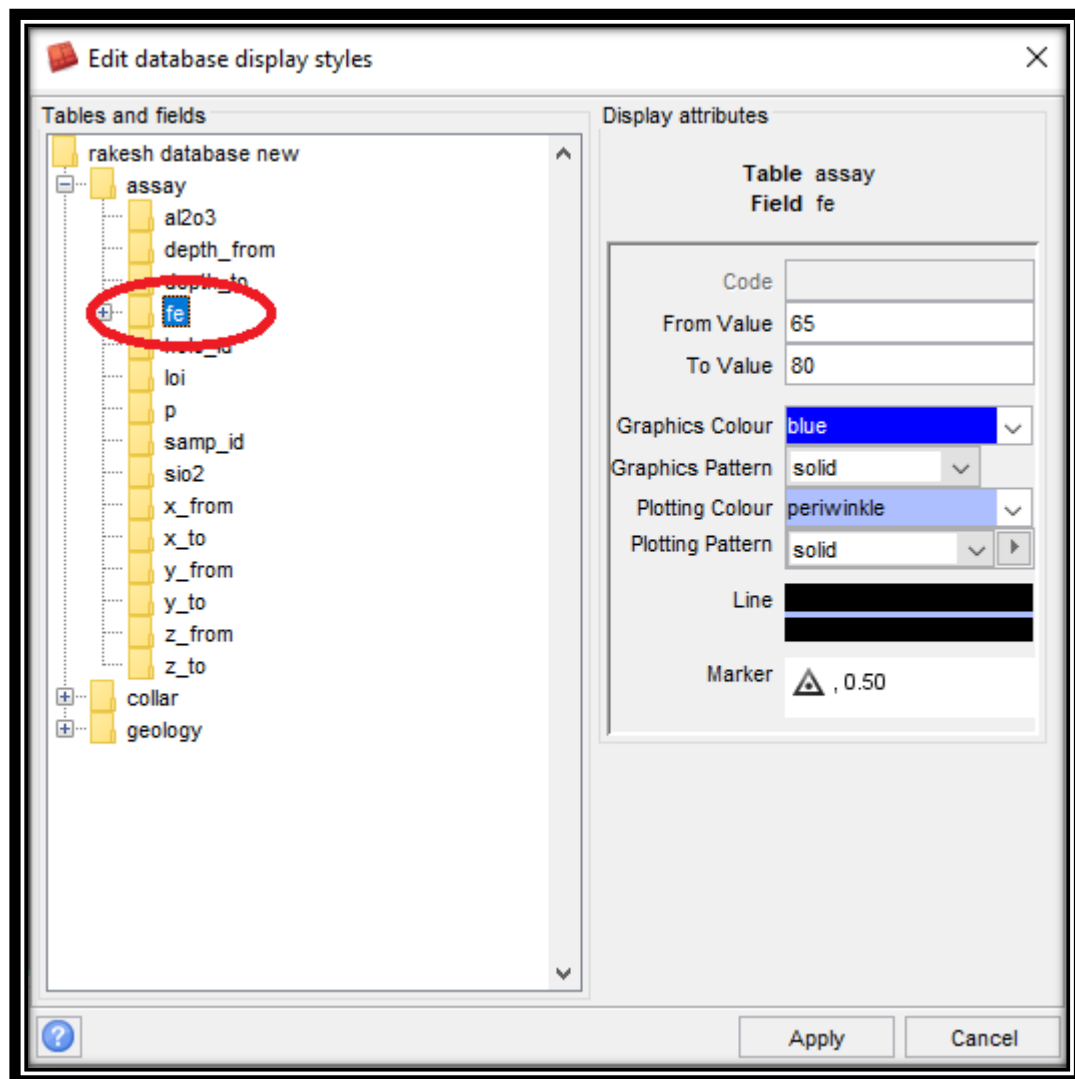


Drillhole display styles:

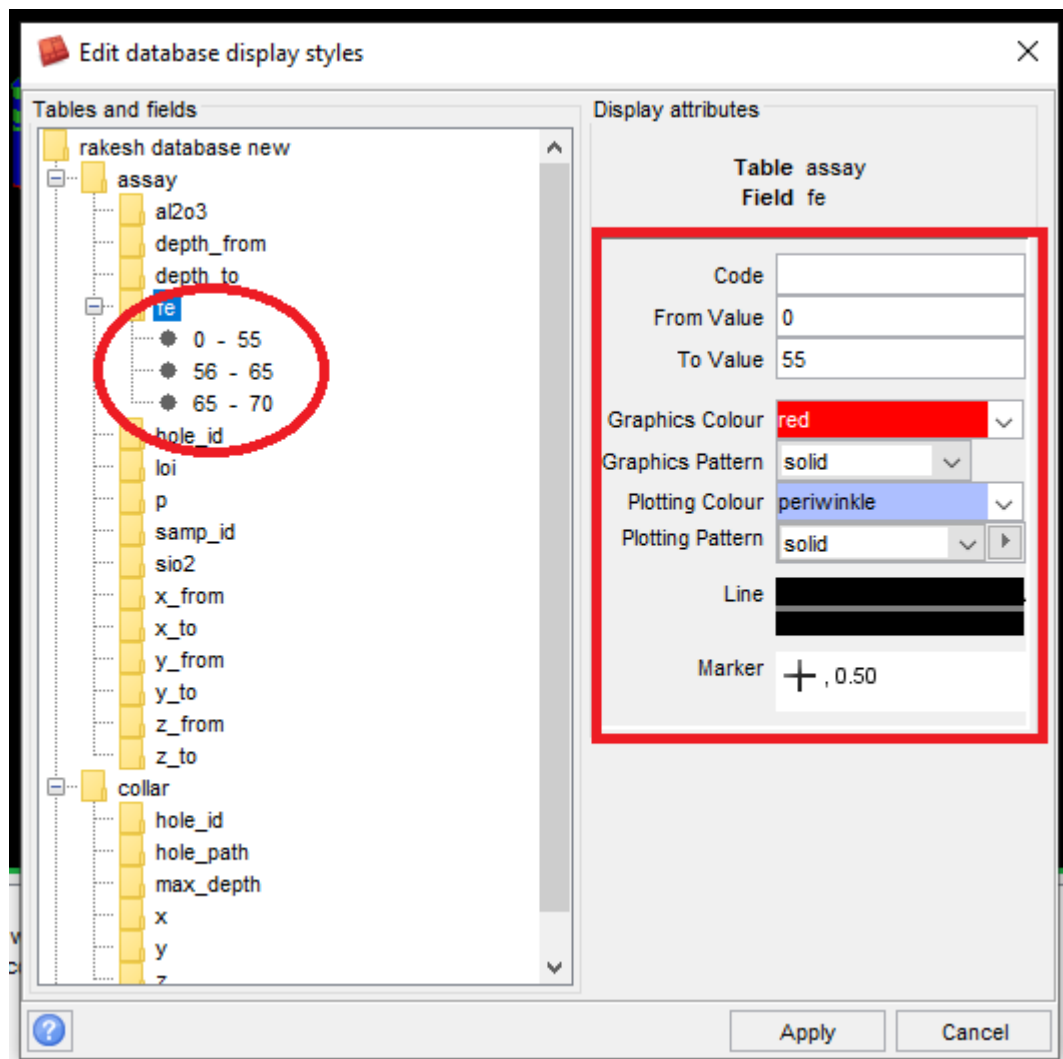
- Styling your bore hole data will help you to get better understanding about the exploration data by differentiating the drill hole with different colours. You get to know where the ore is and where the waste in the drill hole.
- **Connect your Database >> Drillhole display styles**



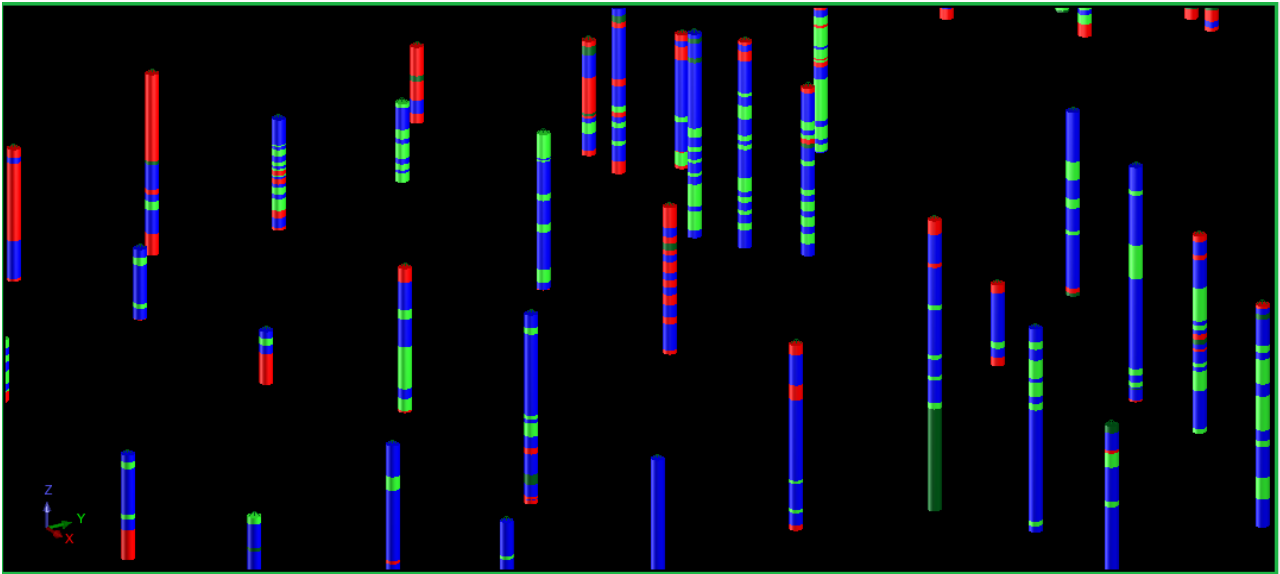
- In the next pop-up Open your database select **assay** and then click on **Fe**



➤ Right click on **Fe** >> Add new Style



- Select the range to the styles respective to your data. Assign a colour to that range and increase the line thickness for better visualisation.
- In this data Fe value majorly falls under 55-65. So the range from 65-70 represents the high grade Fe ore.
- 0-55 represents the waste and 56-65 represents the low-grade Fe ore.
- Click on **Apply**



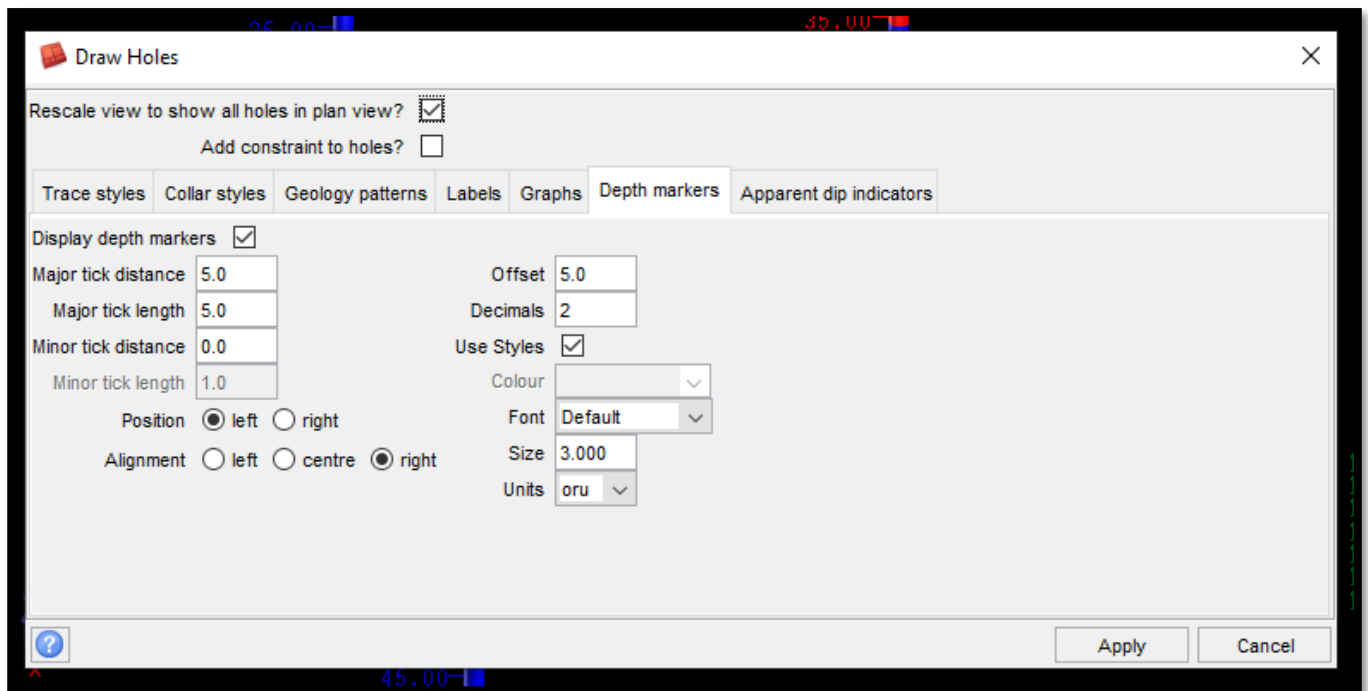
- In this figure red colour in the bore hole represents the waste, blue colour represents low grade ore and green colour represents high grade Fe ore.

Labelling and Depth Marking:

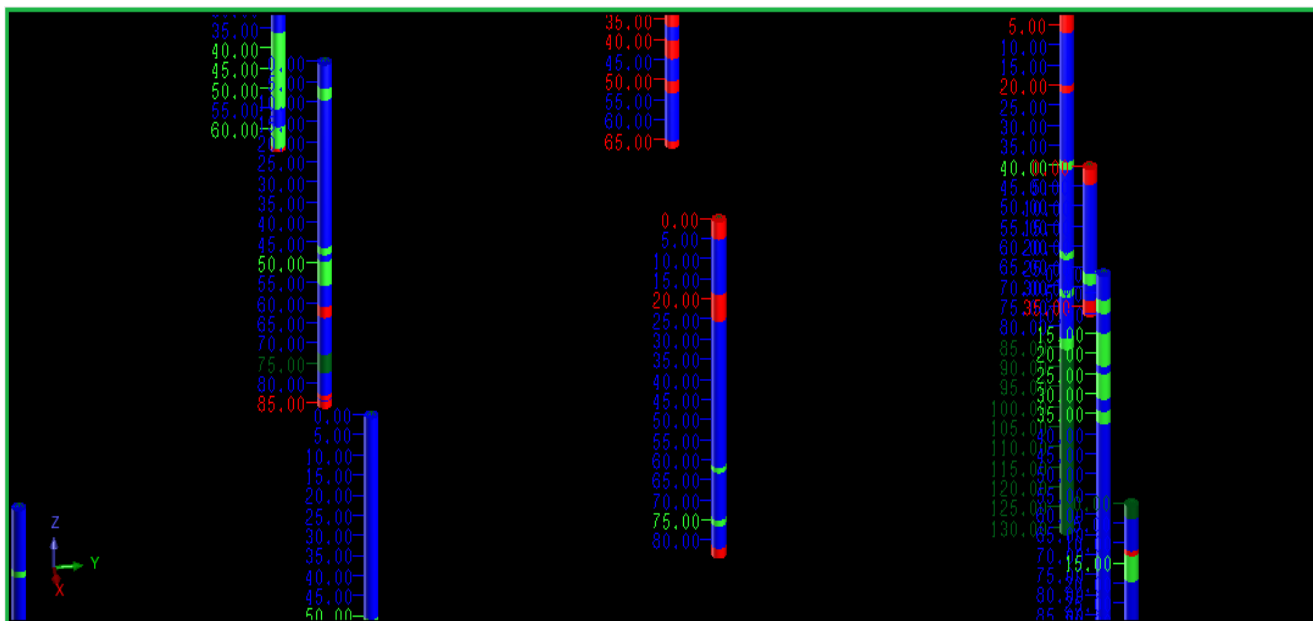
Depth_Marking:

The objective of performing Depth marking is to find out at which depth the high-grade ore is present.

- **Display Drill holes >> Depth markers**
- **Check on Display Depth markers. Check on tick distance, tick length, offset, position, and alignment.**
- Click on **Apply**



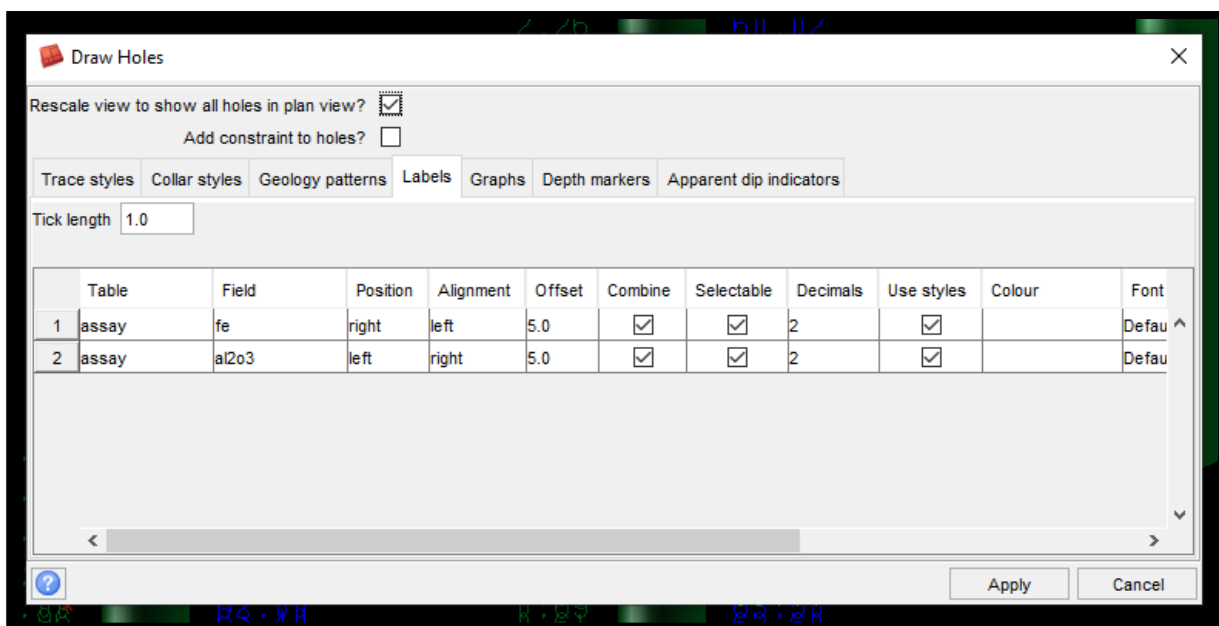
- Then the drill holes will display the depth according to the defined alignment.
Your result should look something like the below figure.



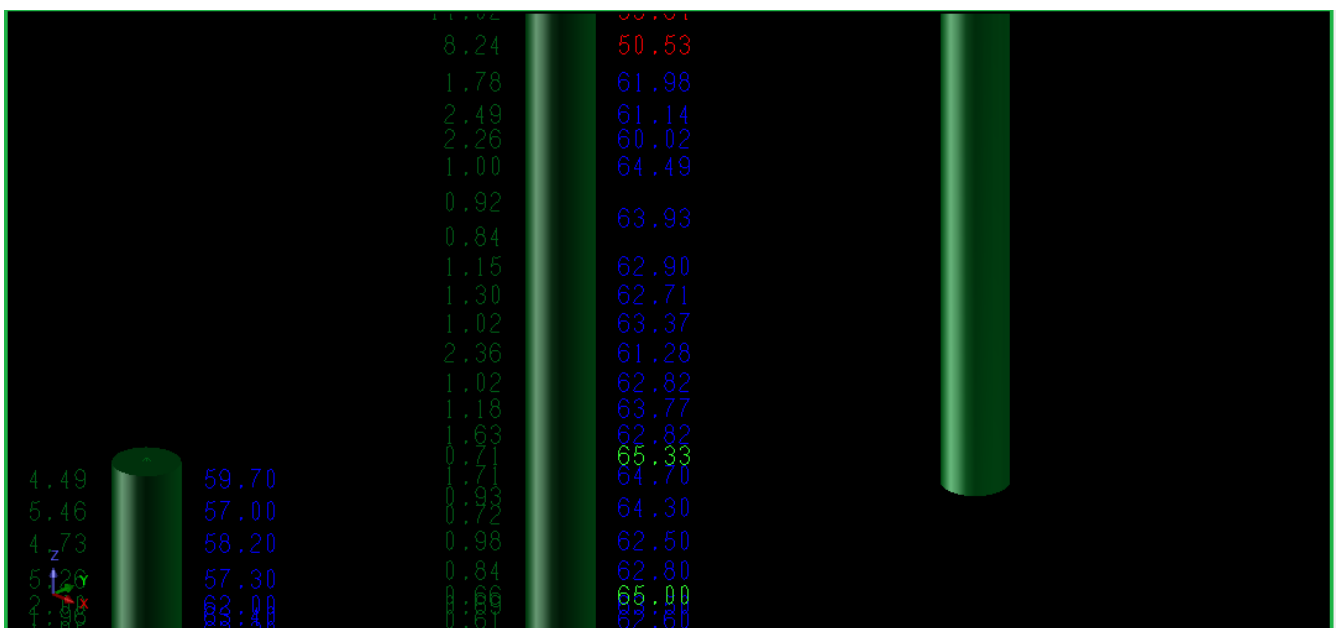
Labelling:

Labelling operation is used to display two fields from the same table or different table at same time.

- **Display drill holes >> Labels**
- **Select the table and field to be displayed. Always Keep the alignment opposite to each field.**
- **Click on [Apply](#).**



- **Ater applying, all the drill holes will display the fields on the both sides as shown in the below figure.**



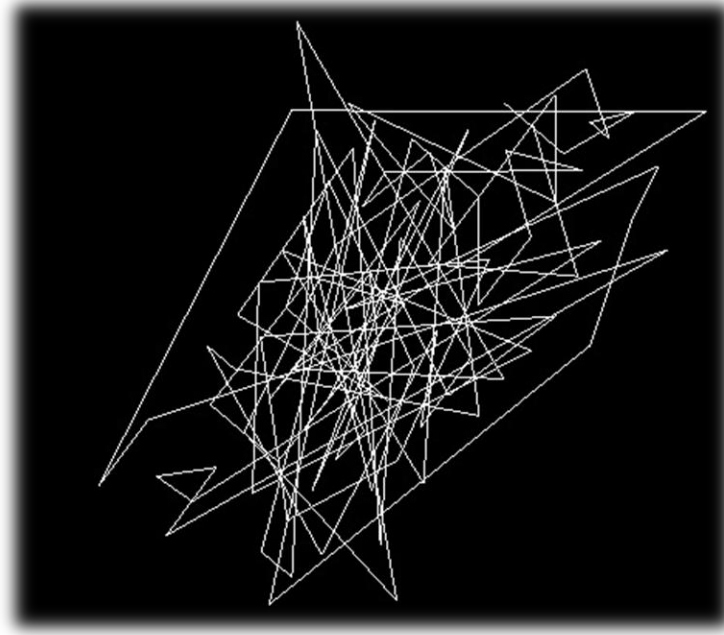
Drill hole layout:

The objectives of getting a drill hole layout is to create a Topographical DTM from it, to get the surface boundary layout and also the same drill hole layout file will be used in contouring to get the contours.

- **Go to Extract >> Drill hole layout**
- Then a pop-up will appear. define the name of your drill hole layout.

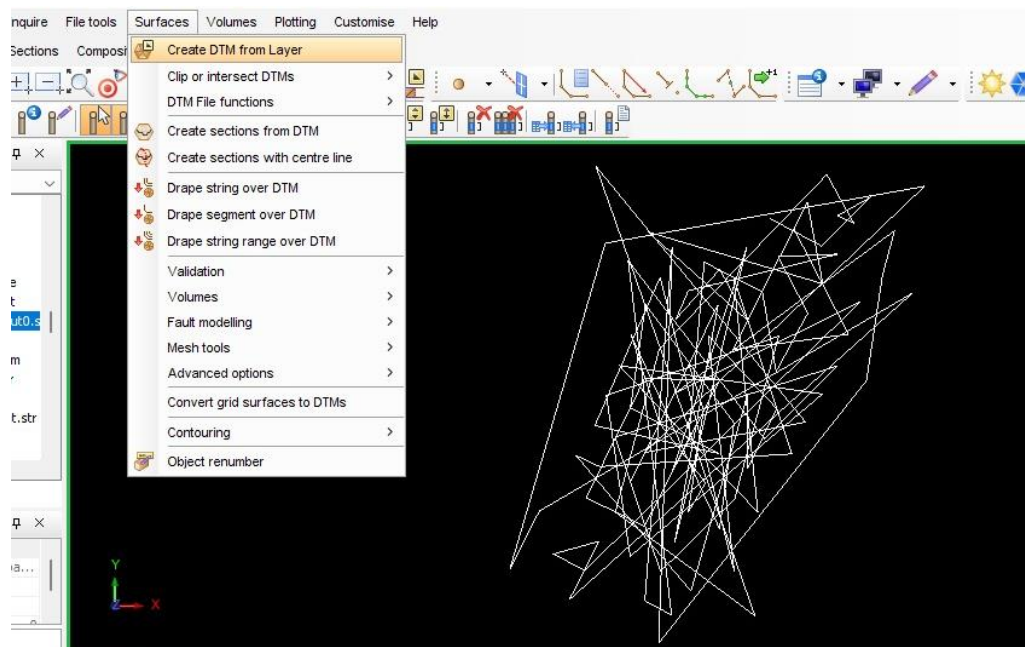


- **Select the field Z.** because we want the drill hole layout with respect to the elevations of the drill holes.
- **Click on Apply**
- **Click on Apply**
- A string file will be saved in your work directory. Bring that string file to the main graphics layer.
- Then your drill hole layout will look something like below figure.

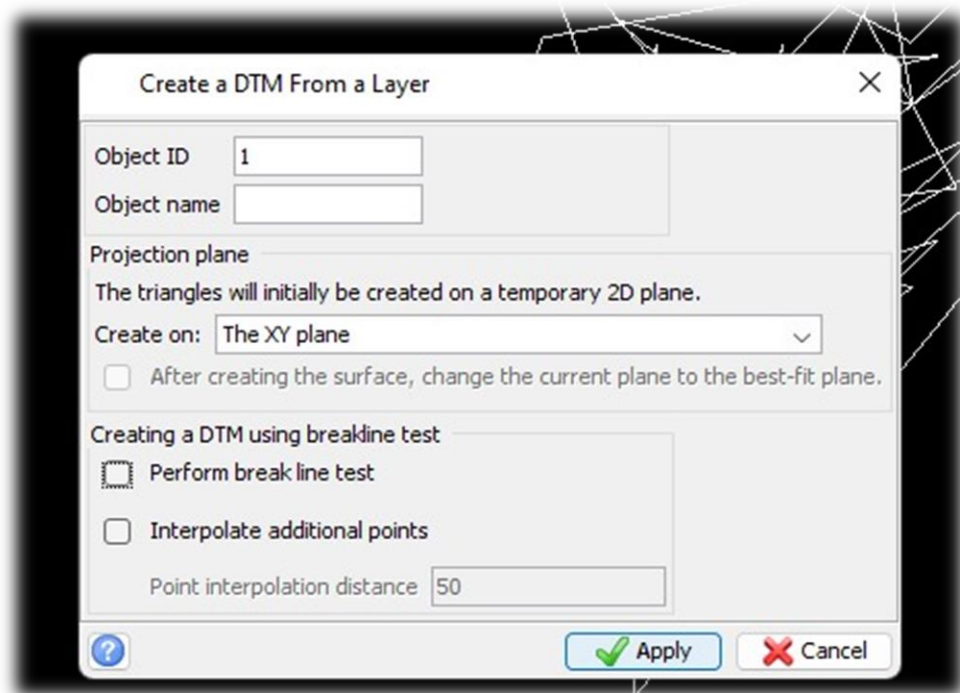


Creating a Topographical DTM:

- Bring the drill hole layout string file to the main graphics layer



- **Go to Surfaces >> Create DTM from layer >> Pop-up >> Apply**

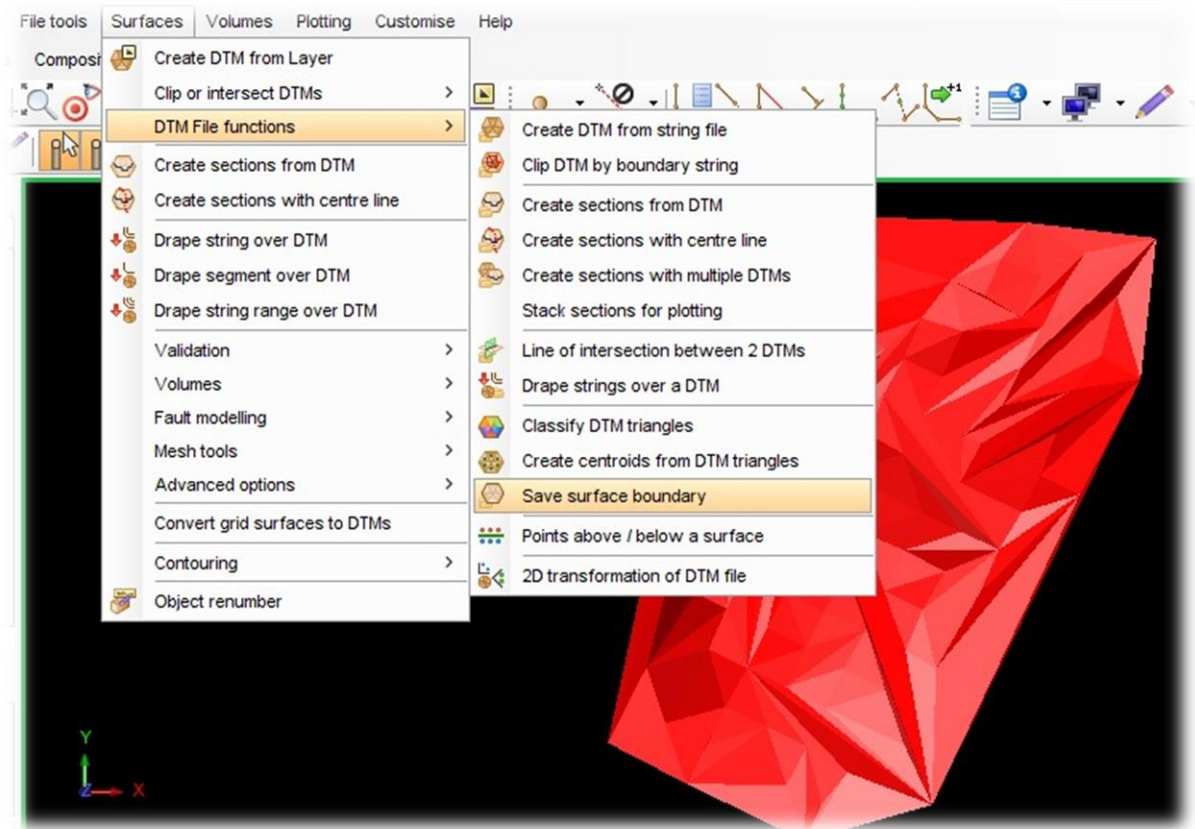


- The object ID will give the colour difference in the DTM. Different object ID will have different colours.

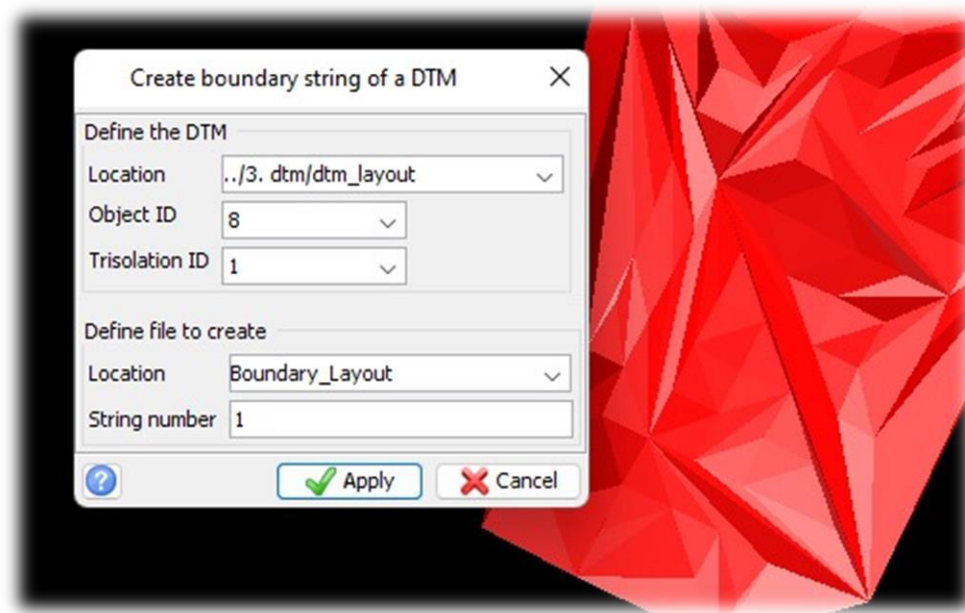


Extracting Surface Boundary:

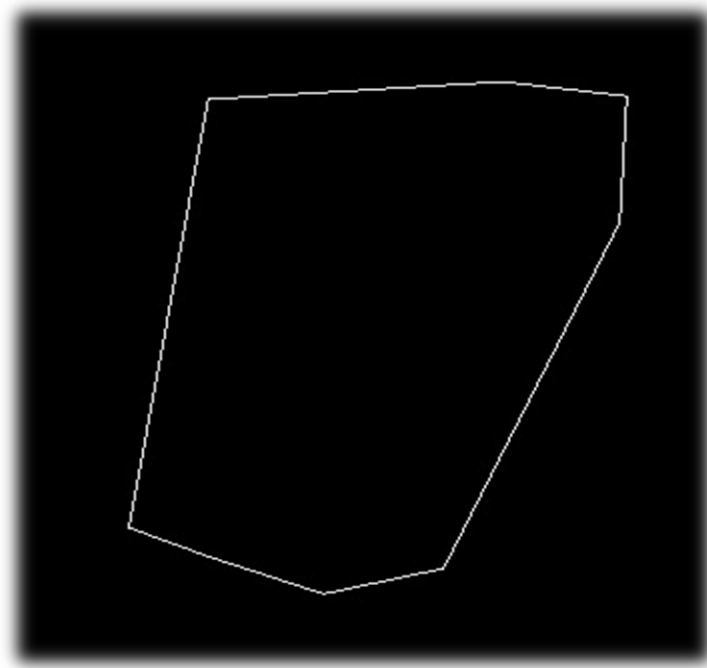
- Surface boundary gives you the extent of your drill hole data.
- **Go to Surfaces >> DTM File Functions >> Save Surface Boundary**



- **Pop up >> Select the Topographical DTM >> Define a name for the boundary >> Apply**

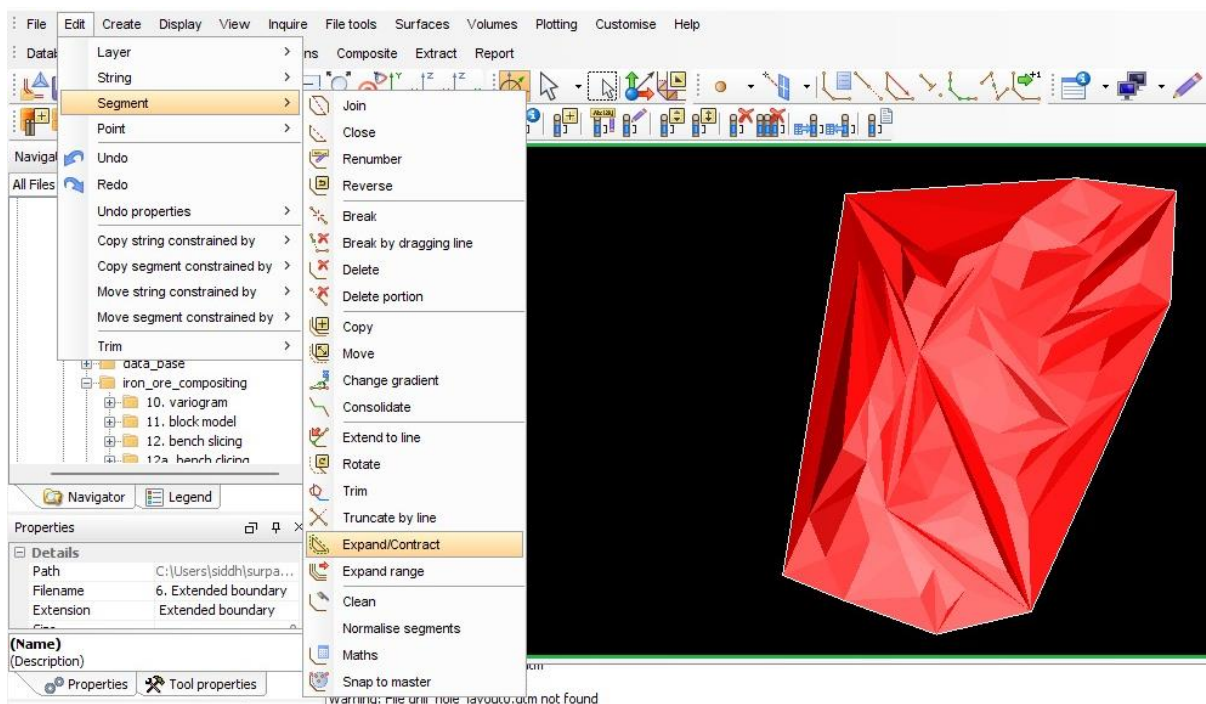


- A string file will be saved in your work directory. Bring that string to the main graphics layer

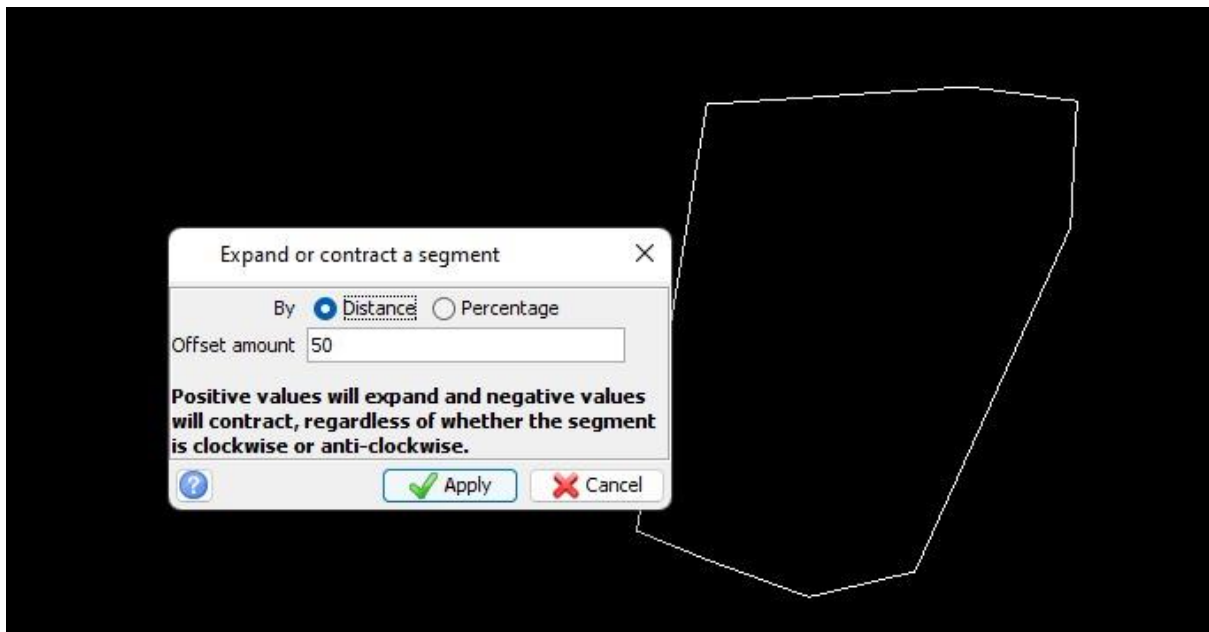


Expanding The Boundary:

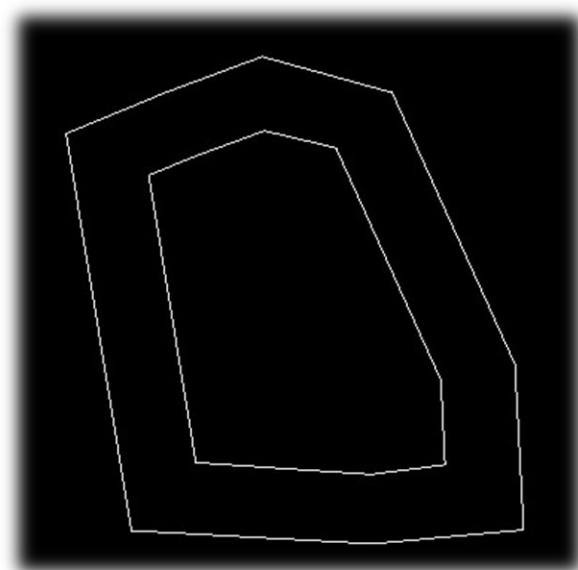
- For designing the surface structures we need to have the extended boundary
- Later we also do the contouring for the extended boundary by inverse distance square method.
- **Go to Edit >> Segment >> Expand/Contract**



- Select any point on the boundary then a new pop up will appear
- The pop up will ask you to expand/contract the segment by percentage or Distance
- Select anyone
- Give a number in the Offset amount of your choice. The offset amount is in meters.
- **Click on Apply**



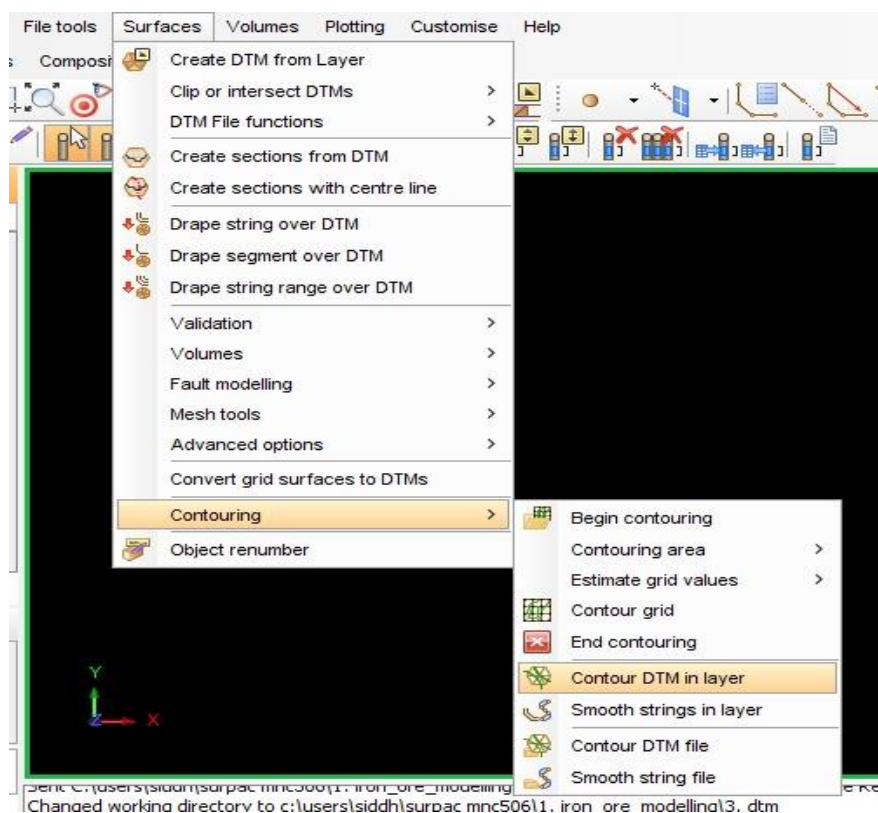
- Then a new extended boundary will be created. Save that layer as a string file.



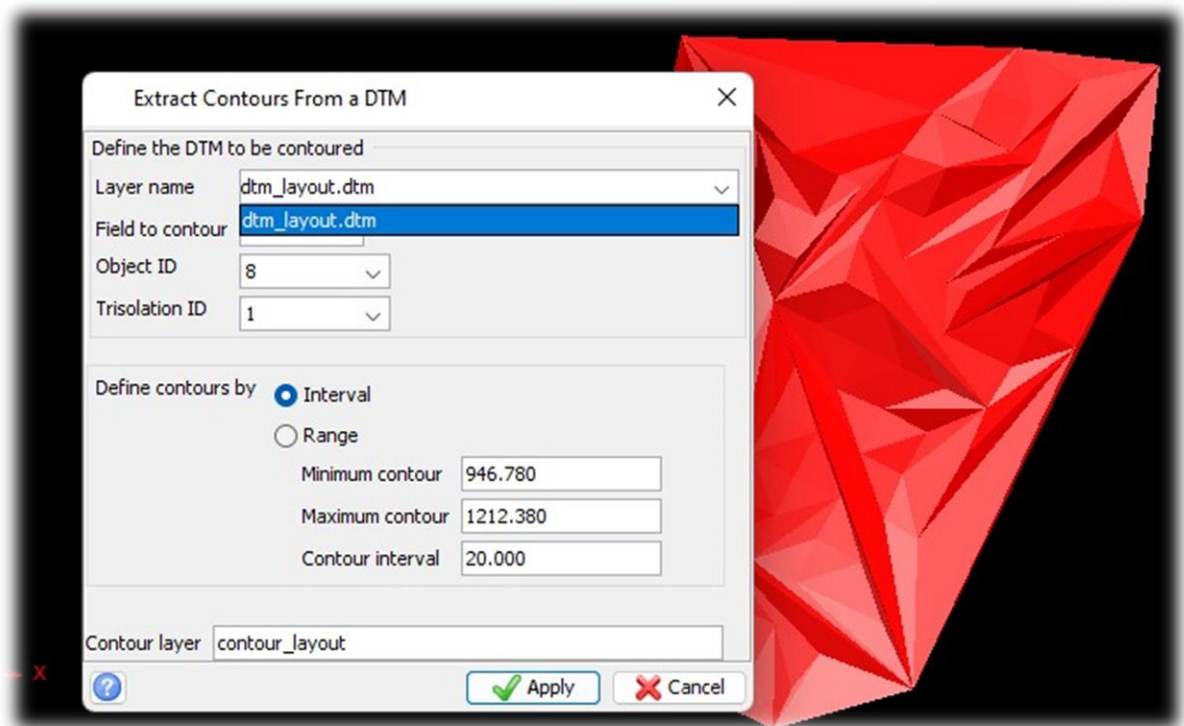
Contouring:

The simplest definition of contouring is the process of showing the elevation differences on your site at regular intervals, from lowest to highest point. Points connected by contour lines share the same elevation. The section becomes steeper as the contour lines move closer together.

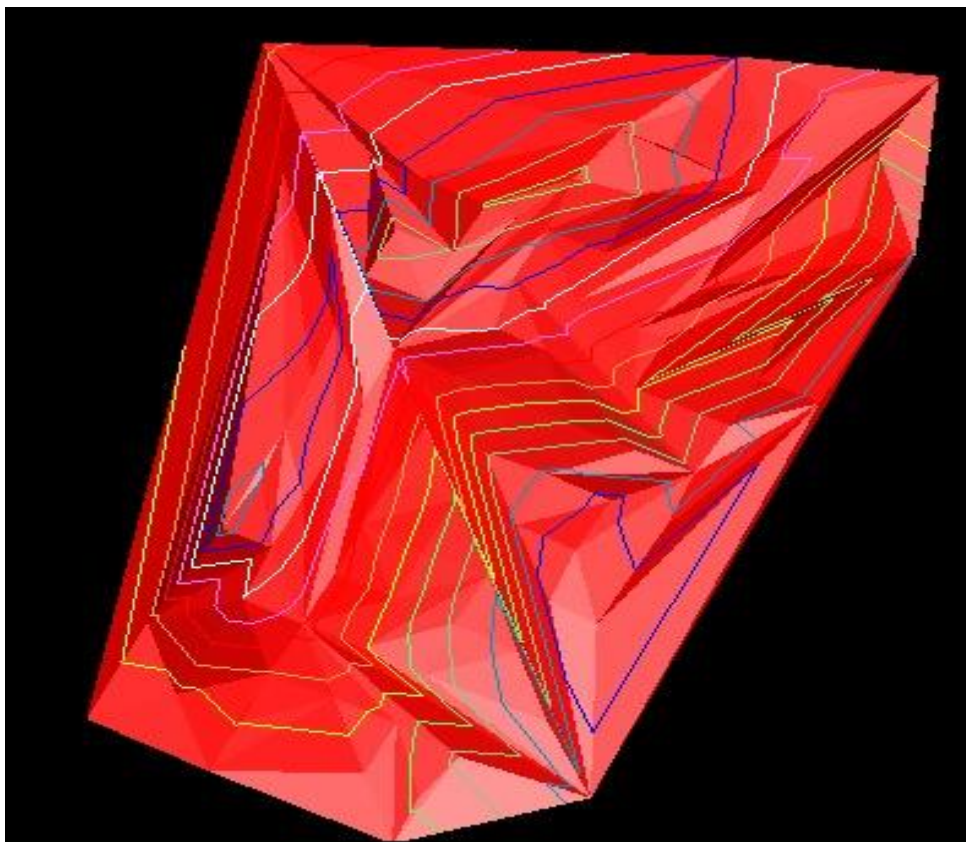
- Bring the Topographical DTM to the main graphics layer.
- **Go to Surface >> Contouring >> Contour DTM in layer**



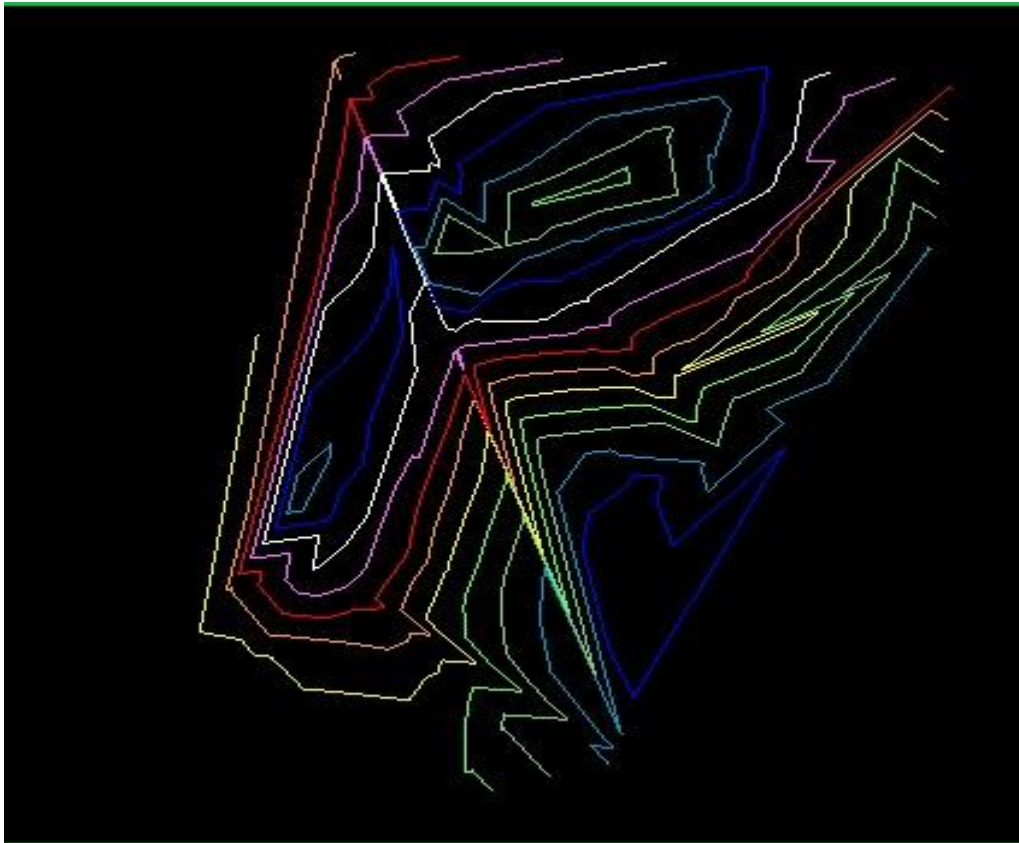
- A new pop up will appear. Define the dtm to be contoured and define the contour interval and **Click on Apply**



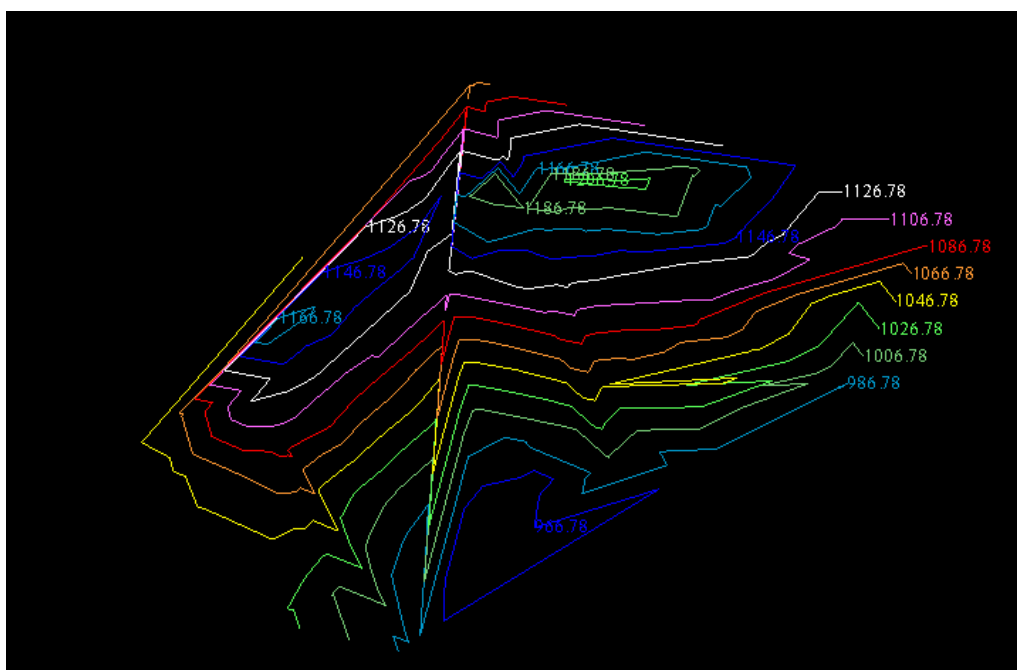
- After clicking on apply, your dtm will be contoured and it should look something like below picture



- Save the new contoured layer. Bring that contoured layer to the main graphics layer.



- Type DRWZ in the command box. You'll find the command box below the description
- A new pop up will appear. Define the contour layer and **click on apply**. This command will show the elevation (Z) of each contour line.



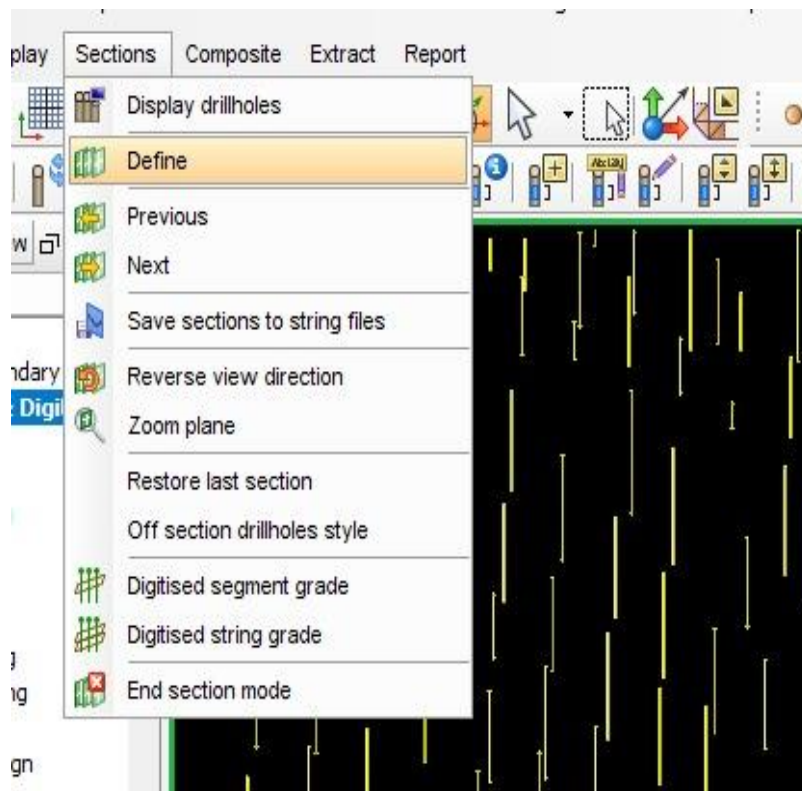
SOLID MODELLING

The development of solid models is a crucial component of mining geology. A solid model not only guarantees accuracy in resource estimation but also aids in more efficient mine planning.

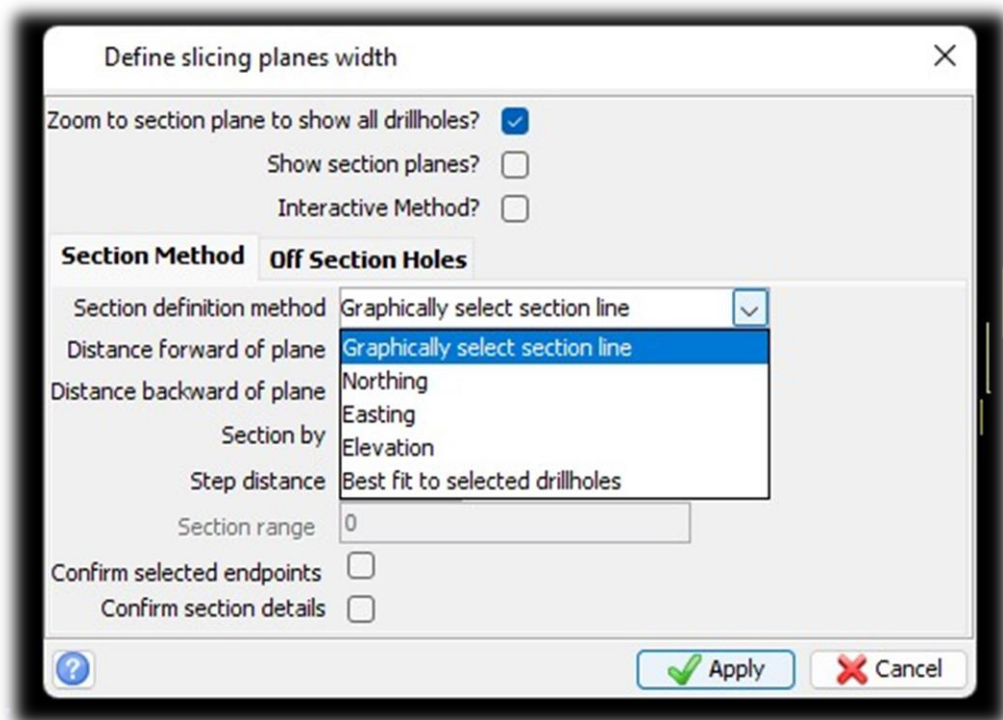
GEOVIA Surpac is an excellent mining industry tool, particularly for resource geologists and mine planners. When it comes to solid modelling in Surpac, there are several triangulation techniques, the most popular and often used of which is creating solids by connecting segments from multiple sections.

Sectioning And Digitising:

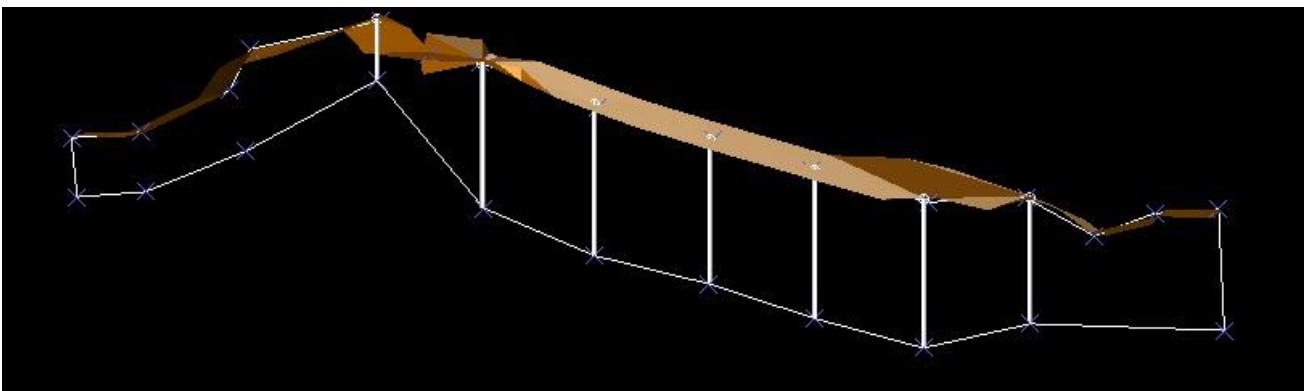
- we need to create sections to create a solid model.
- To create the sections, **connect your database >> Display drill holes >> Bring the topographical dtm to the layer.**
- **Go to Sections >> Define >> pop up**

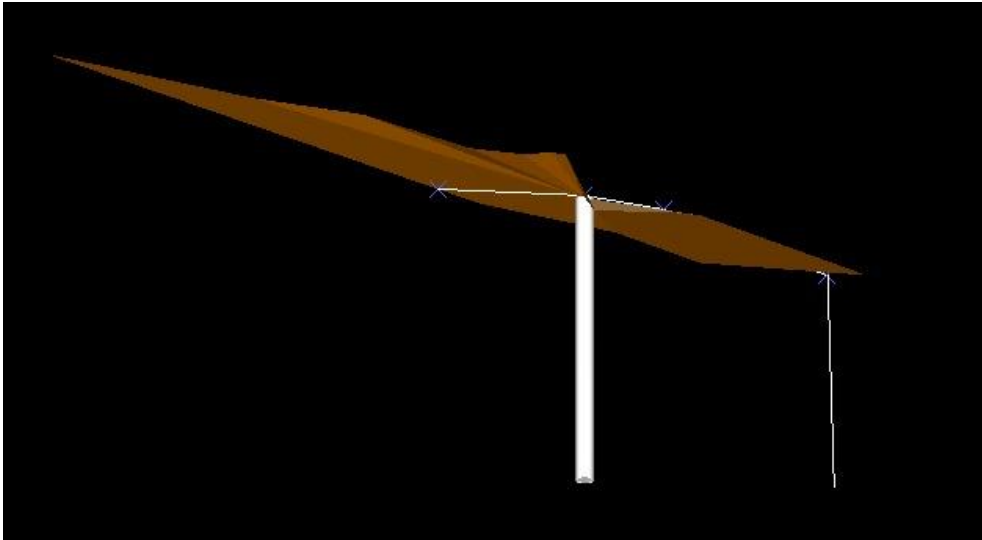


- There are generally 5 types of sectioning methods. You need to choose the best method that suits to your geological data.
- The 5 methods are:
 - 1. Graphically select section line**
 - 2. Northing**
 - 3. Easting**
 - 4. Elevation**
 - 5. Best fit to the selected drillholes**

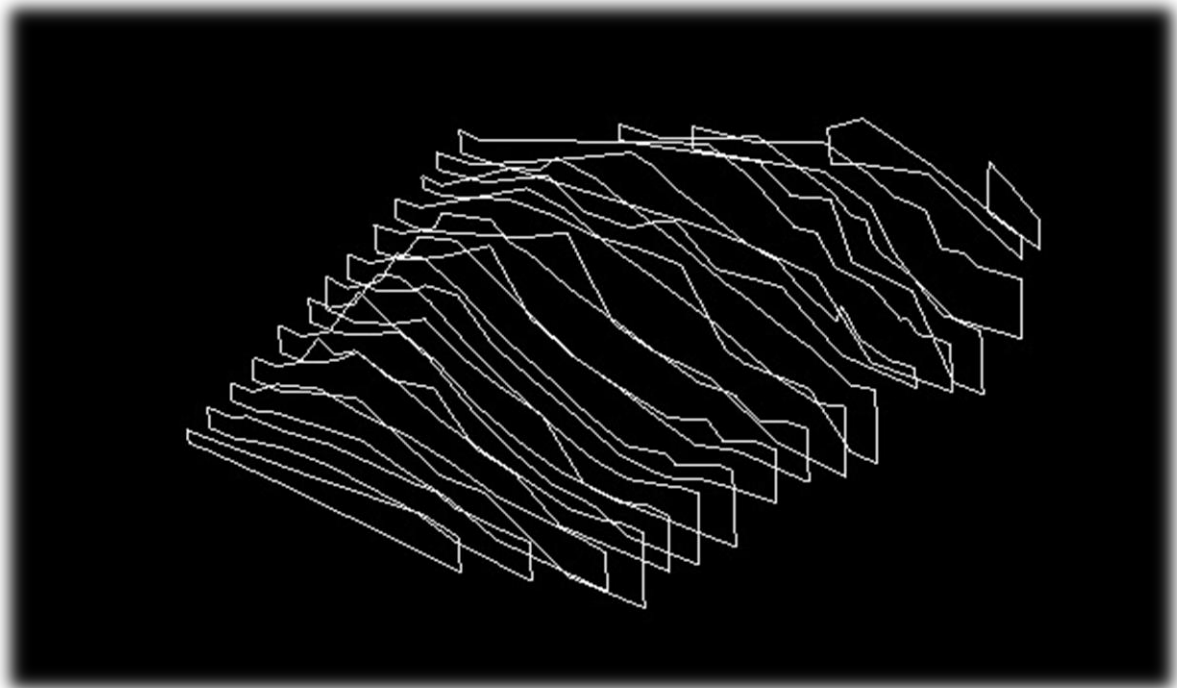


- After selecting the method, digitise the drill hole sections by connecting their top and bottom with a single segment.





- You can also do the sectioning without topographical dtm. But topographical dtm helps to get a better solid model.
- Before digitising the section. Create a new layer and digitise in that new layer.
- Save the new layer after finishing the digitisation.
- Repeat same process for all sections and save all the layers as string files.
- After digitising all the sections bring all the sections in one layer using **Ctrl key**.

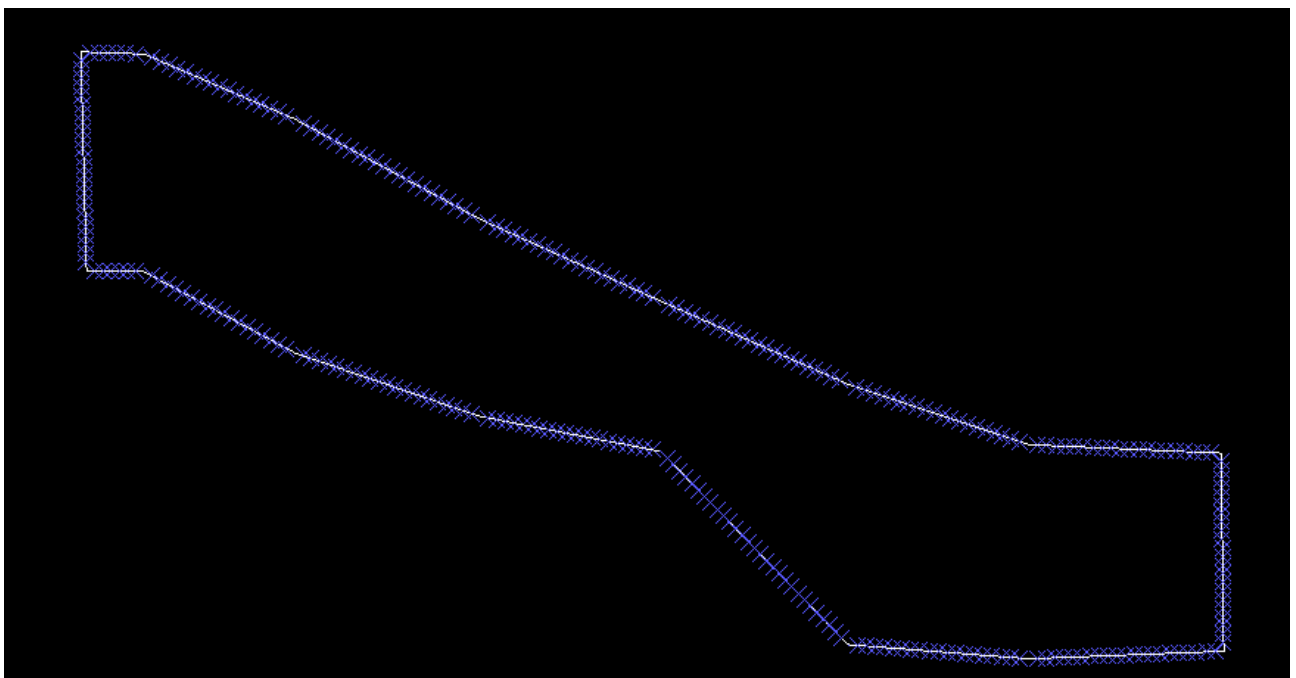


- Save the new layer in which all the sections are combined.

Normalising the sections:

- To normalise the segment, **Go to Edit >> Segment >> Normalise.**
- A popup will appear, select the distance between points of your choice to normalise the segment.
- Click on any segment. After clicking new points will be created on the segment.

Normalised segment:



- Delete all the points which have sharp edges. To delete the points, **Go to Edit >> Point >> Delete.**
- Repeat the same procedure for all the segments. Remove all the sharp edges

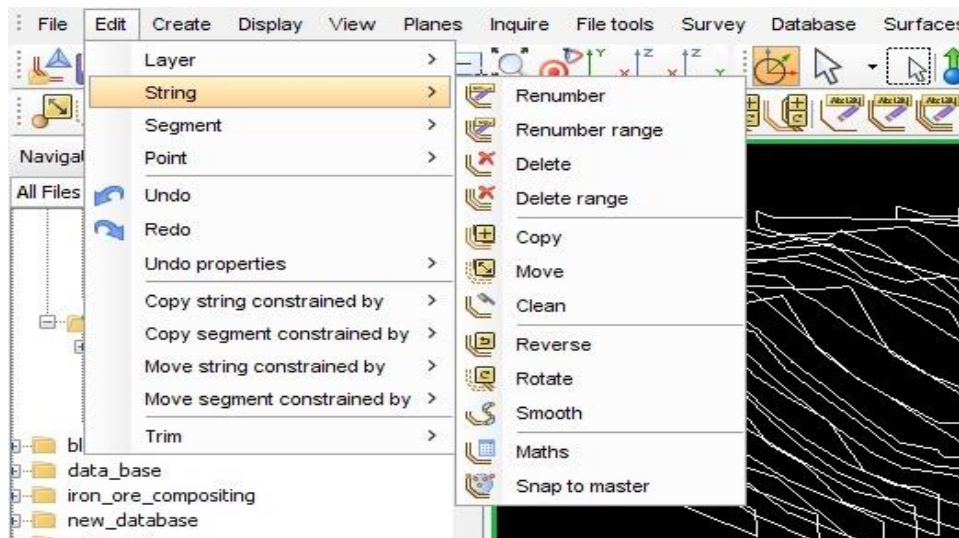
cleaning the sections:

- After normalising the segments, to get a solid model we need to clean the sections. There are 5 operations you need to perform to clean the string. They are

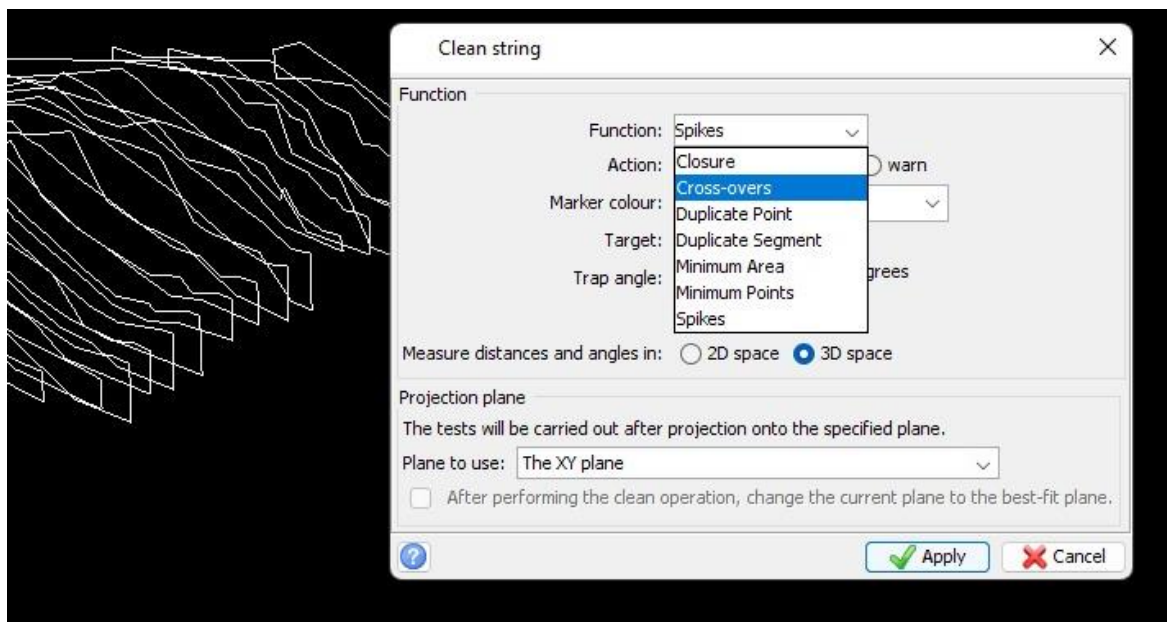
1. **Closure test**
2. **Spike test**
3. **Duplicate point test**

4. Crossover test

- To clean the string **Go to Edit >> String >> Clean >> pop up**

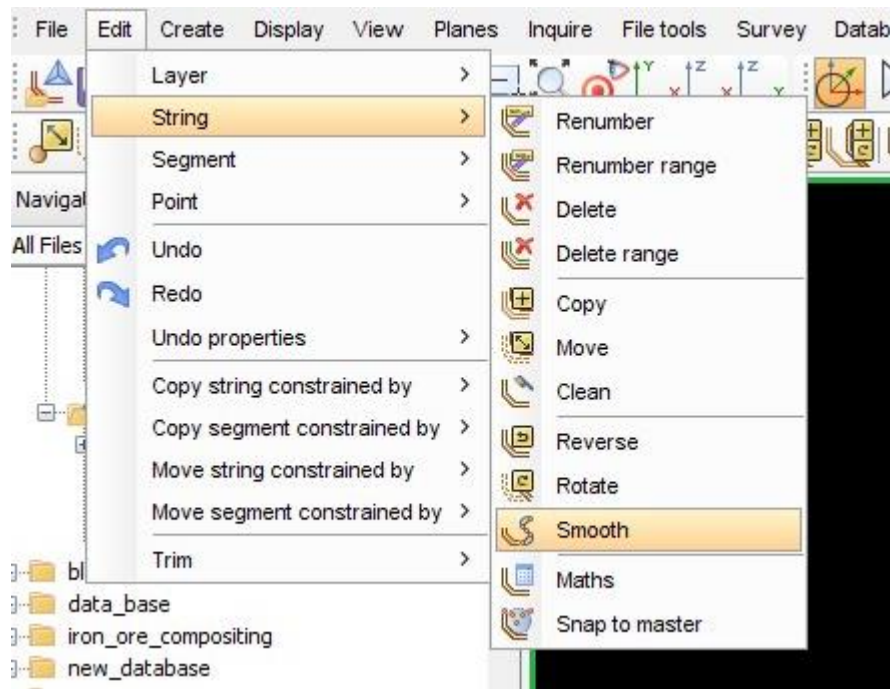


- Select any one operation at a time and perform all the cleaning.



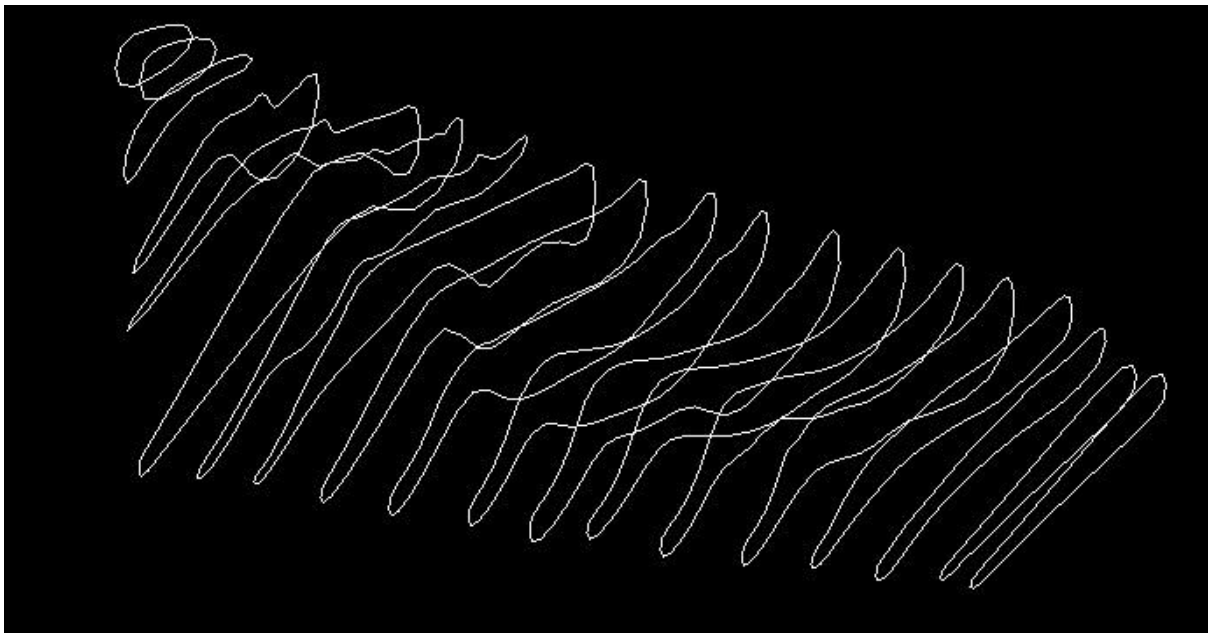
Smoothing:

- **Go to Edit >> String >> Smooth**

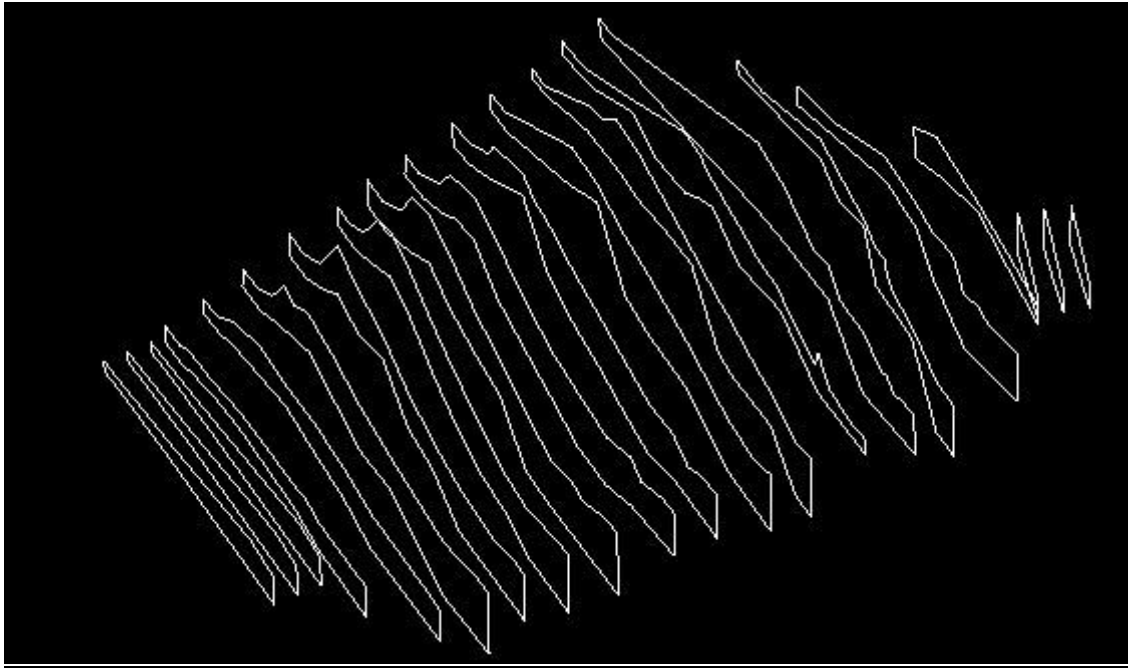


- After smoothing your sections will look something like the below picture.

After Smoothing:



Before Smoothing:



Changing the orientation of the segments:

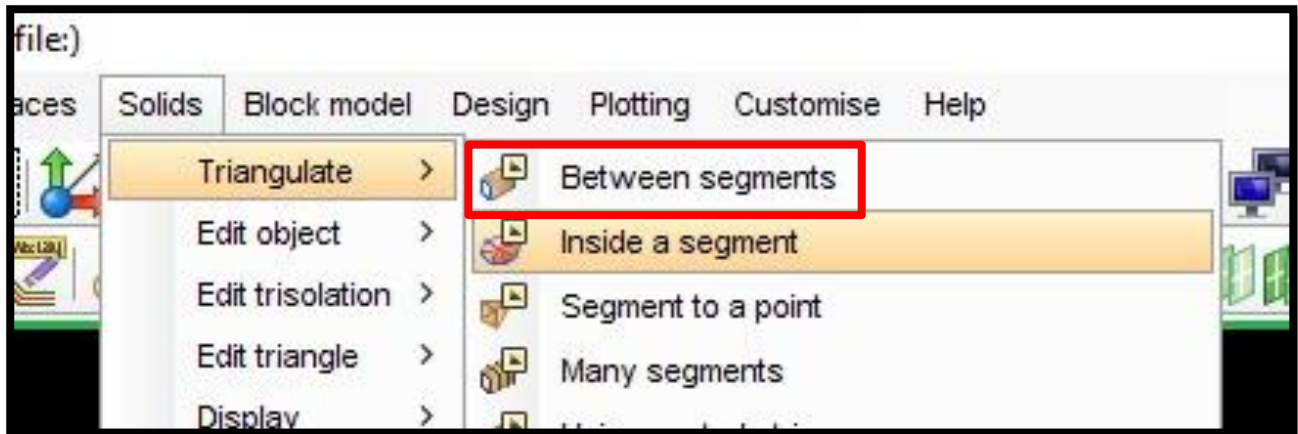
- To get a solid all the segments must be in a clockwise direction. If the segment is in anti-clock direction, it'll create a void instead of creating a solid
- To check the orientation of the segment, **Go to inquire >> Segment properties >> click on the section**
- Check the description box to know the properties. There you'll find the orientation as status. The status must be in clockwise direction

Layer cleaned_sections.str, String 1, Segment 15, #Points 355, **Status=Clockwise**
 Chord Length=1121.34, Slope chord length=1197.807, Arc length=1121.34, Slope arc length=1197.807
 Segment is dipping or not coplanar: horizontal area = 0.065 vertical area = 26855.149 best fit area = 26855.149 section area = 0.065

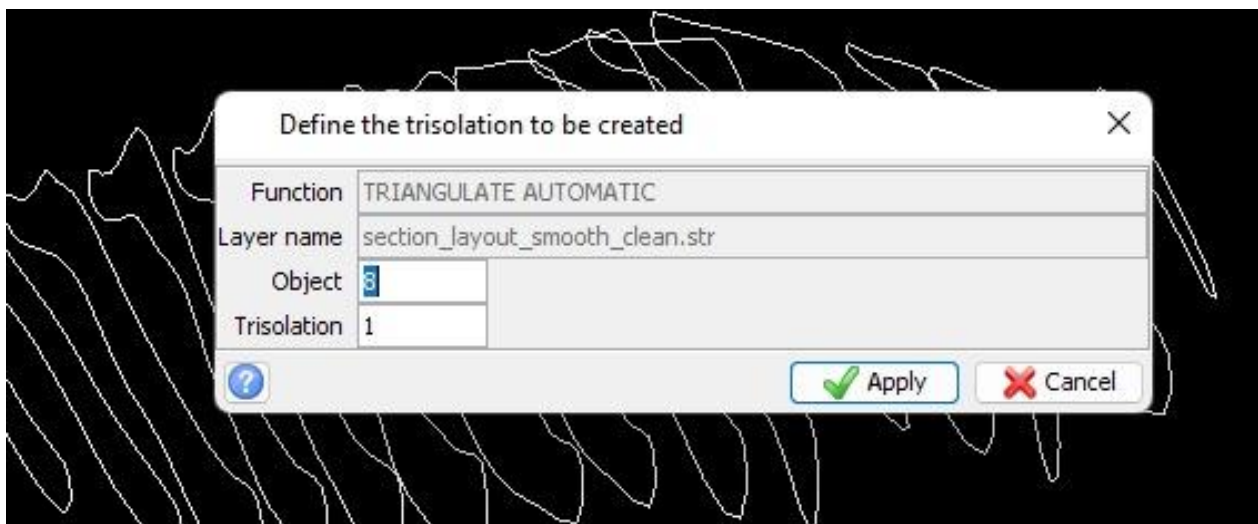
- Check these properties for all the segments in the string.
- If the segment is in anti-clockwise direction. Change the orientation of the segment to clockwise
- To change the orientation, **Go to Edit >> Segment >> Reverse >> Select the segment.**

Creating a Solid dtm:

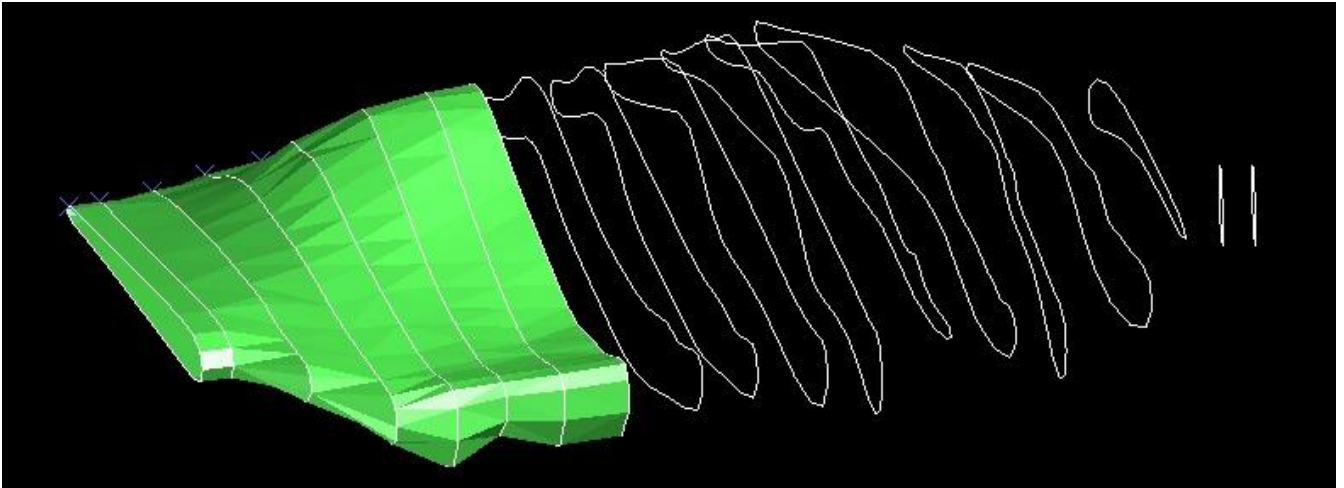
- Bring all the segments to the main graphics layer. **Go to Solids >> Triangulate >> Triangulate between segments**



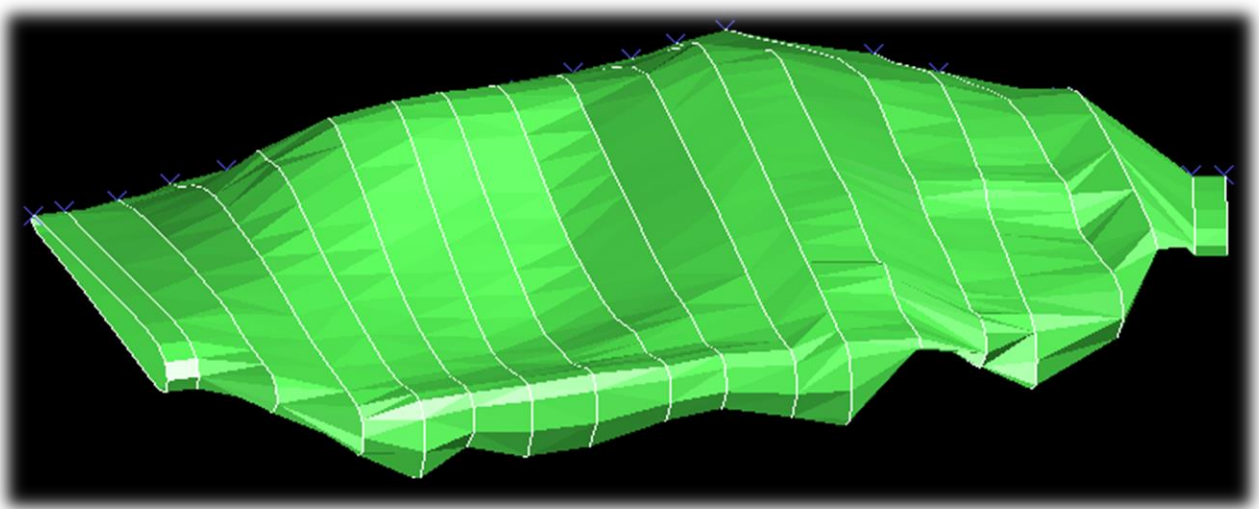
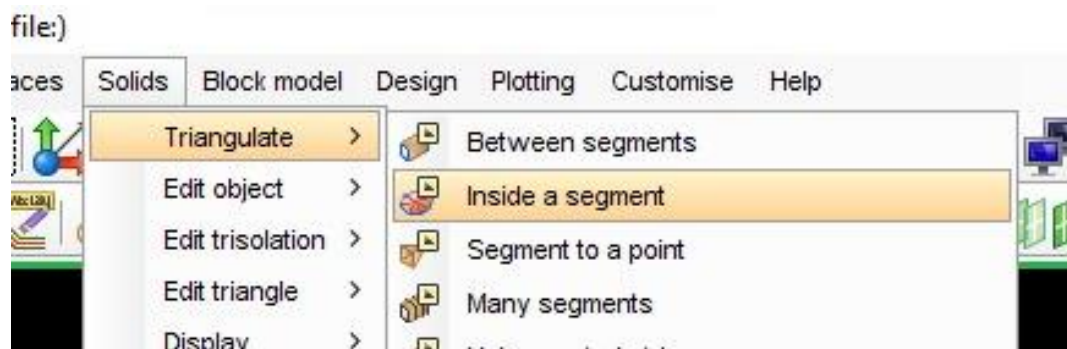
- A new pop up will ask you to define the object id and trisolation. Object id will decide the dtm colour. Different object ids have different colours. Type any number.



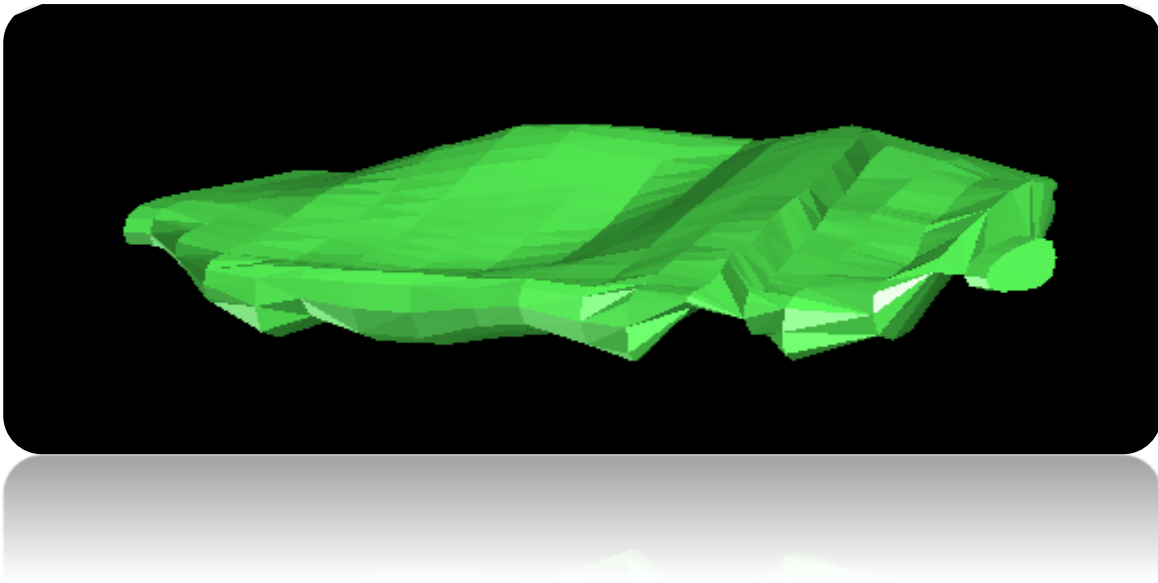
- **Click on Apply**
- Select the portion between the two segments.
- After selecting surpac will create the dtm in between that segments



- Once you done triangulating all the segments. You have to close the start and end segments to finish the solid dtm.
- To close the segments, **Go to Solids >> Triangulate >> Triangulate inside segment >> Select the segment.**

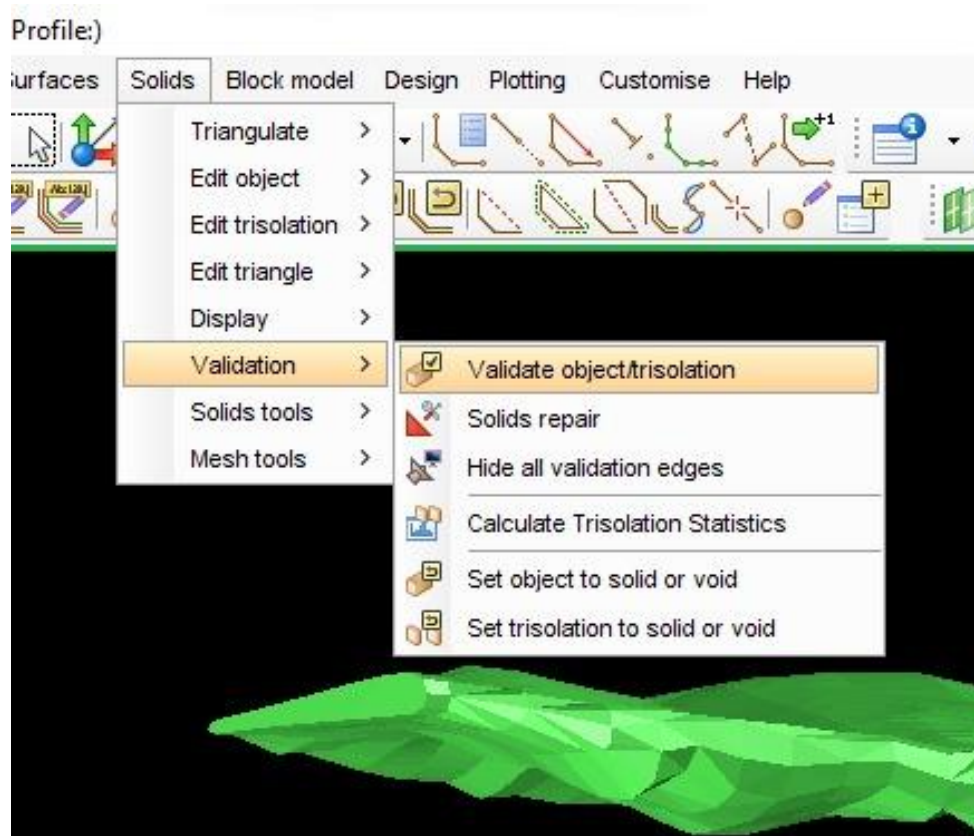


- Save the new layer as solid dtm.



Validating the solid:

- The solid you have created must be validated and should have zero errors.
- If the solid is invalid, that means something is wrong with your sectioning and cleaning. In such case, you have to do all the sectioning, normalising, cleaning, and smoothing all over again.
- To get the solid validation report **Go to Solids >> validation >> Validate object/trisolation**



- A new pop up will appear. Select the dtm and give a file name to the report file.
- Click on Apply
- A report file in .not format will appear on your screen.

solid_validation_report - Notepad

File Edit Format View Help

GEOVIA Jun 29, 2022

Solid validation report

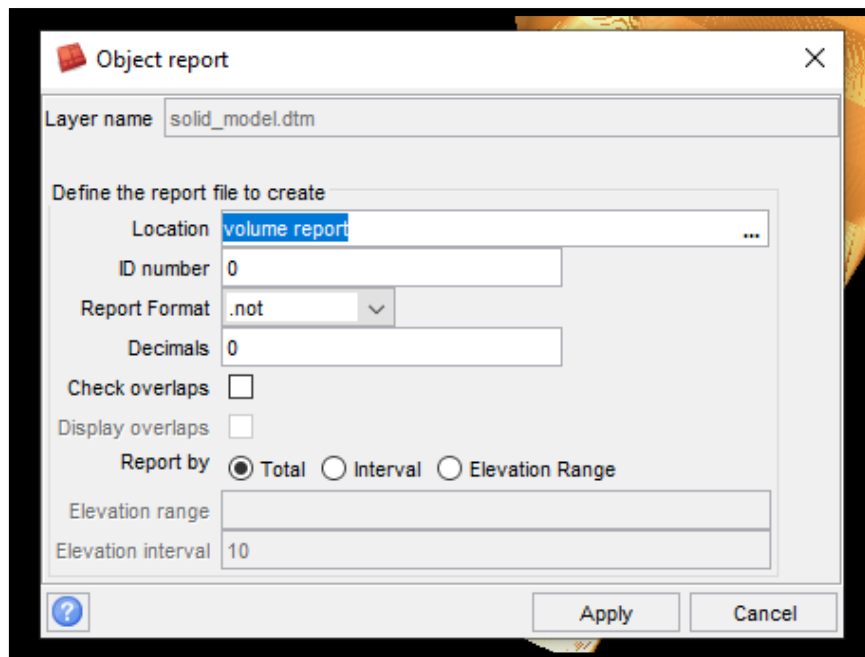
Layer:solid_model.dtm

Object	Trisolation	Valid	Open/closed	Connected	Duplicate (removed)	Invalid Edges	Intersecting	Reversed
7	8	Valid	Closed	Connected	0	0	0	0
Totals					0	0	0	0

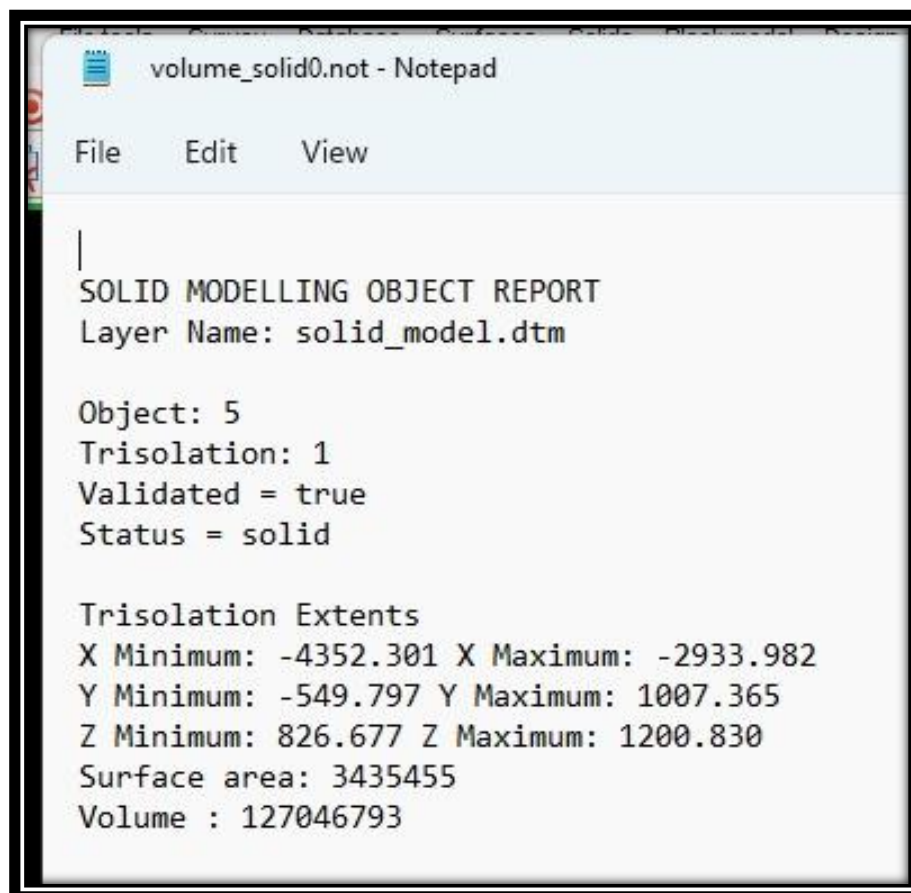
Solid validation report 1/1

Solid Volume Calculation:

- To get the volume of the solid, **Go to Surfaces >> Volumes >> Report volume of solids >> define the report file name >> Click on Apply**



- Volume report also gives the min and max extents of X,Y and Z coordinates.
- Solid Volume report should look something like the below picture.



COMPOSITING

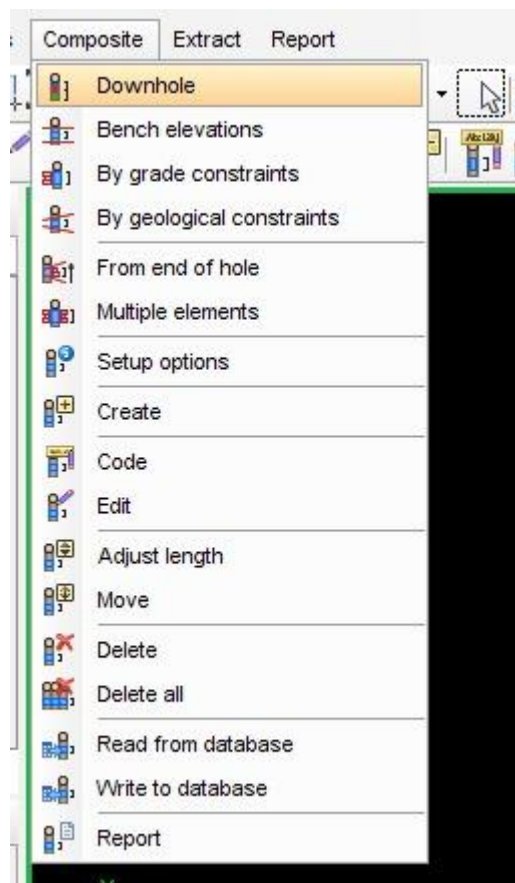
The process by which the values of adjacent samples from boreholes are combined to determine the value of the longer down-hole intervals is known as "compositing".

The weighted average of the grades from the initial sample is used to generate the grade for each new interval. These are typically weighed by length, with the possibility of adding weight from specific gravity and core recovery.

➤ There are many methods to do the compositing, but the most famous ones are,

1. Down hole compositing
2. Ore zone compositing
3. Compositing by bench height

➤ To do the compositing, bring the geological profile on to your screen.



- **Go to Composite >> Down hole >> Pop up**
- Define the name of the composite in location. Min % of samples to be included would be 75. Click on no selection in zone selection method.
- Select the assay table and field type fe, **Click on Apply**

Composite downhole

Define the string file to create

Location: down_hole_composite

ID number: 0

String: 1

Composite length: 1

Determine composite length by: fixed length

Minimum % of sample to be included: 75

Define the zone selection method:

- ☒ NO SELECTION
- ☐ MULTIPLE ZONES
- ☐ ZONE FROM TO

Dilute negative samples: ☐

Table name: assay

Fields to be composited

	Field Name
1	fe

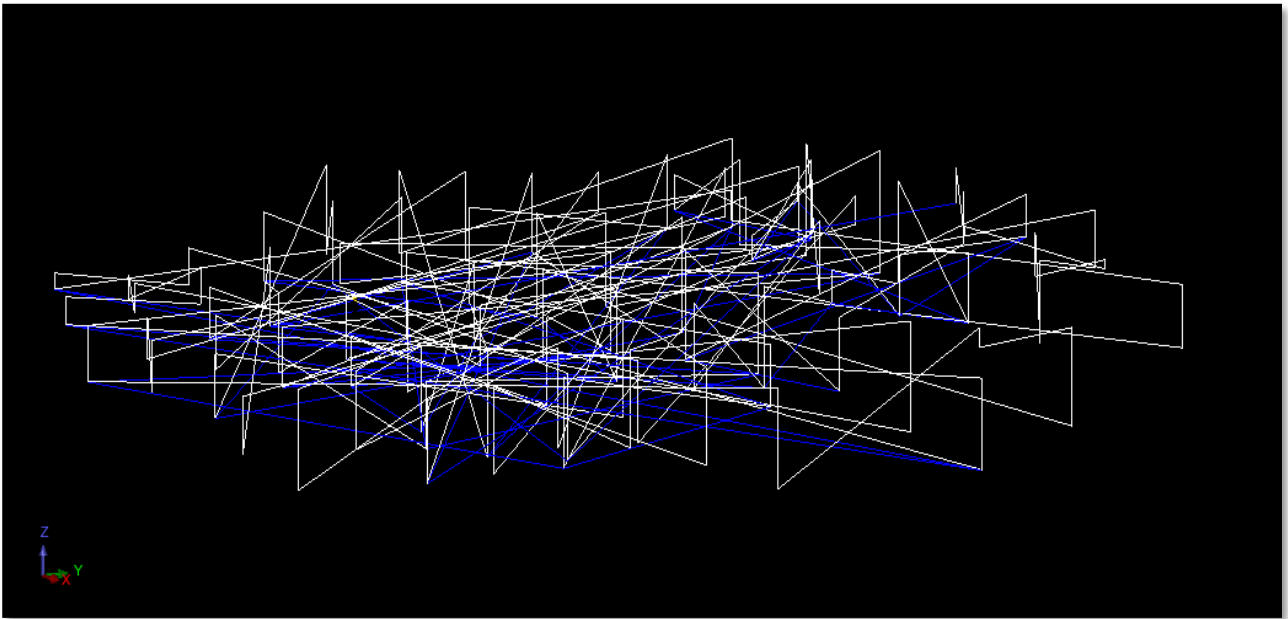
Optional weighting fields

	Field Name	Default	Include Limit
1		1	

Apply Cancel

- Look in the description box to find out how many raw samples and how many holes were processed. Note the sample data's mean, standard deviation, and variance as well.
- In your work directory, a string file will be generated and saved.

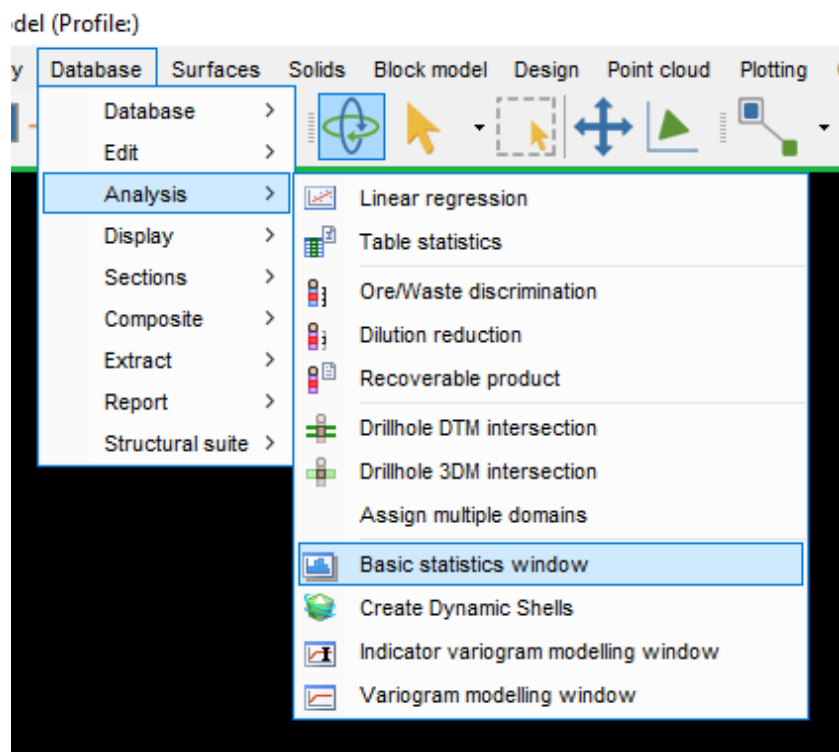
- Bring that string file to the main graphics layer. It should look like the below picture
- This string file will be used in basic statistics and variogram modelling of your geological data.



STATISTICAL ANALYSIS

Basic statistics:

- Connect the database
- **Go to Database >> Analysis >> Basic statistics window**



- **Pop up**
- **New window will appear on your screen**
- **Go to File >> Load data from String file >> Pop up**

Basic statistics

Basic Advanced

Location: downthe_hole_composite

Id range: 0

String range:

Description Fields for Data

	D Field	Name	Minimum value	Maximum value	Weighting Field
1	D1	fe	0	99	

Histogram Bins

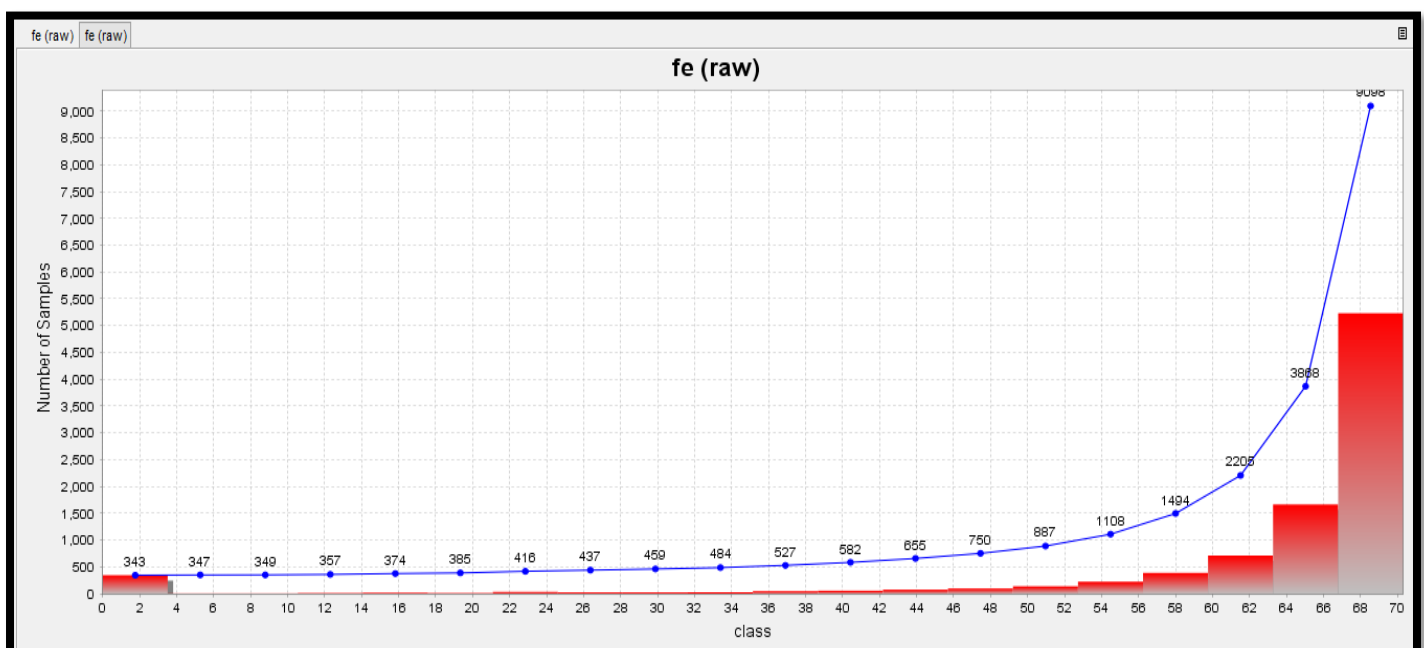
Number: ☒ 20

Width: ☐ 2

Include Coordinate Graphs? ☐

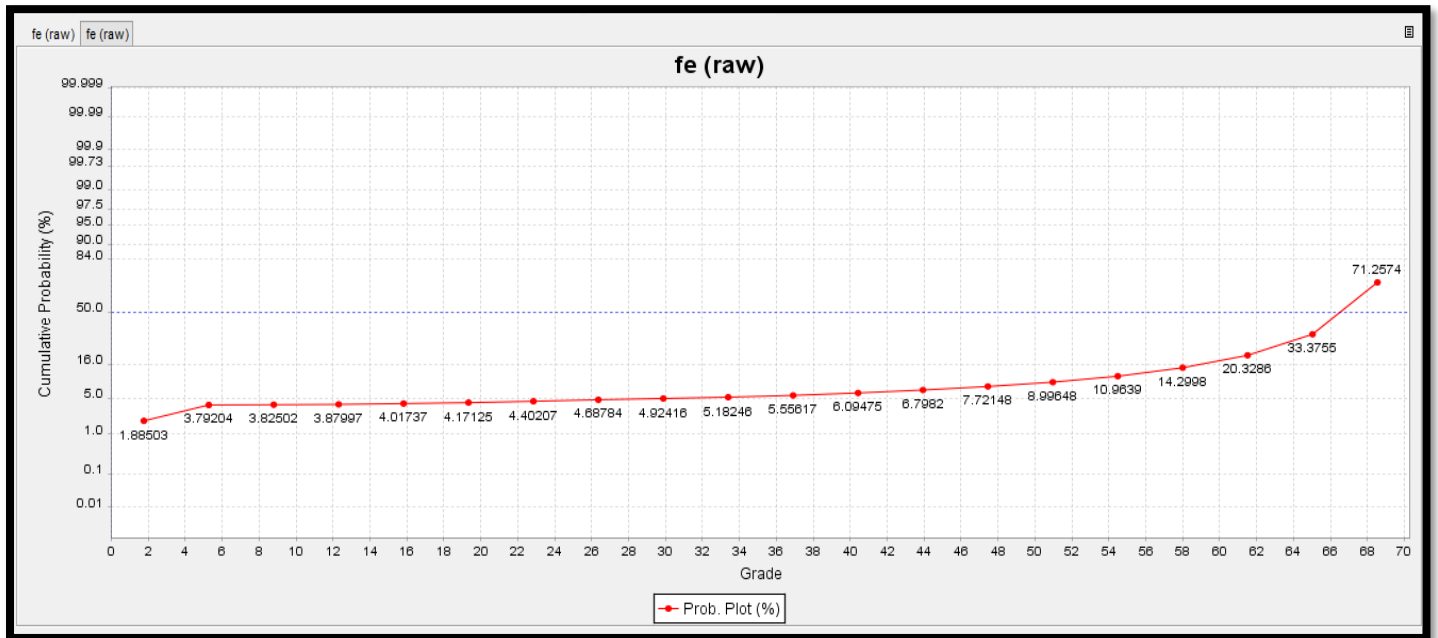
Apply Cancel

- In the location select the down hole compositing string file
- In description fields select fe and set min and max value as 0 and 99
- Select the number of histogram bins and width
- **Click on Apply**
- Now the basic statistical histogram and cumulative frequency curve will display on your screen

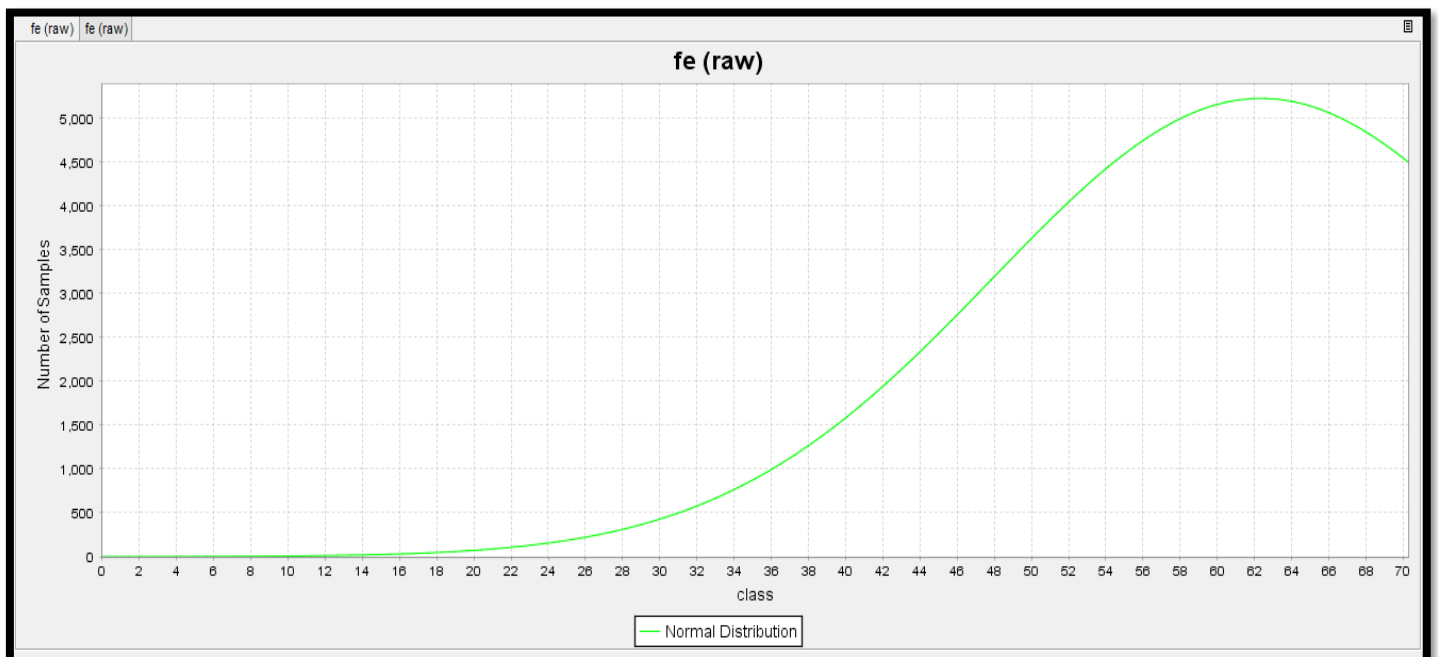


- After loading the histogram in the basic statistical window, we can perform some basic statistical operations. Some of those are,

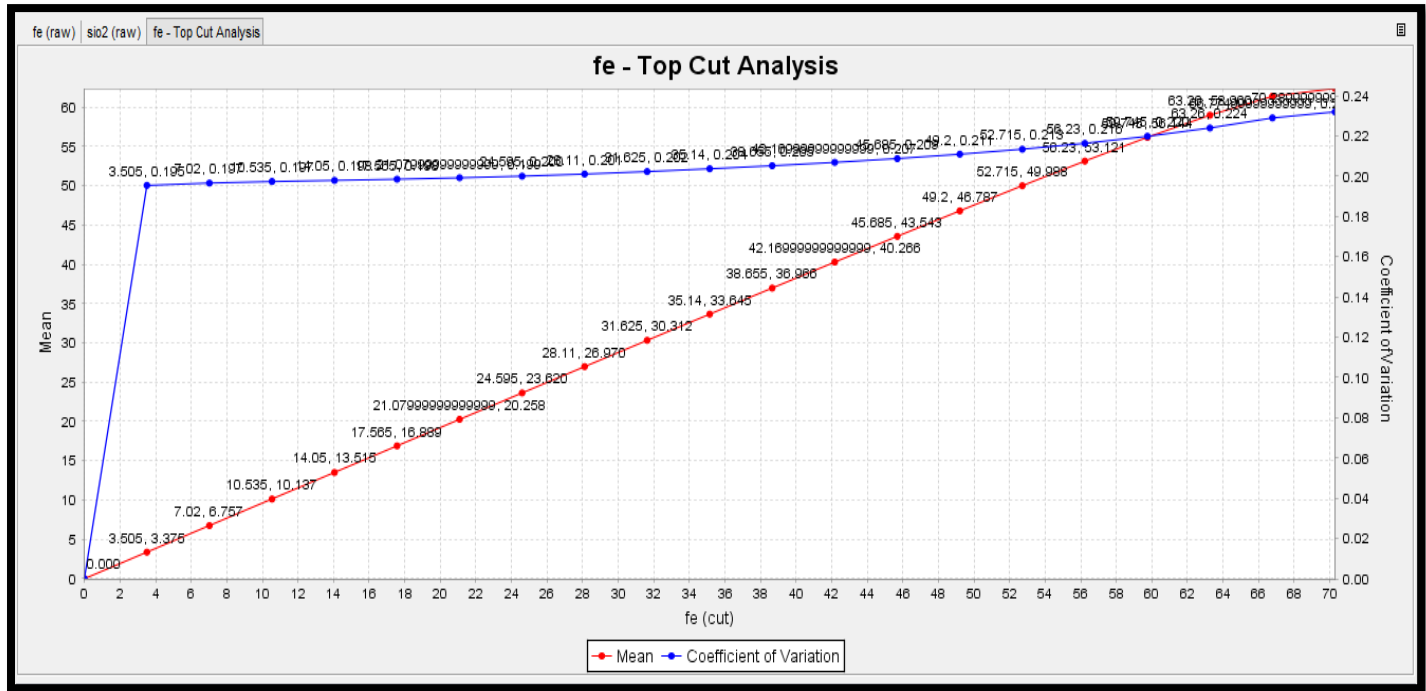
Probability curve:



Normal Distribution Curve:

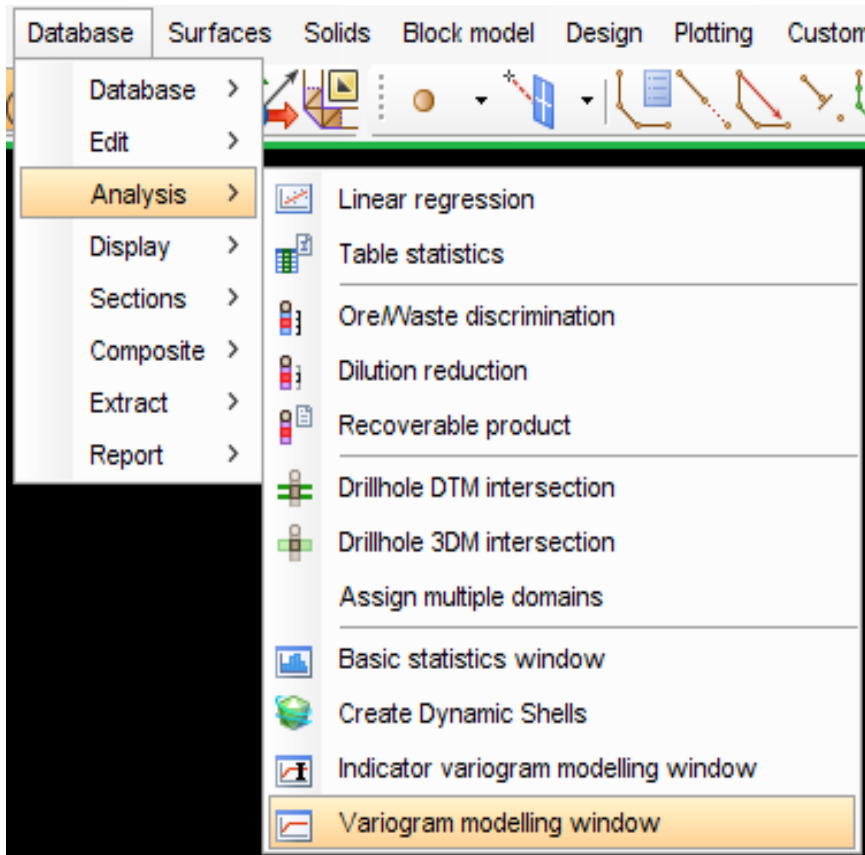


Mean and co-efficient variation curve:



Variogram modelling

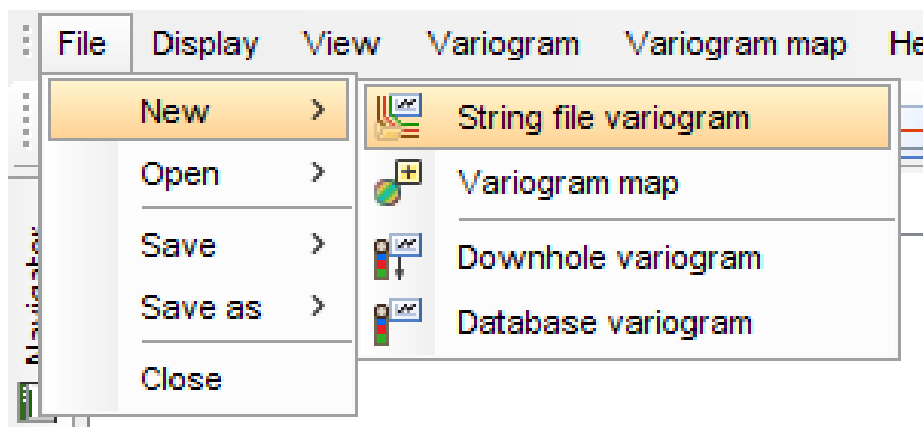
- Compositing.str file will be used for variogram modelling.
- **Database >> Analysis >> Variogram modelling window**



- New Experimental variogram modelling window opened

Now,

File >> New >> String file variogram



- Variogram map calculation window will open

1. In Location attach compositing.str file
 2. Number of variogram : Choose from drop down list
 3. Lag : According to data
 4. Maximum distance : (it's a max range of data)
- and fill all other information according to the available data. >> **APPLY**

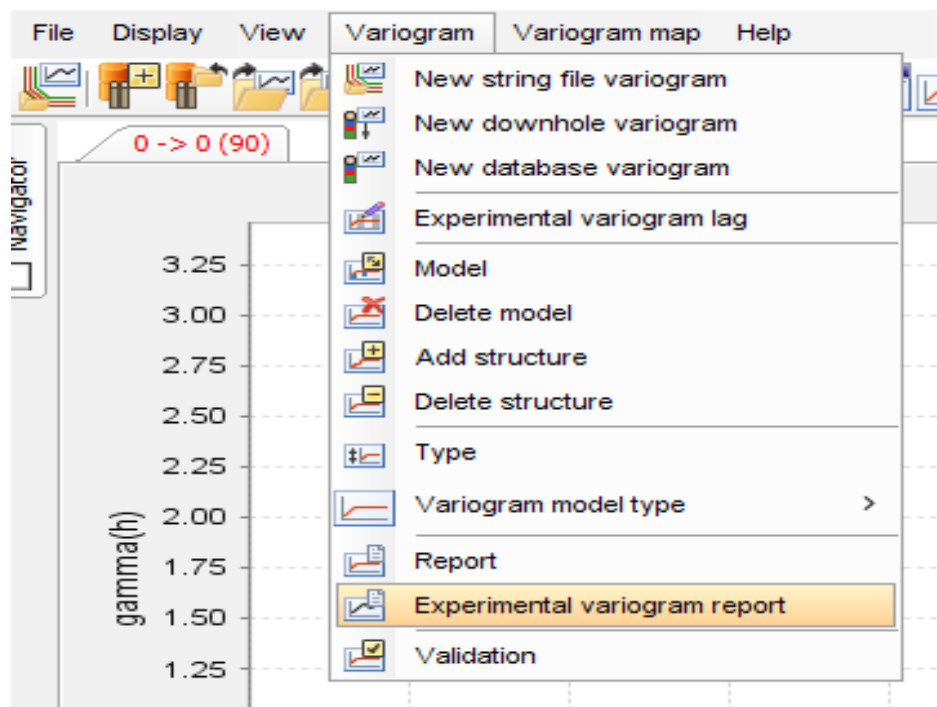
Variogram map calculation

Basic **Advanced**

Location: whole_compositing
 Id range:
 String range:
 D field: D1
 Minimum value:
 Maximum value:
 Data Selection
 Plane dip: 0
 Dip direction: 0
 Number of variograms: 36
 Angular increment: 10
 Spread: 90
 Spread limit:
 Lag: 50
 Maximum distance: 1500
 Output report file name: map_variogramn
 Output report file format: .not - Surpac Note File
 Display report: ☒

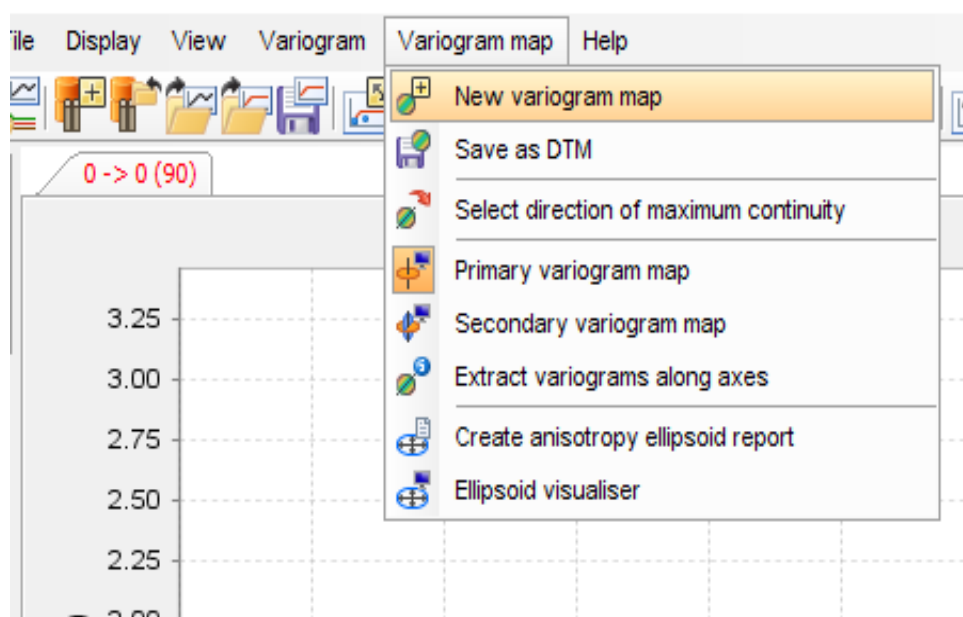
- The Experimental variogram will open and .not report file also saved in the folder.
- This window we can also see in tile and tab form.
- We can create the different variogram models.
 - Spherical model
 - Exponential model
 - Gaussian model
- For Experimental variogram report

**Variogram >> Experimental variogram Report >> Enter File Name >>
APPLY**

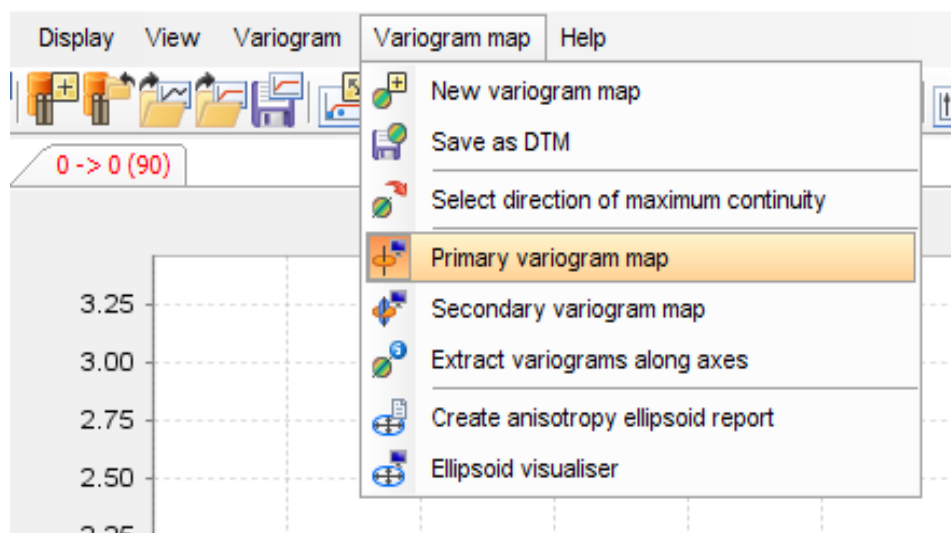


Variogram

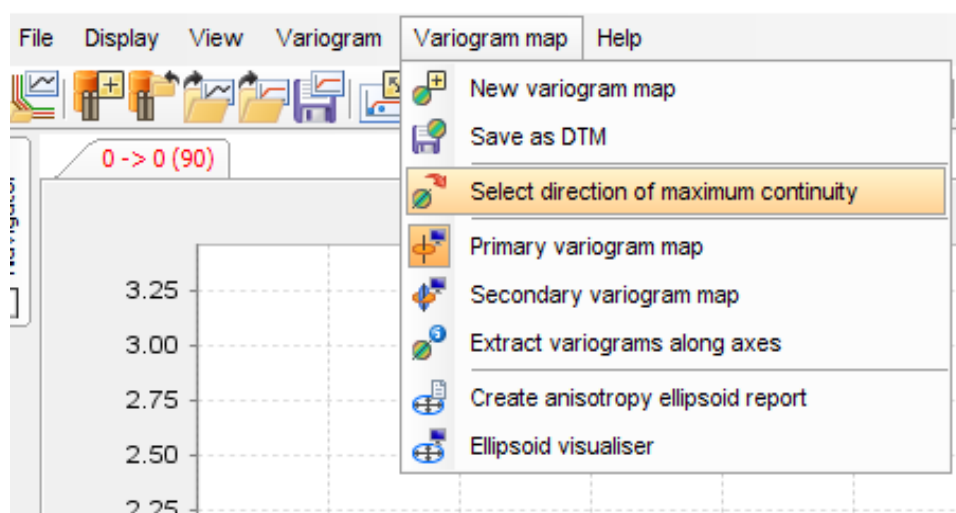
- **Variogram map >> New Variogram Map**



- **Variogram map >> Primary variogram map**

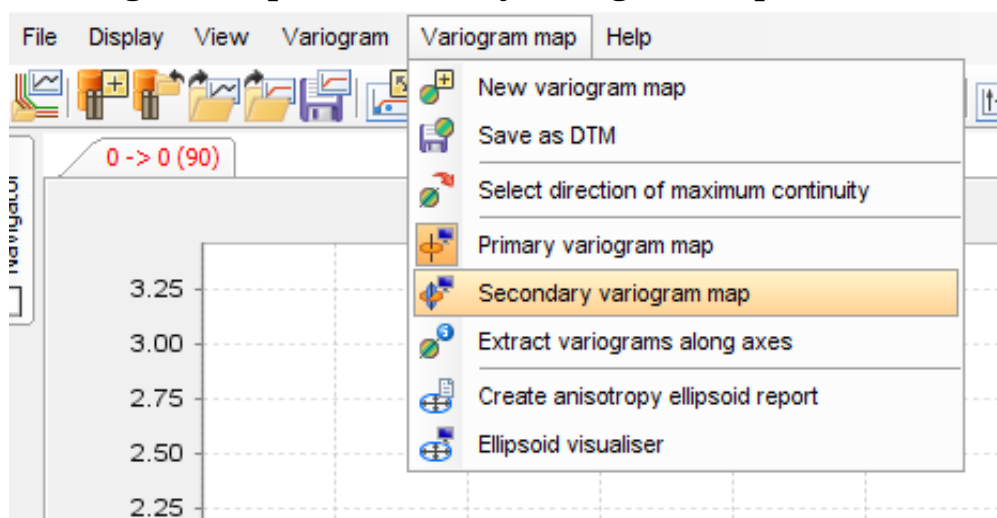


- **Variogram map >> Select direction of maximum continuity**

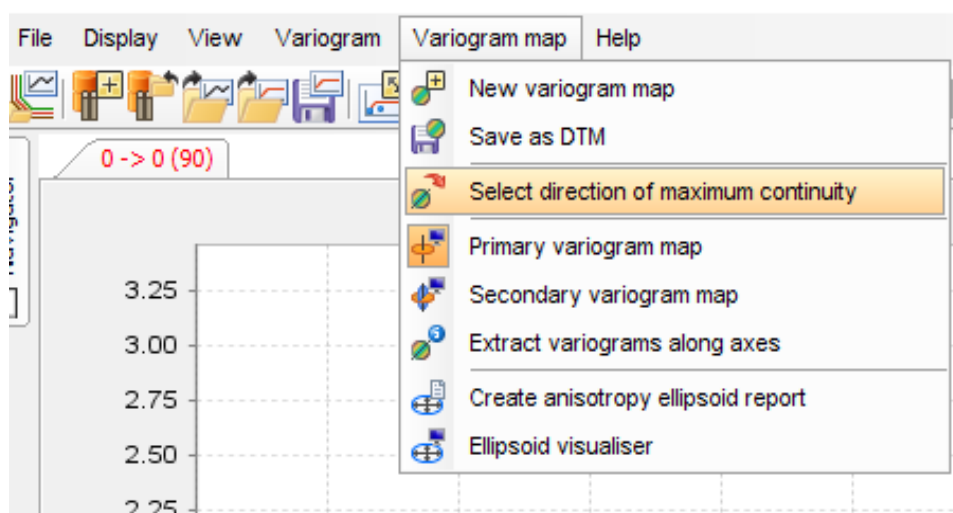


Now, the process will happen and at the right bottom graph select the line and set at the maximum continuity of graph.

- **Variogram map >> Secondary variogram map**

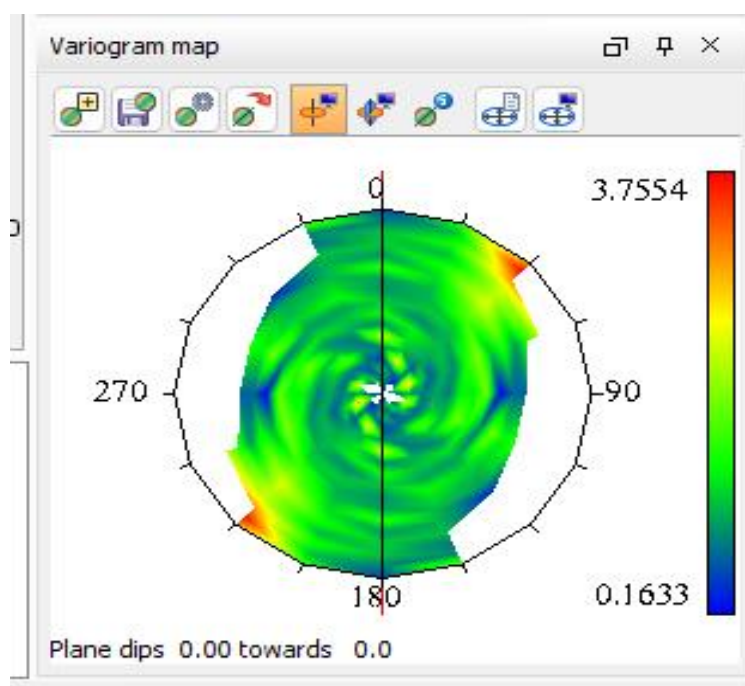


- **Variogram map >> Select direction of maximum continuity**



Now, the process will

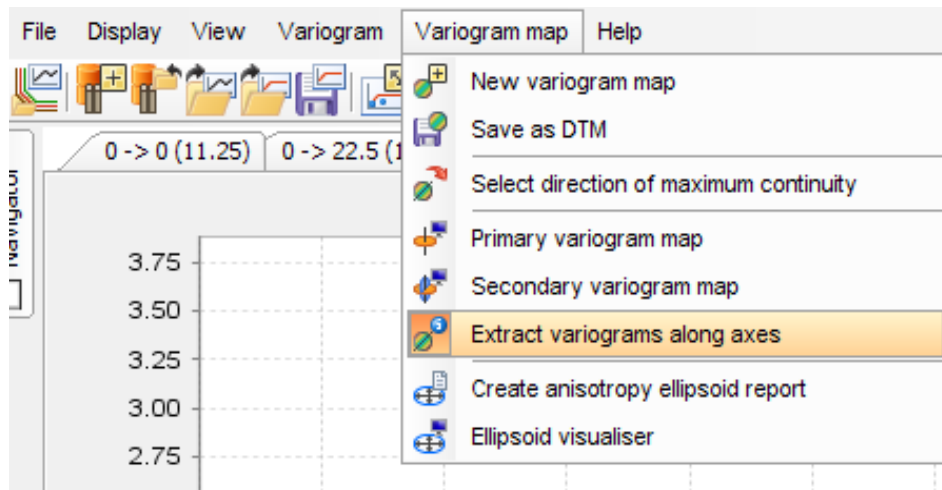
line



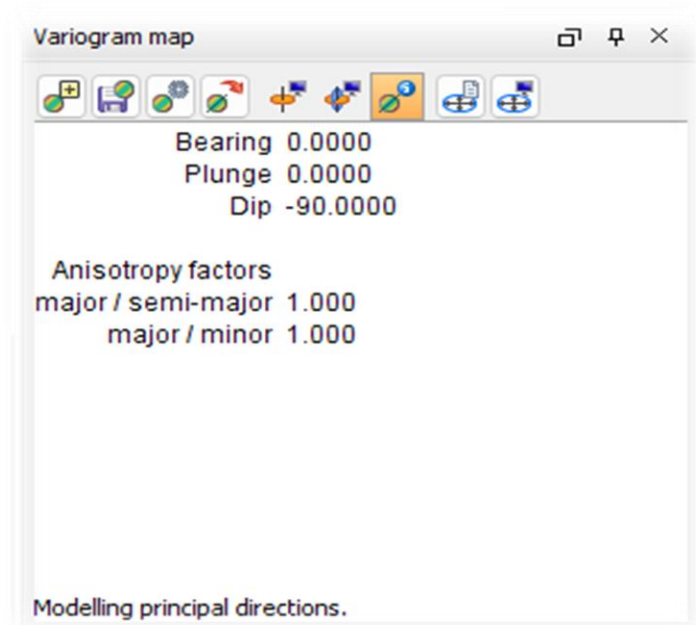
happen and at the right bottom graph select the and set at the maximum continuity of graph.

This red line move to adjust at the maximum continuous graph

- **Variogram map >> Extract Variogram along axis**

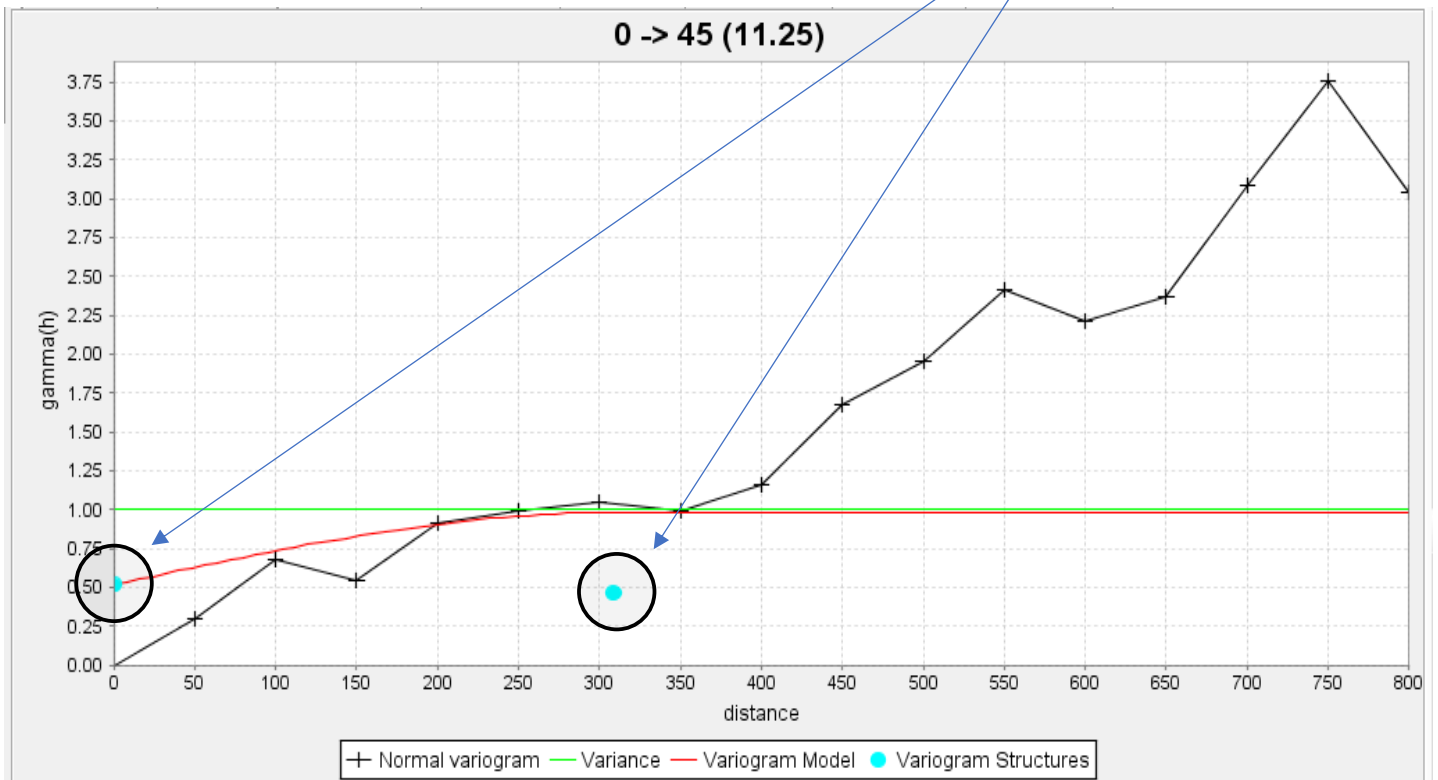


After this process at the right bottom most corner there will be **Anisotropy Report**. Note down for further use .

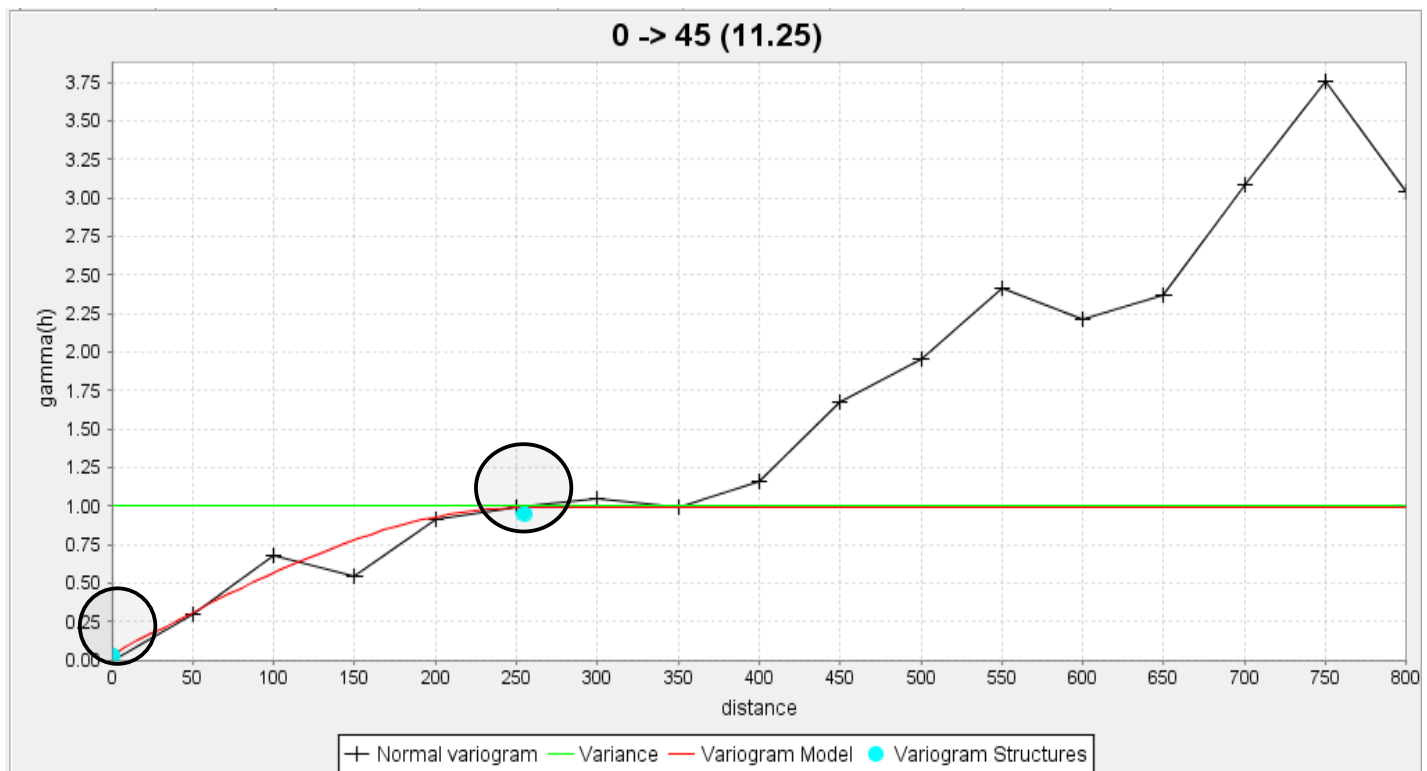


- **Variogram >> Model**

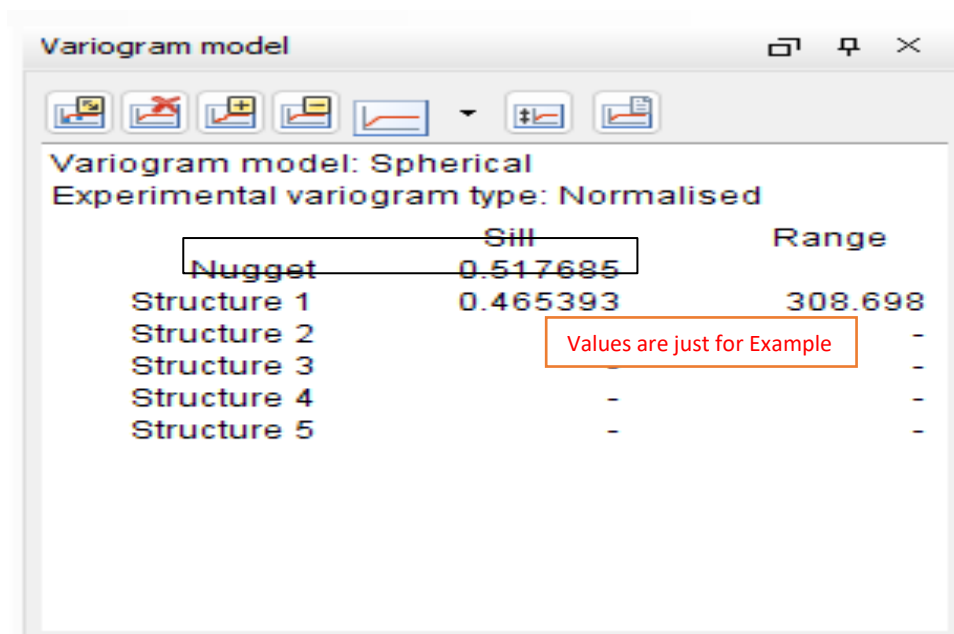
1. The variogram window will show the skyblue points to adjust the perfect variogram map.



2. Now, adjust the point perfectly and create the perfect variogram map.



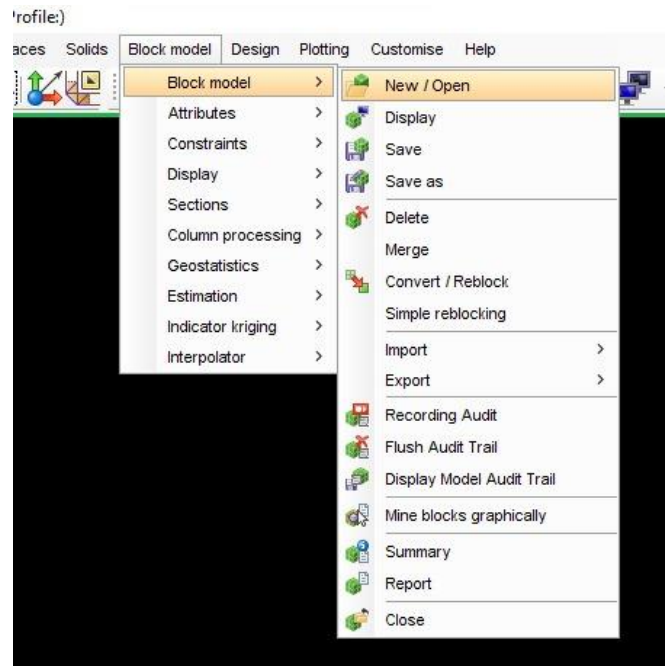
3. After the adjustment we will get the **Nugget, Sill, Range** value.



BLOCK MODELLING

Creating a new block model:

- **Go to block model >> New/Open**
- **Pop up**



- **Define a name for the block model, Click on Apply**
- **Pop up**

Creating new block model definition

Model name fe_block_model.mdl

Description

Define model using ☒ Min/Max coords
☐ Origin coords/extends

Extents **Rotation**

☐ Get extents from string file ?

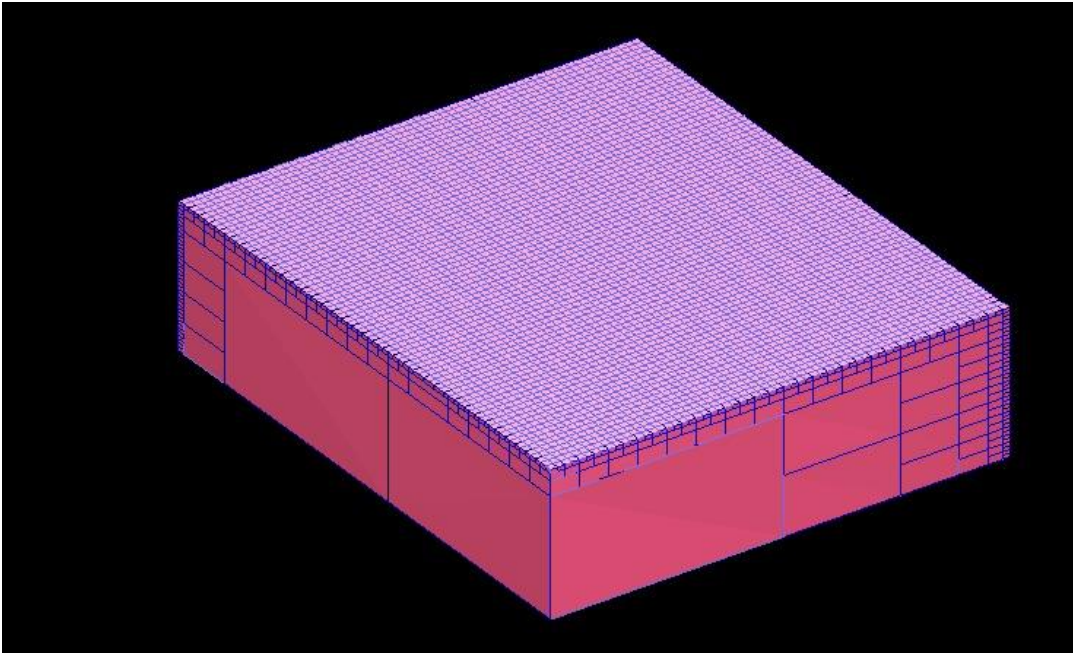
Coordinate extents		User block size	
	Minimum coordinates		Maximum coordinates
Y	0.000	Y	0.000
X	0.000	X	0.000
Z	0.000	Z	0.000

Sub blocking None

☒ Maintain audit trail

Apply Cancel

- Define a description for you block model, say this is the new block model.
- To get the X,Y and Z coordinates extents, Click on Get extents from string file.
- Select the solid model string file and define the block sizes in X,Y and Z direction.
- **Click on Apply**
- **A new block model will be created**

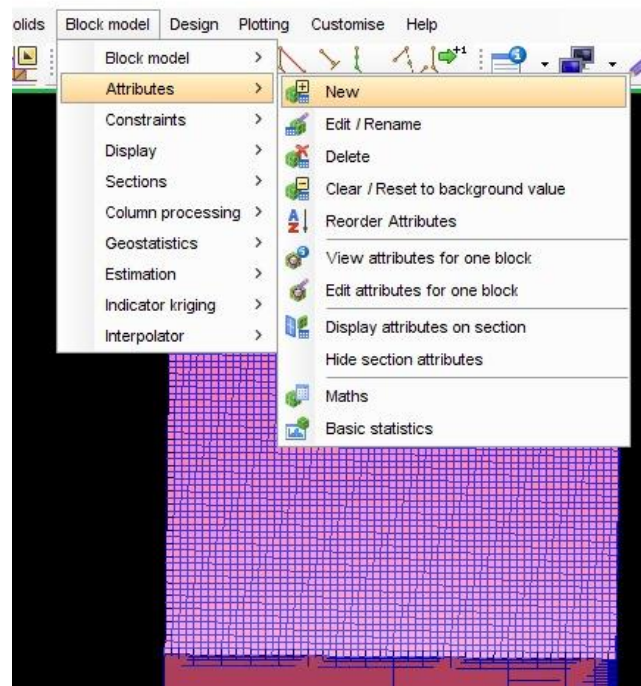


Attributes:

- Attributes is the values that will be assigned to each block in the block model.
- There are generally 3 attributes are there, they are
 1. Partial Percentage (pp)
 2. Specific gravity (sg)
 3. Fe (assay value)
- Partial percentage is an attribute for percentage of volume representation in the block
- Specific gravity is an attribute used for tonnage calculation in resource estimation
- Fe attribute is an attribute to represent the grade value of Fe in each block.

Partial Percentage:

- To assign the partial percentage attribute, **Go to Block model >> Attributes >> New**
- **Pop up**
- Give the attribute name as pp, attribute type is real, decimals 3, type partial percentage in the description.
- **Click on Apply**

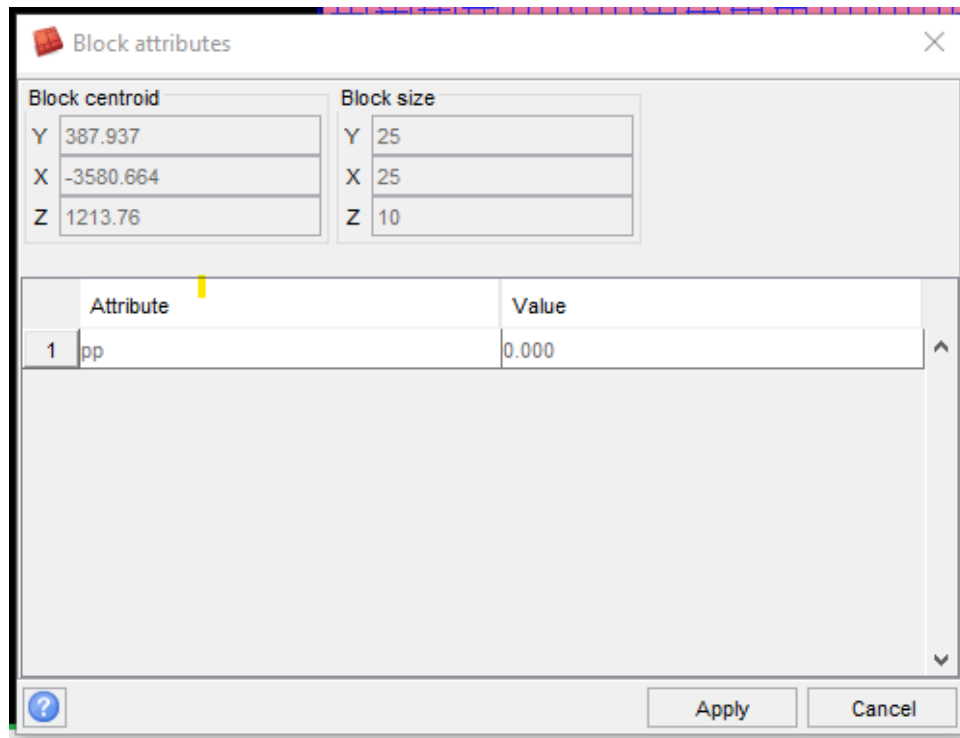


Add attributes

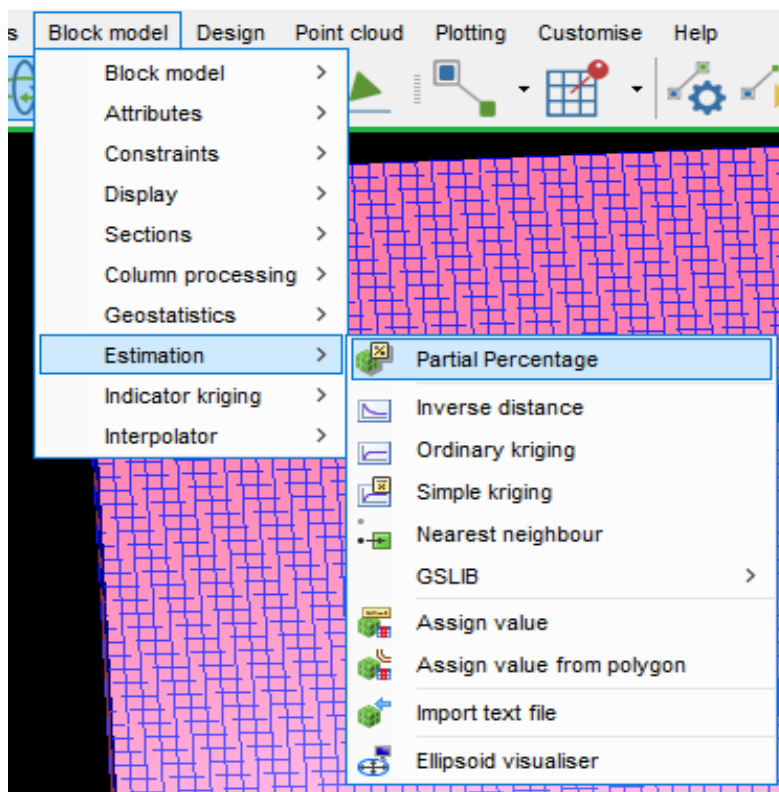
	Attribute Name	Type	Decimals	Background Value	Description / Expression
1	pp	real			Partial percentage

Apply Cancel

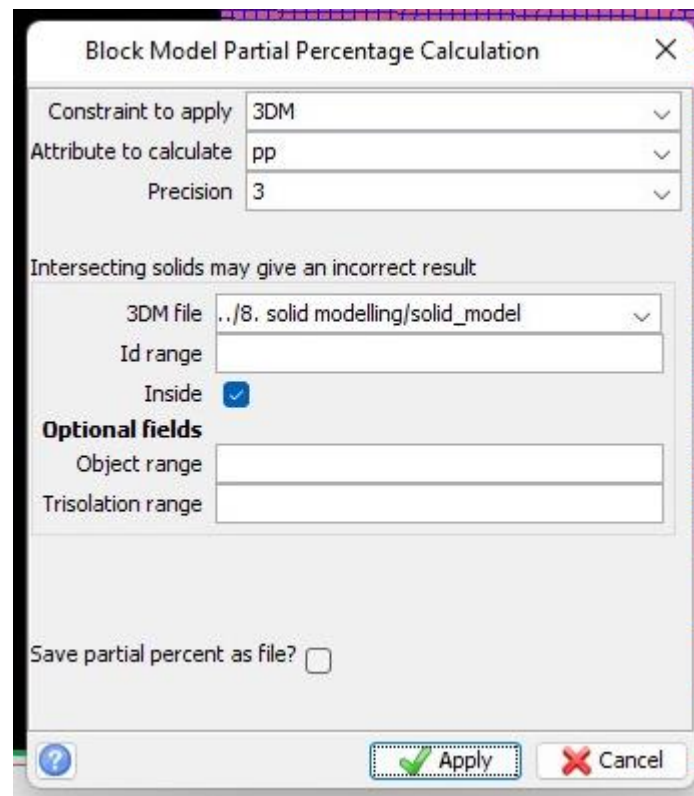
- To check whether the attribute is added or not to the block model, **Go to Block model >> Attributes >> view attribute for one block >> Click on any block.**



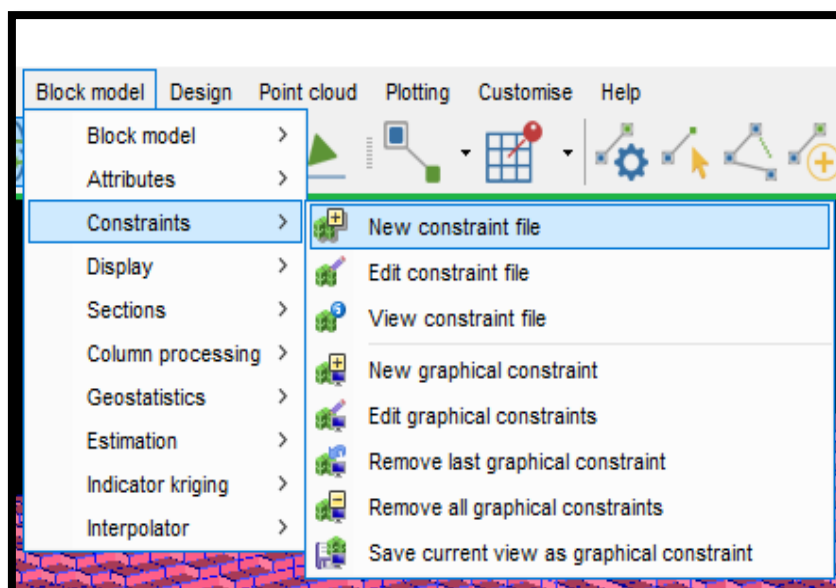
- The pp attribute is added to the block model, but it is not showing any value in it.
- To get the partial percentage of each block. We need to do the estimation
- **Go to Block model >> Estimation >> Partial percentage**



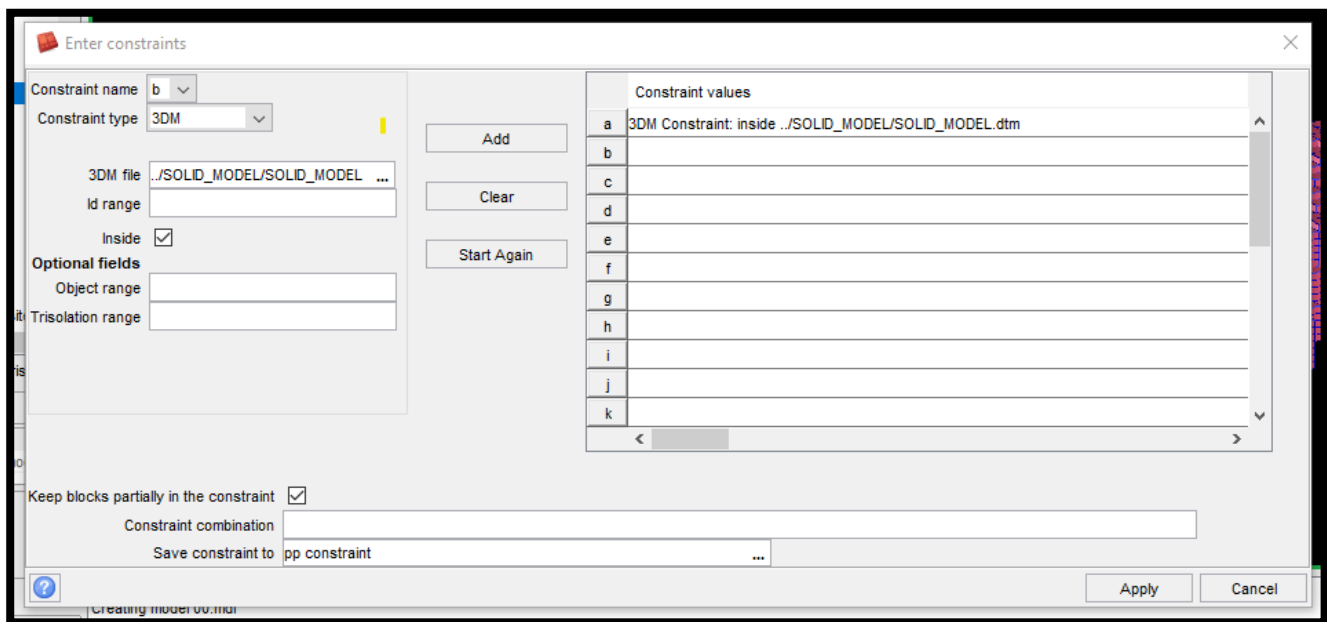
➤ **Pop up**



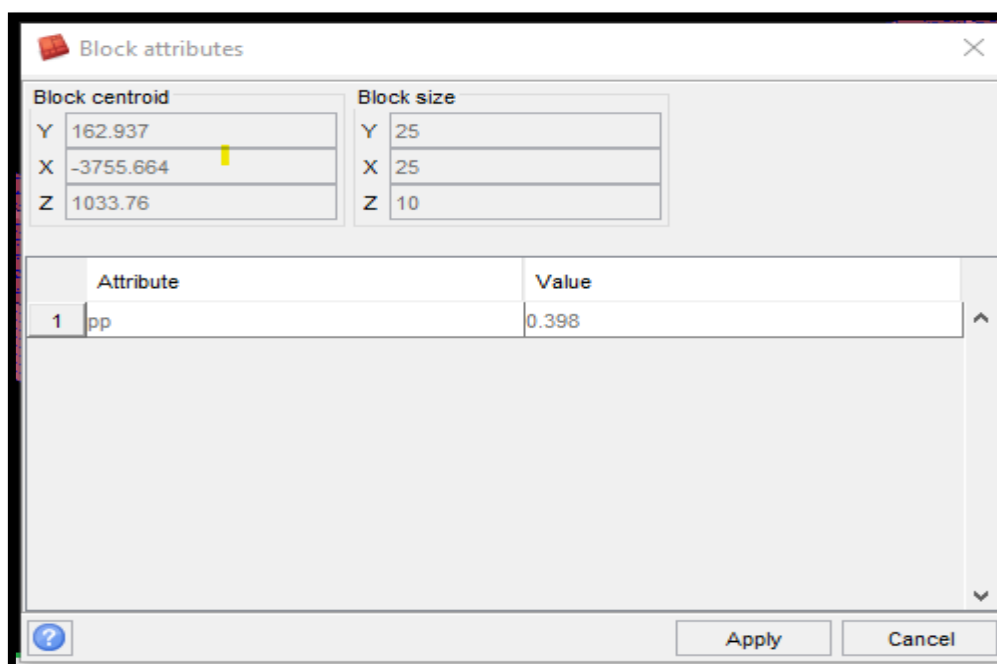
- Define the constraint to apply as 3DM , select attribute to calculate is pp, precision is 3.
- Select the solid model dtm in 3DM file
- **Click on Apply**
- After estimation. We need constrain the block model to get the attribute values.
- **Go to block model >> constraints >> New constraint file**



- A new pop up will appear on your screen



- Select the constraint type as 3DM and select the solid model dtm in 3DM file, **click on add**
- Click on keep blocks partially in the constraint
- Give a name to the constraint file
- **Click on Apply**
- Now the value of pp attribute is added to your block model
- To check the value of the partial percentage, **Go to Block model >> Attributes >> view attribute for one block >> Click on any block.**



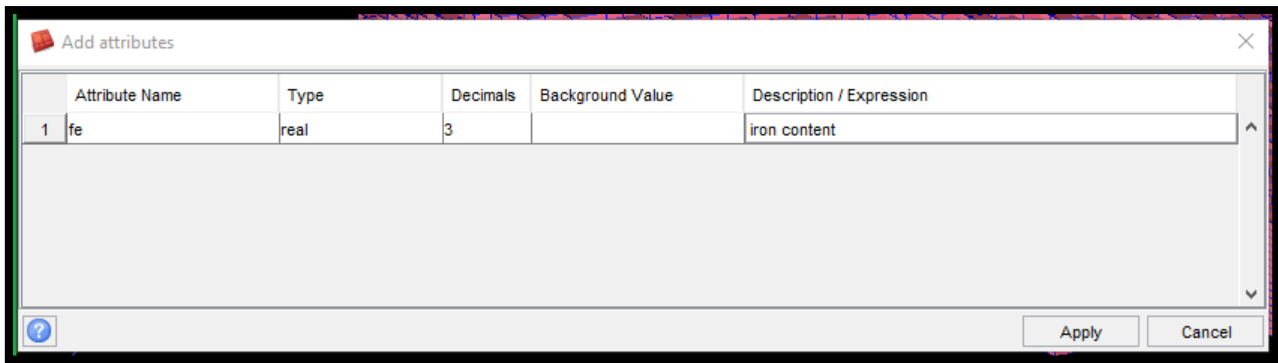
Specific Gravity:

- To assign the value of specific gravity, **go to attributes >> new attribute >> sg >> real >> background value is 3.5 >> Type specific gravity in description >> Click on Apply.**
- we are manually assigning the background value as 3.5 because Fe specific gravity is in between 3 to 3.5
- we don't need to do the estimation and constraining the block model for sg attribute as we already assigned the background value.
- To check whether the sg attribute is added or not, **Go to Block model >> Attributes >> view attribute for one block >> Click on any block.**

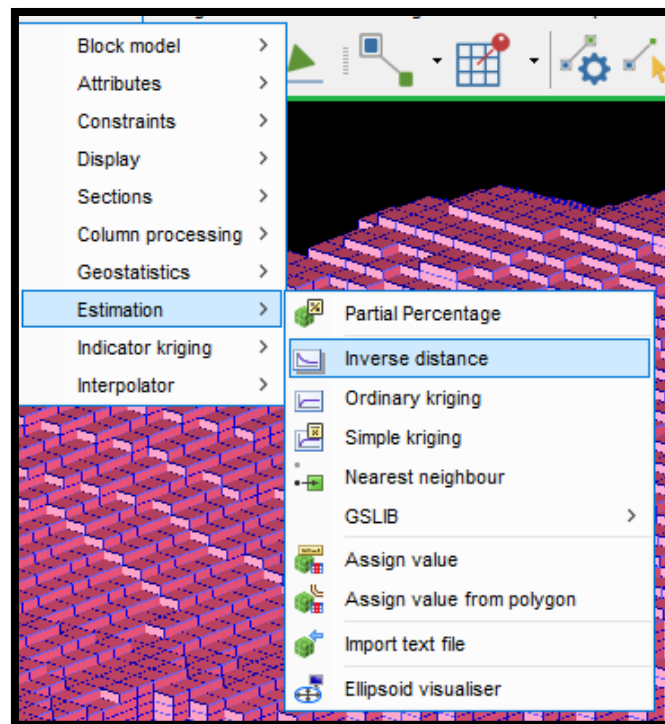
Block attributes		
Block centroid		
Y	237.937	
X	-3705.664	
Z	1023.76	
Block size		
Y	25	
X	25	
Z	10	
	Attribute	Value
1	pp	0.000
2	sg	3.500

Fe attribute (Assay Attribute):

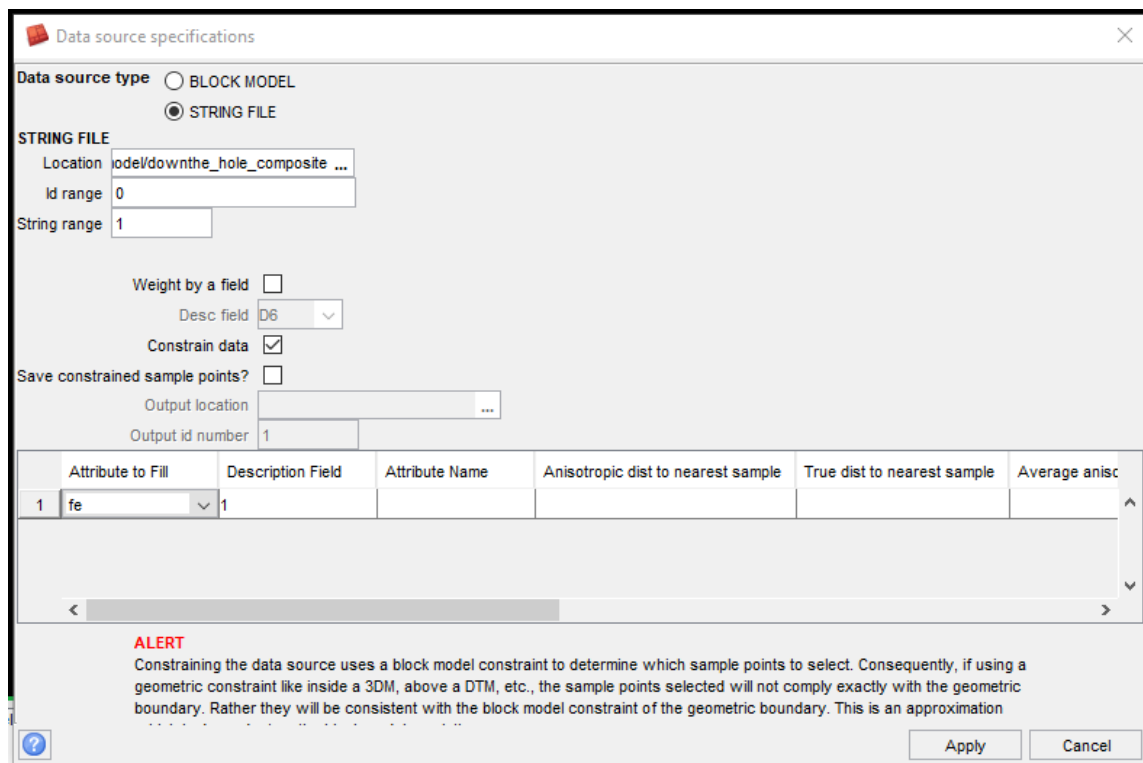
- To assign the assay value to the block model. Fe attribute is used



- Create and add the new fe attribute to the block model, same as the previous 2 attributes.
- To get the assay values of each block, **Go to estimation >> Inverse distance >> Pop up**



- Select string file, upload downhole composite file in location, select **fe** as attribute to be filled, description field 1, **Click on Apply.**



Data source specifications

Data source type ☐ BLOCK MODEL ☒ STRING FILE

STRING FILE

Location: model/downthe_hole_composite ...

Id range: 0

String range: 1

Weight by a field: ☐

Desc field: D6

Constrain data: ☒

Save constrained sample points?: ☐

Output location: ...

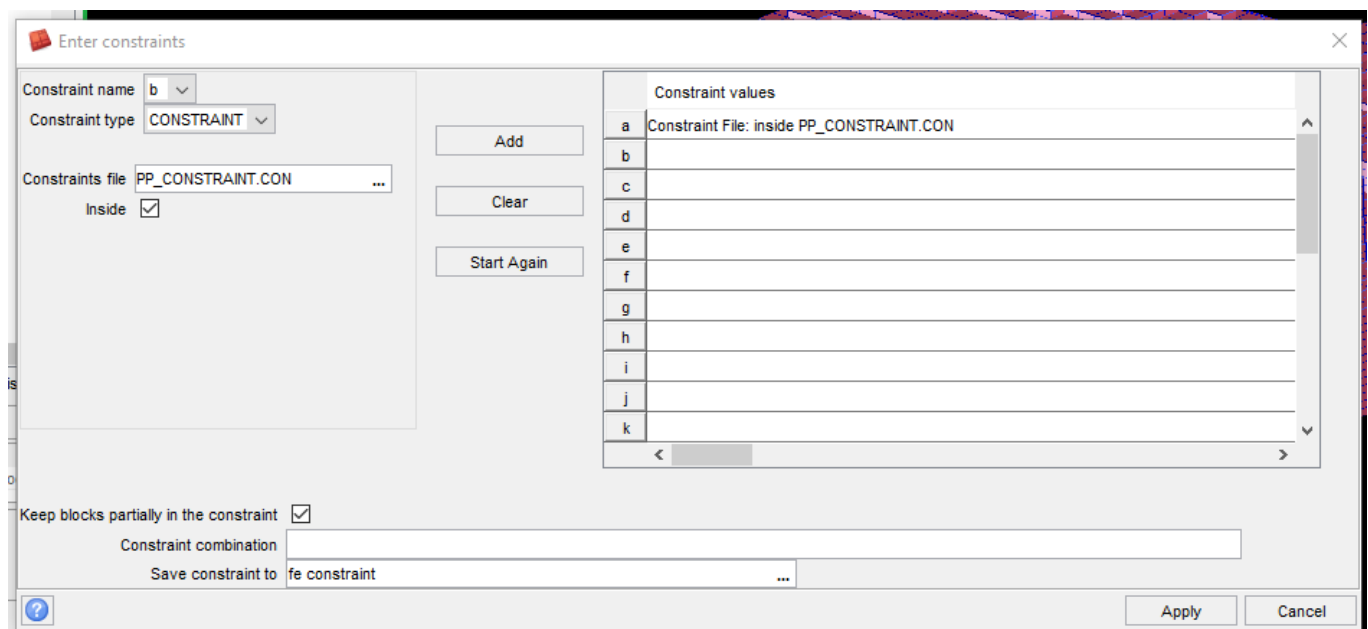
Output id number: 1

	Attribute to Fill	Description Field	Attribute Name	Anisotropic dist to nearest sample	True dist to nearest sample	Average anisc
1	fe	1				

ALERT
 Constraining the data source uses a block model constraint to determine which sample points to select. Consequently, if using a geometric constraint like inside a 3DM, above a DTM, etc., the sample points selected will not comply exactly with the geometric boundary. Rather they will be consistent with the block model constraint of the geometric boundary. This is an approximation

Apply Cancel

- After clicking on apply a new pop up will ask you to edit constraints



Enter constraints

Constraint name: b

Constraint type: CONSTRAINT

Constraints file: PP_CONSTRAINT.CON

Inside: ☒

Add

Clear

Start Again

Constraint values

a	Constraint File: inside PP_CONSTRAINT.CON
b	
c	
d	
e	
f	
g	
h	
i	
j	
k	

Keep blocks partially in the constraint: ☒

Constraint combination:

Save constraint to: fe constraint

Apply Cancel

- Select the pp constraint file for constraint file, click on add, click on keep blocks partially in the constraint and save the constraint as **fe constraint**.
- Click on apply
- Pop up

Search parameters

Standard Grade dependent

Search type ☒ Ellipsoid ☐ Octant

Minimum number of samples to select 3

Maximum number of samples to select 15

Maximum search radius 800

Maximum vertical search distance 10

Constrain by drill hole? ☐

Desc field containing hole id D2

Maximum number of samples per drill hole 15

Estimation Block Size (X, Y, Z) 25, 25, 10

Search Ellipsoid Specifications

Ellipsoid Orientation

Bearing 9.54

Plunge 0

Dip 10.82

Anisotropy Ratios

major / semi-major 1

major / minor 1

Rotation Convention Surpac ZXY LRL

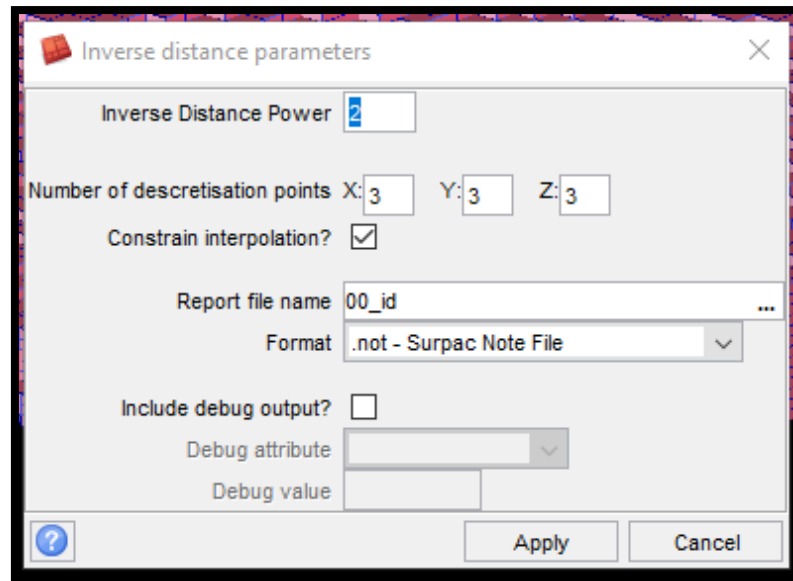
☒ Z ☒ X ☐ Y ☒ X ☐ Y ☐ Z

☐ L ☐ R ☐ L ☒ R ☐ L ☐ R

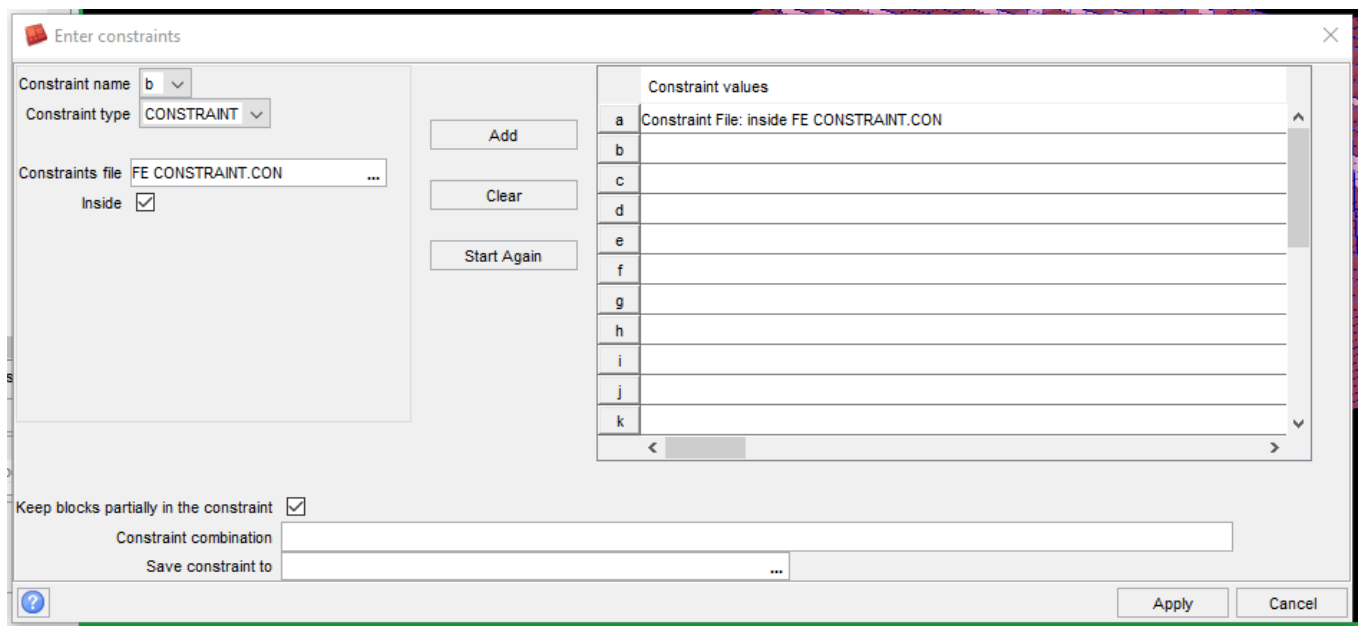
Ellipsoid Visualiser

Apply Cancel

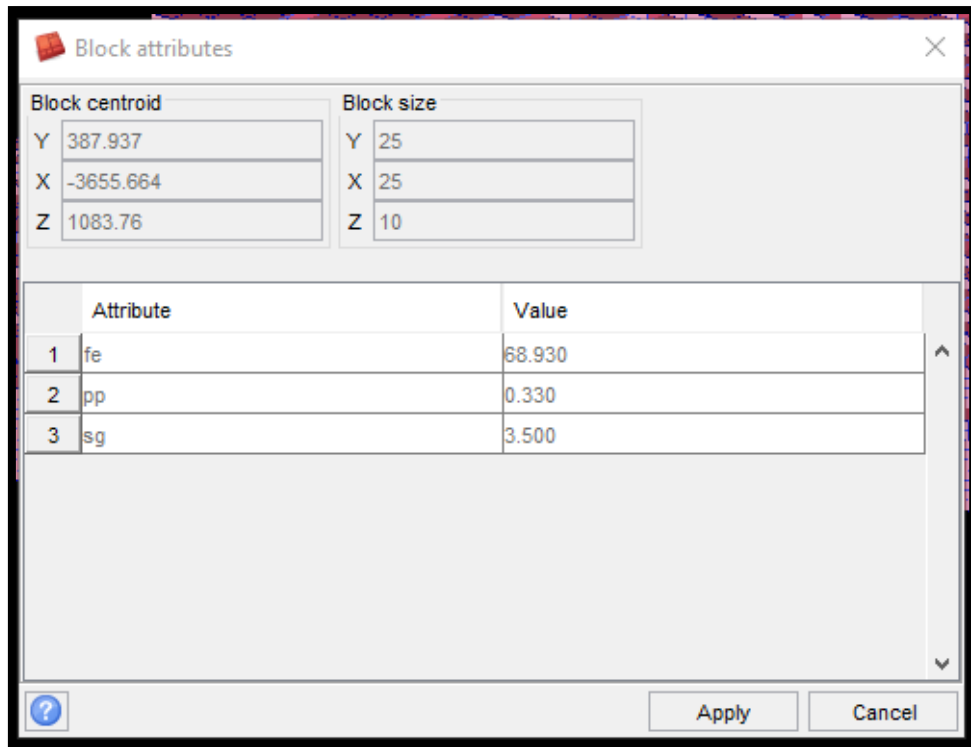
- Search type will be ellipsoid, minimum number of samples is 3 and maximum number of samples will be 15.
- Enter the bearing and dip from the anisotropy report.
- Click on Apply
- Pop up



- Select the inverse distance power and define a name for report file name
- **Click on apply**
- **Pop up**

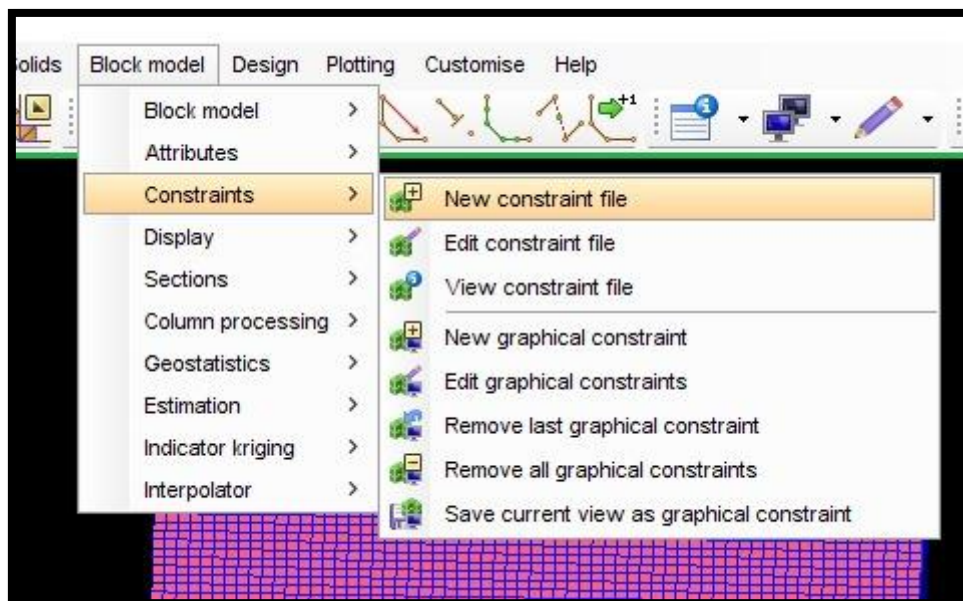


- New pop up will ask you to enter constraints, select **fe constraint** for constraint file, click on add, click on keep blocks partially in the constraint.
- **Click on Apply.**
- Now the assay value has been added to the block model.
- To check the assay value of each block, **Go to Block model >> Attributes >> view attribute for one block >> Click on any block.**



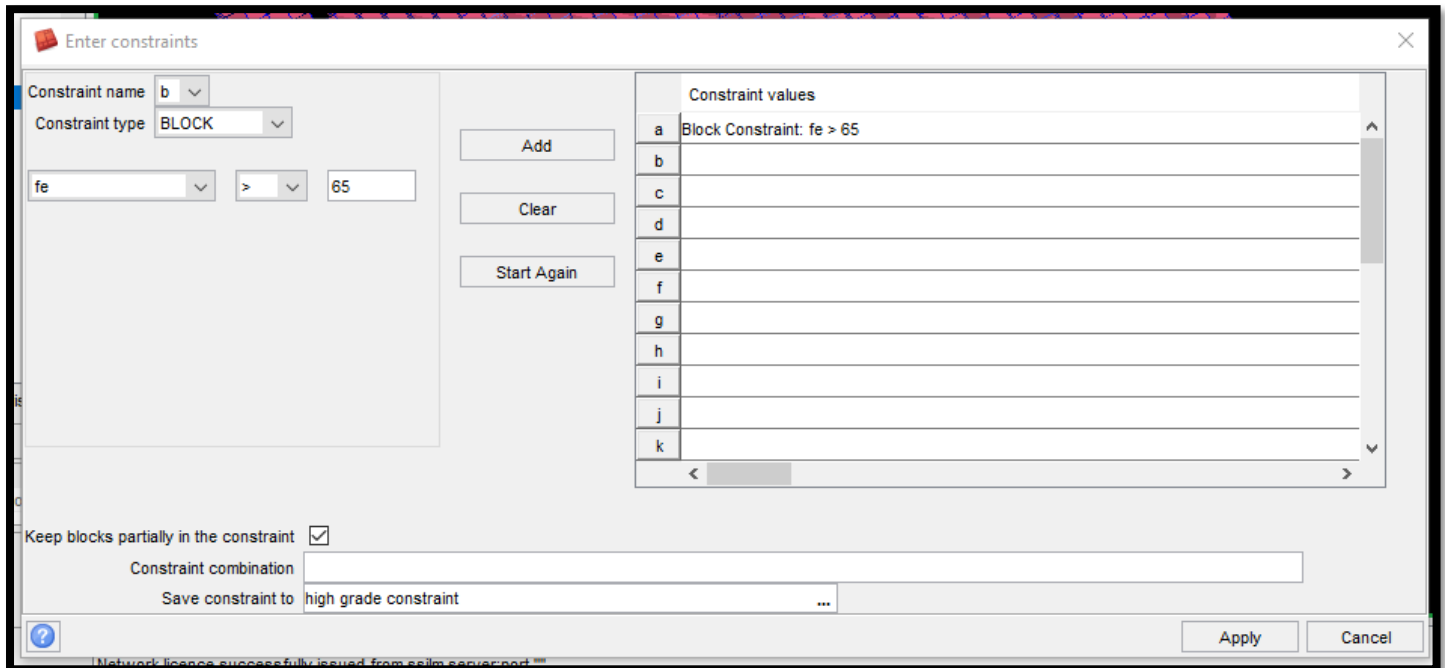
Adding high grade ore Constraint:

- Constraining the block model with high grade ore
- This high-grade constraint will eliminate all the low-grade blocks and only high-grade ore blocks will be displayed.
- **Block model >> constraints >> new constraint file >> Pop up**



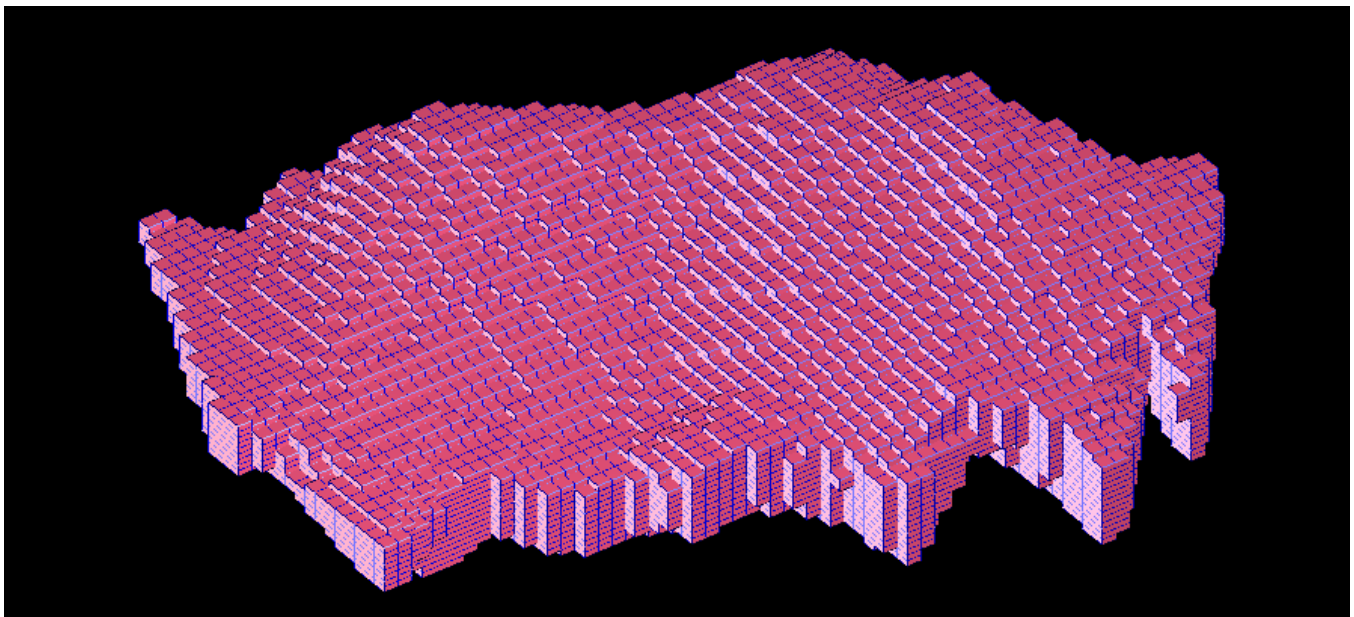
- Select block in constraint type, select fe grade >65, Click on add.

- Click on keep blocks partially inside the constraint, give a name to the output file as high-grade ore constraint.
- **Click on Apply**

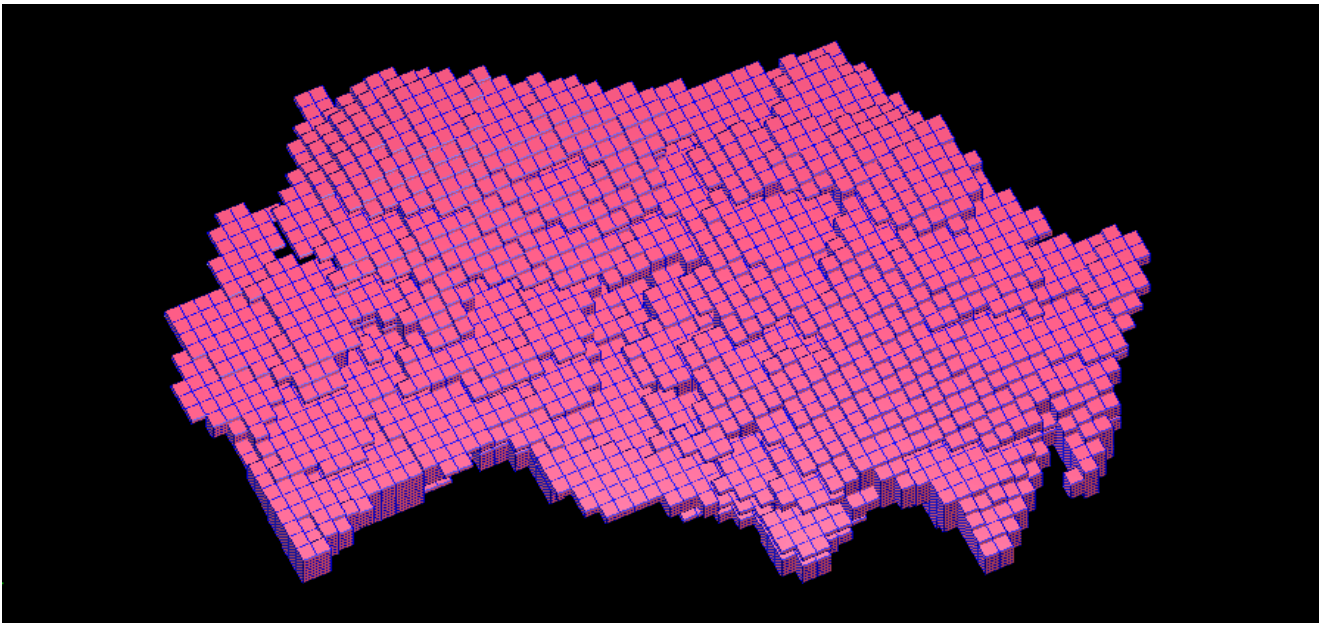


- A new constraint file will be saved in your work directory. drag that constraint to the main graphics layer.

Before Constraining with high grade ore:



After Constraining with high grade ore:



Volume calculation of block model:

Reserve volume calculation:

- **Connect the block model**
- **Apply the fe and pp constraints to the block model.**
- **Go to Block model >> report >> Pop up**

Block model report format file

Format file name ...

Output report file name ...

Output report file format ▾

Report type ☒ Standard report
☐ Multiple percent report
☐ Indicator kriged model
☐ Modify format
☒ Constrain ?

- Give the output report file name as reserve report. **Click on Apply**
- **This report will be your reserve report because we are considering the whole low grade and high-grade block**

Block model report

Report description

Format the headers ☒

Remove the lines with zero volume ☒

Report volume and tonnes to 0 decimal places

Report attributes	Display the attribute	Low cut	Upper cut	Weight by	Report	Expression
a fe	☒			Mass	Average	

Volume adjustment

Use volume adjustment ☒

Attribute pp

Density adjustment

☐ None

☒ Attribute sg

☐ Value

Geometric grouping

Group geometrically None

Grouping attributes	Numeric range
1	

Fill all cells for the group attribute ☐

Enable pivot table compatibility ☐

Use partial percentage ☐

Precision 3

Partial percentage attribute

Apply Cancel

- **Select fe as report attributes, click on use volume adjustment and select pp attribute, density attribute will be the sg attribute.**
- **Click on Apply**
- **Pop up**

Enter constraints

Constraint name: **b**

Constraint type: **CONSTRAINT**

Constraints file: **FE CONSTRAINT.CON**

Inside: ☒

Buttons: Add, Clear, Start Again

Constraint values:

	Constraint values
a	Constraint File: inside FE CONSTRAINT.CON
b	
c	
d	
e	
f	
g	
h	
i	
j	
k	

Keep blocks partially in the constraint: ☒

Constraint combination:

Save constraint to:

Buttons: Apply, Cancel

- Select the constraint type as constraint and upload the fe constraint file, click on add
- Click on apply

```

reserve report - Notepad
File Edit Format View Help
GEOVIA
Block model report
Block Model: 00.mdl

Constraints used
a.  INSIDE CONSTRAINT FE CONSTRAINT

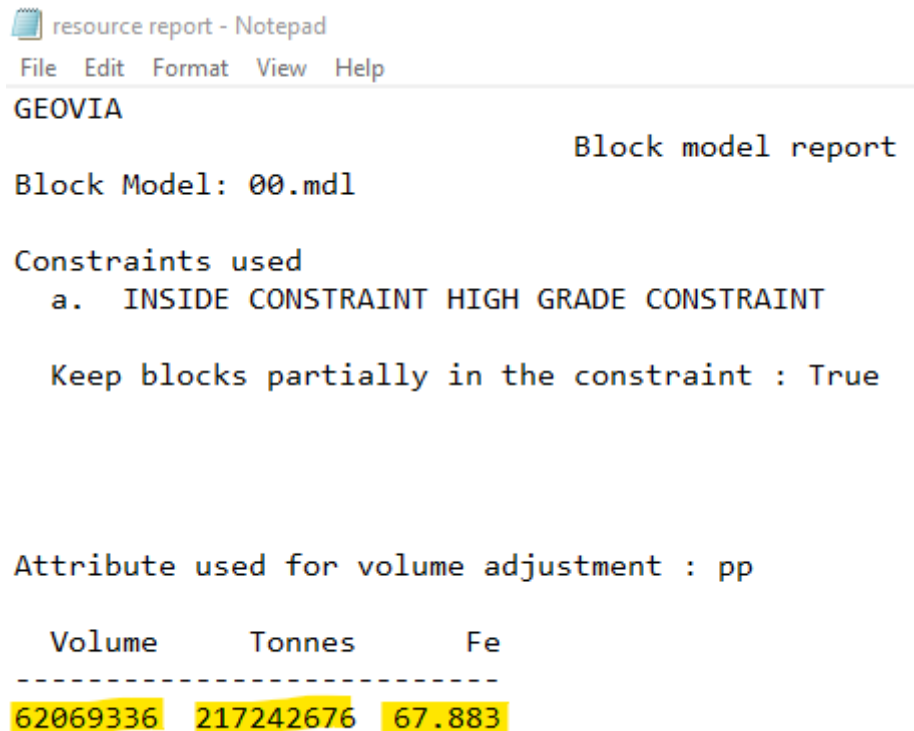
Keep blocks partially in the constraint : True

Attribute used for volume adjustment : pp

Volume      Tonnes      Fe
-----
88930029    311255103    63.453
  
```

Resource volume calculation:

- For resource calculation follow the same process as reserve calculation but upload the high-grade constraint file in the place of fe constraint.
- This report will be your resource report because we considered only high-grade ore blocks in the constraint.



```
resource report - Notepad
File Edit Format View Help
GEOVIA
Block model report
Block Model: 00.mdl

Constraints used
a. INSIDE CONSTRAINT HIGH GRADE CONSTRAINT

Keep blocks partially in the constraint : True

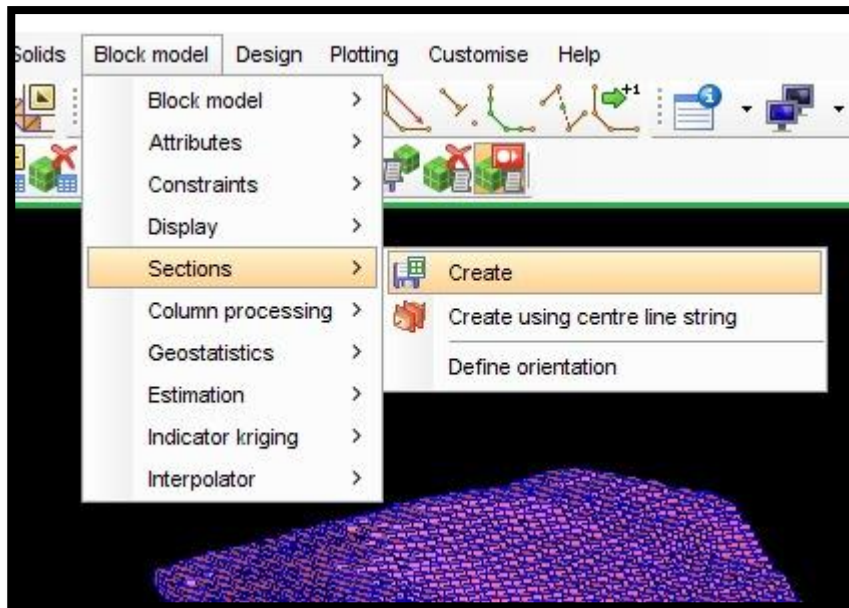
Attribute used for volume adjustment : pp

Volume      Tonnes      Fe
-----
62069336    217242676    67.883
```

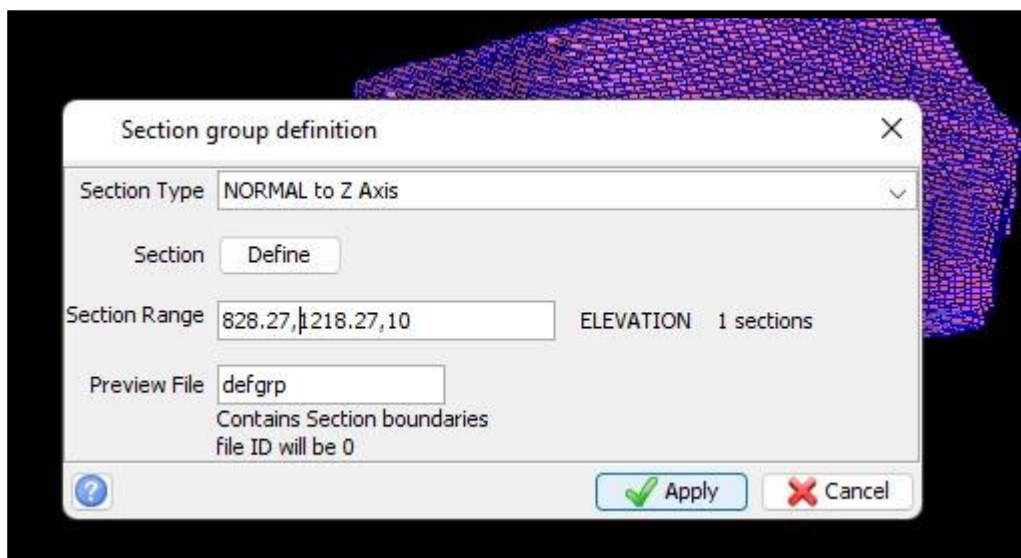
- The fe value showing in the report file is the average grade value of the whole blocks present in that constraint.

Extracting bench slices from block model:

- **Go to Block model >> Sections >> create**

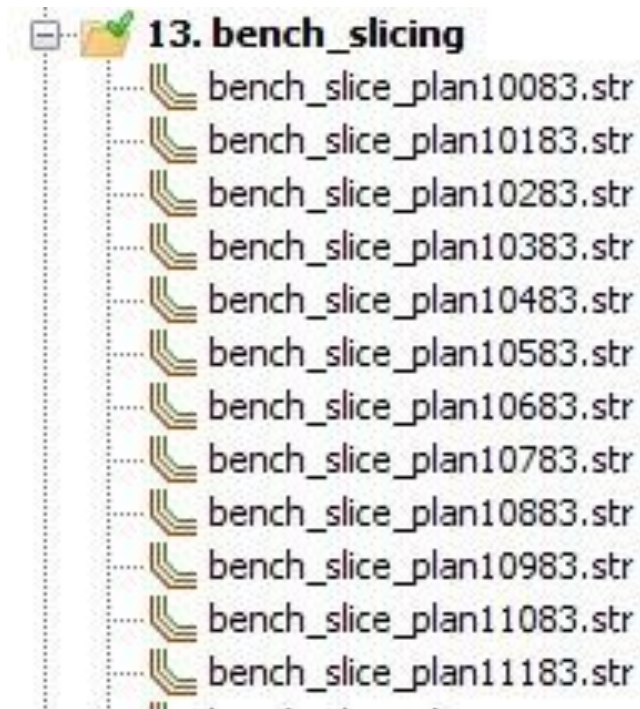


➤ **Pop up**

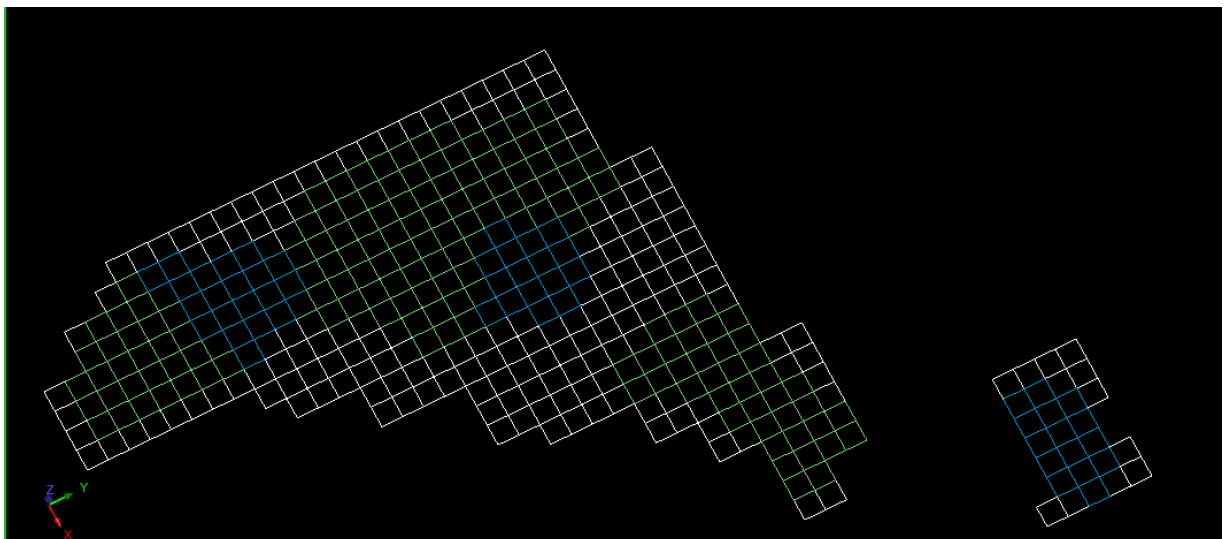


- Select normal to z axis in section type, because we want slices from block model with respect to elevation.
- In section range type your minimum and maximum extents of Z, using comma.
- In section range type min z value, max z value, bench height (for this pit it is 10m).
- **Click on Apply**
- **Pop up**
- **Define** a name to the slices as **bench_slice_plan**, **Click on Apply**

- Pop up
- Check on constrained model, model attribute is fe, numeric range will be 0;45;50;60;70;75
- Click on Apply



- Bench slice plans are created as string files in your work directory.



- Bring the bottom slice plan to the main graphics layer and type DRWDESC command in command box.
- This command shows the assay value of each block in that slice plan.

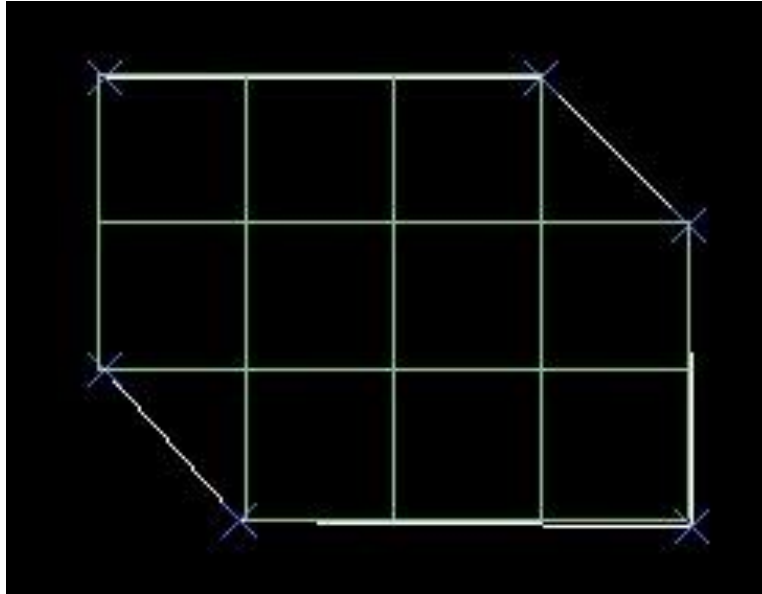
		0.000	0.000	0.000	0.000	65.578	65.577		
		0.000	0.000	65.579	65.578	65.572	65.567		
723	66.726	66.726	66.723	65.579	65.577	65.566	65.543	65.575	0.000
726	66.767	66.749	66.725	66.721	65.578	65.576	65.575	65.577	0.000
726	66.745	66.740	66.724	66.721	0.000				

- If the block shows 0, that means it is waste and we don't consider that waste blocks in pit design.
- If all your slice plan doesn't show any assay value in the blocks, it means you didn't do the estimation and inverse distance properly.
- You have to do it all over again, if you didn't find any coloured blocks in your slice plan.

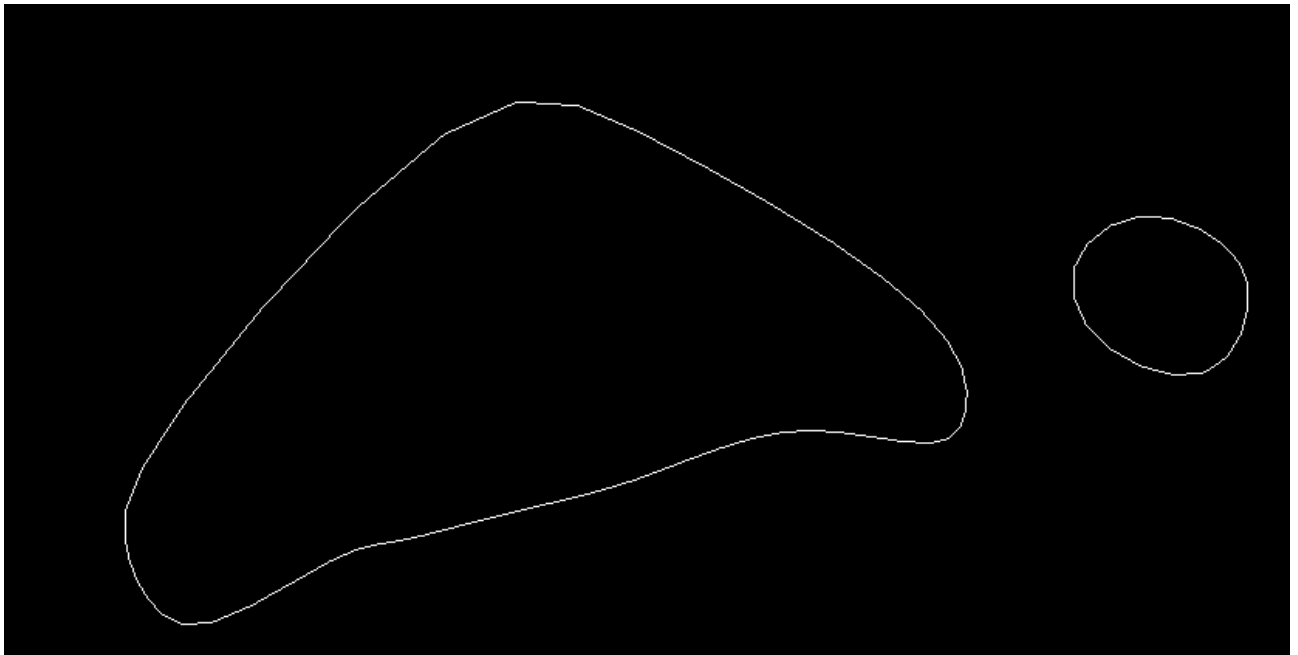
PIT DESIGN

- Create a new layer and name that new layer as pit design
- Bring the bottom slice and digitise the slice with ore blocks.
- Blocks that have assay value more than 0, will be displayed in different colours in the slice plan.

- Waste blocks will be displayed as white blocks.
- Only digitise the coloured blocks

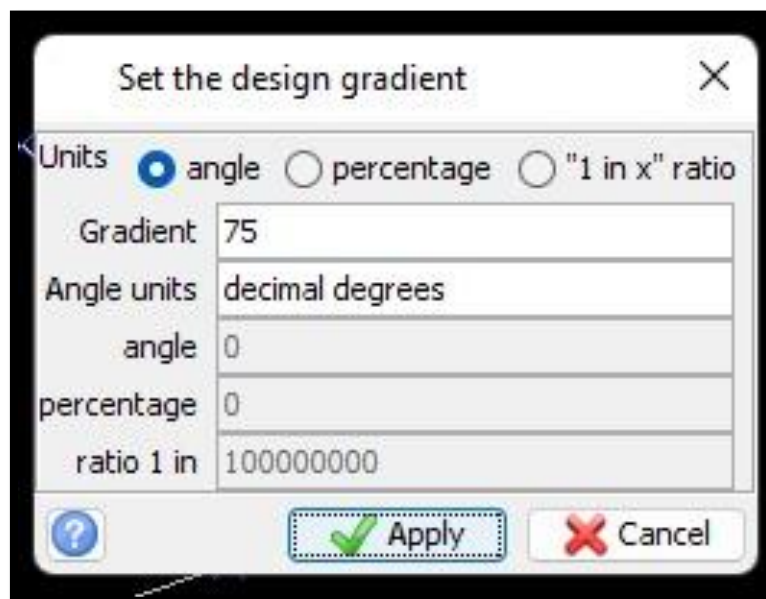
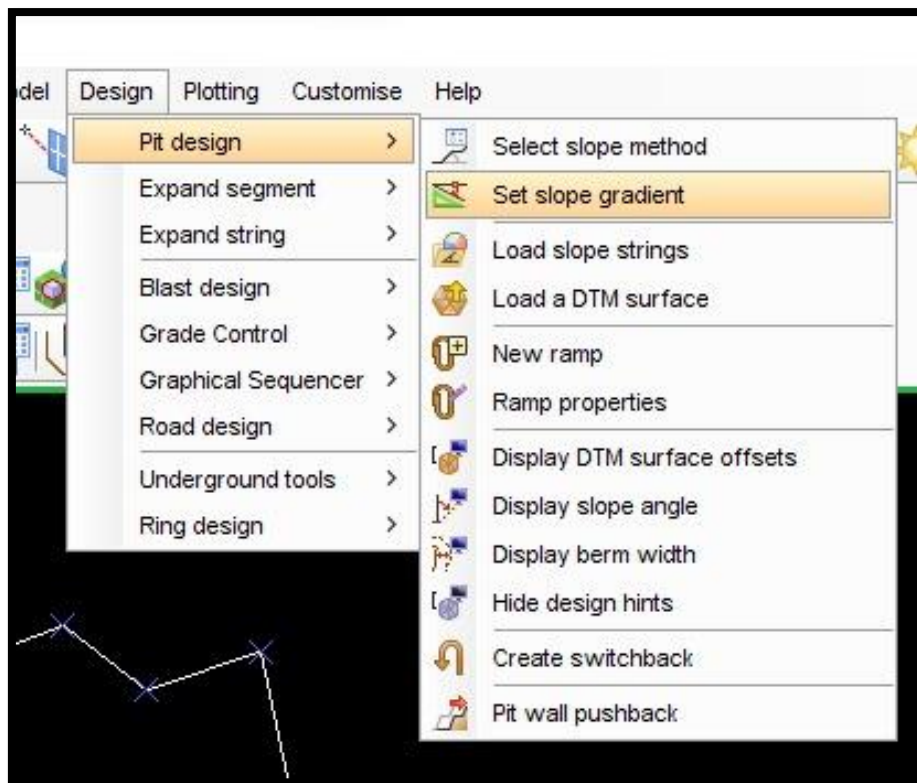


- Save the digitised layer
- Smooth the digitised segment, **Edit>> Segment>> Smooth.**



Set slope Gradient:

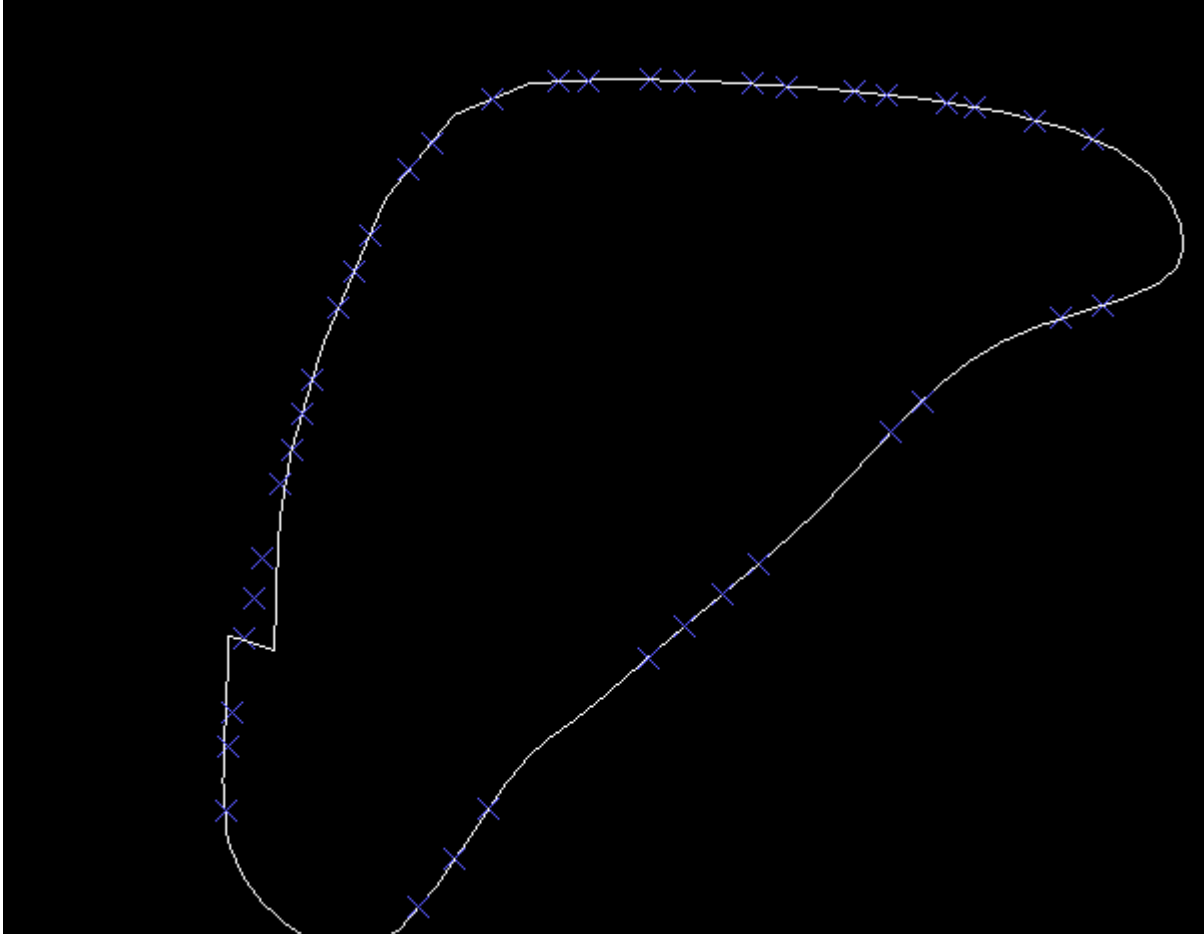
- Define the slope gradient, **Go to pit design >> set slope gradient.**
- **Pop up**



- Units- angle, Gradient-75
- **Click on Apply.**
- Before designing the ramp, must check the properties of the segment that you digitised.
- If the elevation Z is showing 0 in the description. The z value can be assigned manually
- **Go to string>> Maths >> Enter the elevation of your slice plan**
- **Click on Apply.**

Creating a ramp:

- **Before creating a ramp, you must digitise the ramp assuming the best location for your pit**



- **Normalise the segment**
- **Edit >> point >> move**
- **Move the points to design the ramp.**
- **Save the layer.**

- Go to pit design >> new ramp >> pop up

Define a new ramp

Define the ramp characteristics

Ramp name: Ramp2

Ramp String: 102

Ramp type: anti-clockwise

Ramp width: 20

Ramp gradient 1: 16

Gradient method: ☒ inside edge
☐ outside edge
☐ centre of ramp

Define the berm crossing characteristics

exit at: ☐ crest
☒ toe
☐ crest and toe
☐ neither

Berm taper distance: 15

Define the switchback characteristics

Switchback angle: 180

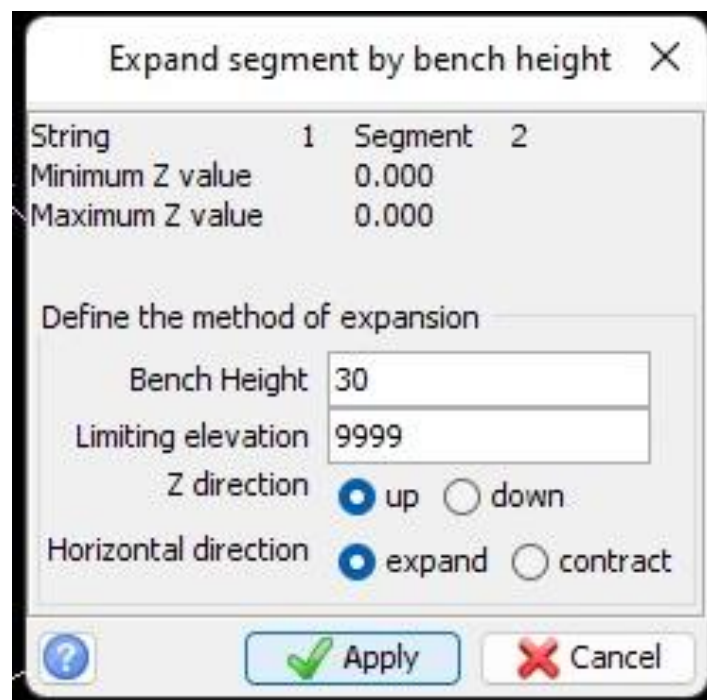
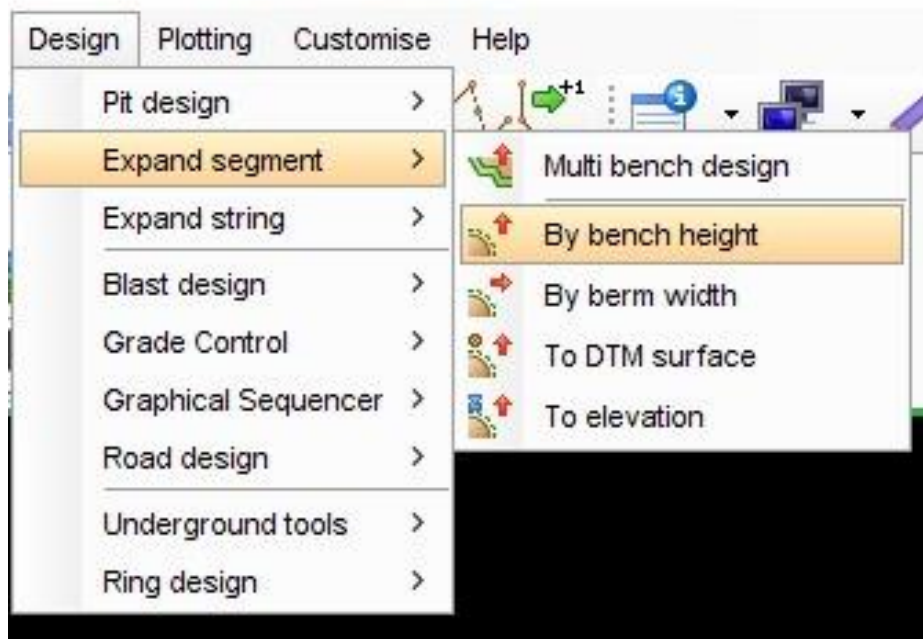
Switchback inner radius: 0

? Apply Cancel

- Give a name to the ramp
- Ramp string number should match the digitised string number.
- Choose the ramp type Clockwise/anti-clockwise
- Ramp width 20m, ramp gradient will be 1 in 16
- Gradient method will be inside edge.
- Click on Apply.
- Select the first and second points of the ramp.
- A new ramp will be created.

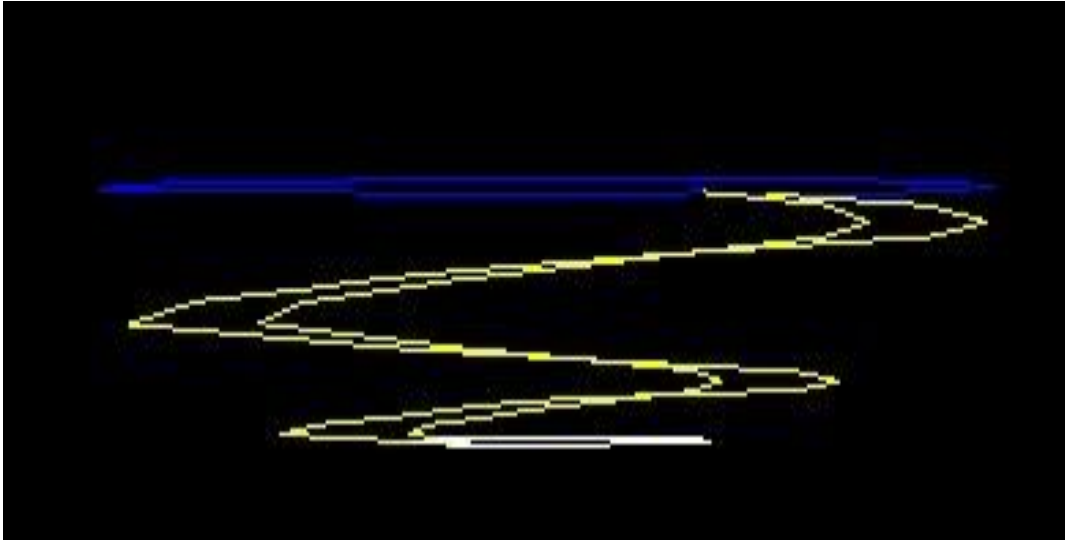
Expanding by bench height:

- The new ramp will not show up until you expand it by bench height.
- **Go to pit design >> expand segment >> by bench height**
- **Pop up**



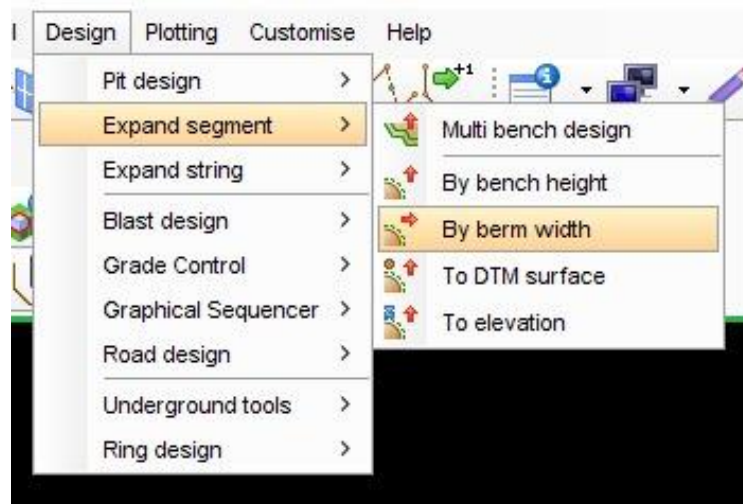
- **Define the bench height**
- **Z direction should be upwards because we are doing the pit design from bottom to top.**
- **Click on expand**

- Click on Apply
- The new ramp will be expanded by the bench height



Expanding the pit by berm width:

- Go to pit design >> expand segment >> by berm width



- Pop up

Expand Segment by Berm Width [X]

String 4 Segment 1

Minimum Z value 888.2700000000002

Maximum Z value 888.2700000000007

Define the method of expansion

Berm creation method

Limiting elevation

Delta Z limit

Berm width method ☒ fixed width

☐ descriptions

☐ slope strings

☐ block model

Berm width attribute

Default berm width

Berm gradient (%)

Horizontal direction ☒ expand

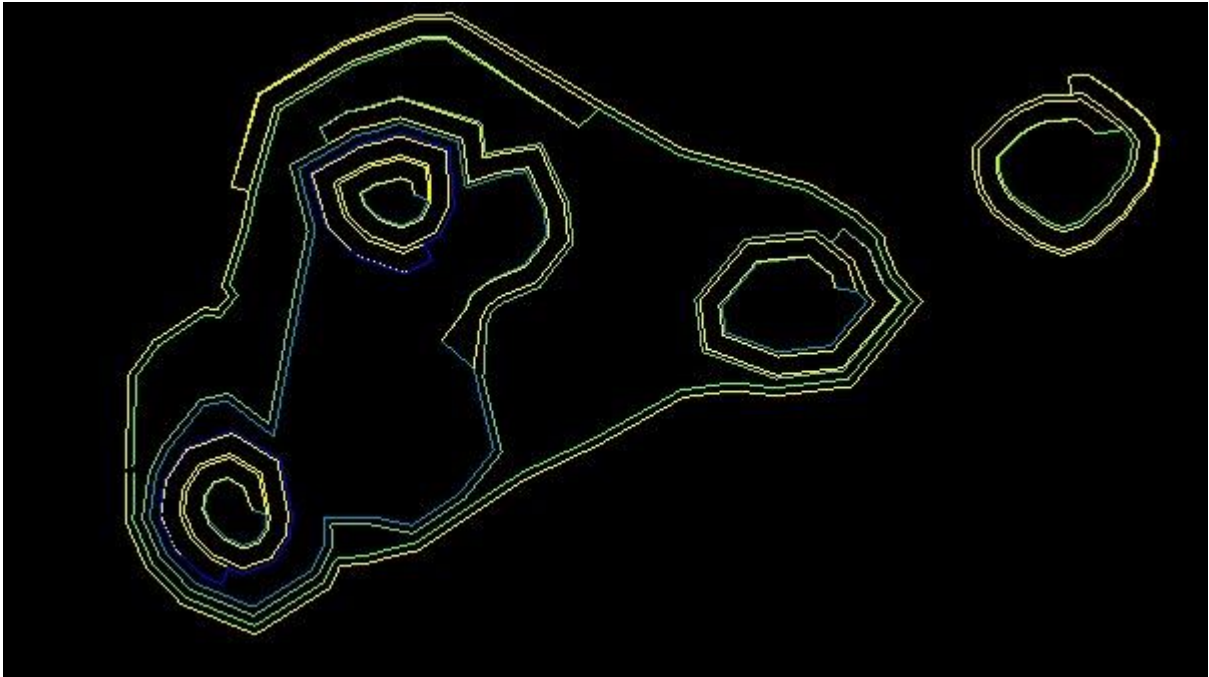
☐ contract

Switchback characteristics

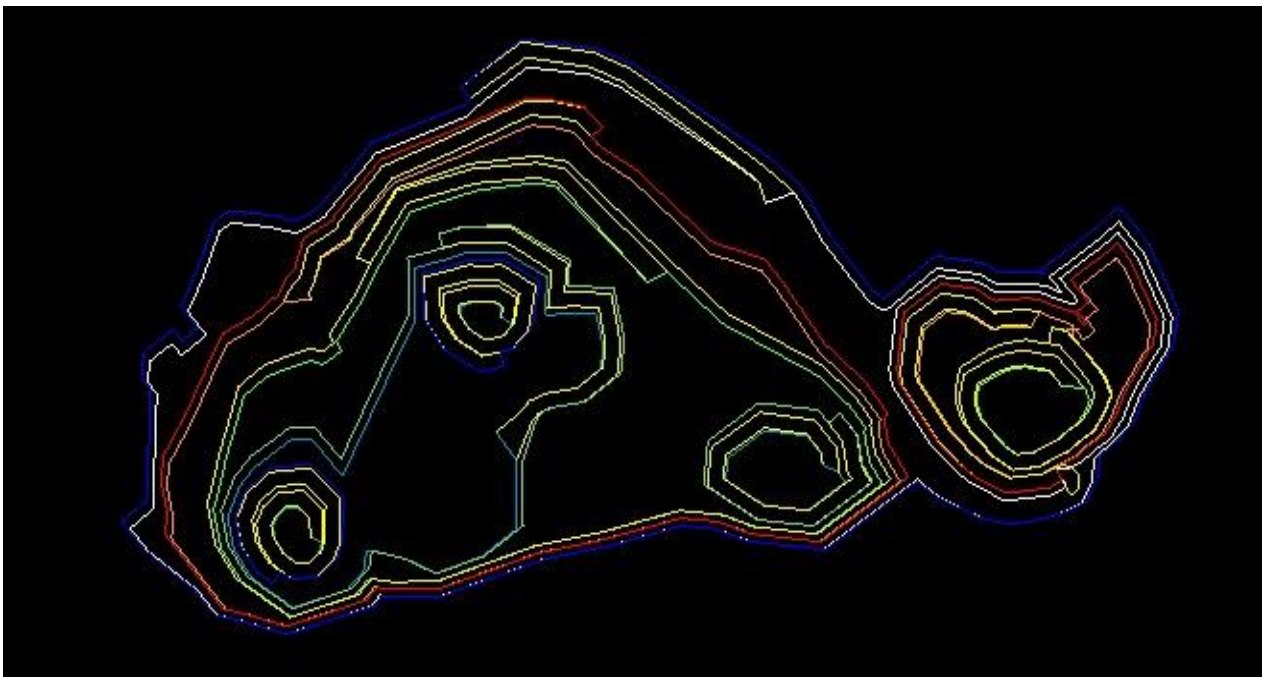
	Add switchback	Ramp name	Switchback angle	Switchback inner radius
1	☐	Ramp3	180	0

[?] [Apply] [Cancel]

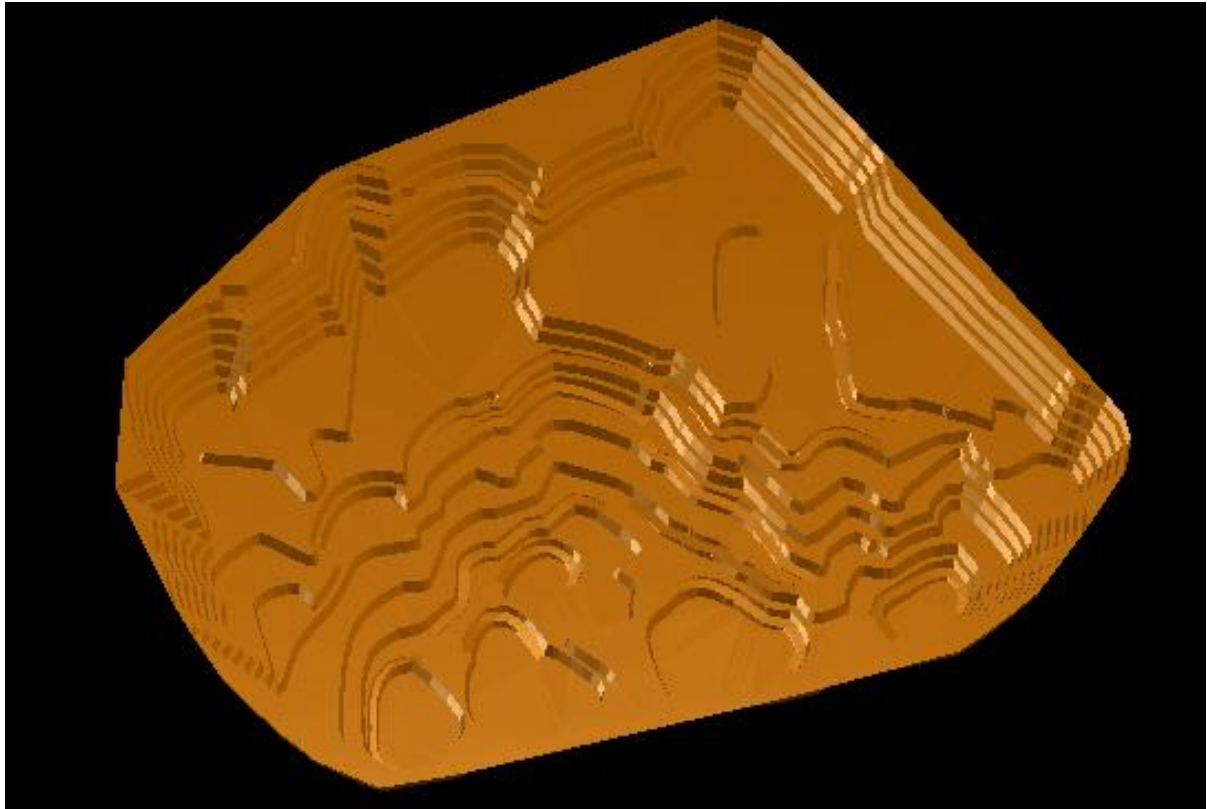
- Select the berm width method as fixed width
- Define the berm width
- In horizontal direction Click on expand
- Click on apply.
- A new segment will be created
- Now design the next ramp in that expanded segment to reach next bench.
- Follow the same procedure for all the slice plans
- Digitise >> ramp design >> expanding by bench height >> expanding by berm width >> next bench
- In some slice plans you can have more than one pit, in such case digitise both pits in same layer, and design the ramps for both separately.



➤ Both pits will be merged when designing the next bench



Ultimate pit configuration:



Overall pit clipped with topography:

